

IDENTIFICATION CONSTRAINTS IN
BAYESIAN FACTOR MODELS

A COMPARATIVE STUDY OF BAYESIAN FACTOR MODELS
UNDER POSITIVE LOWER TRIANGULAR AND UNORDERED
GENERALIZED LOWER TRIANGULAR IDENTIFICATION
CONSTRAINTS

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Lay Abstract

When researchers collect data with many measurements, such as economic indicators, gene expression levels, or survey responses, it is often the case that a smaller number of hidden factors derive the observed data. Factor analytic models attempt to uncover these hidden factors from the observed measurements. The drawback to factor models is that they fail to obtain a unique solution without additional constraints, making them difficult to interpret. Different structural constraints may be imposed to resolve this issue. However, it remains unclear which constraints are robust to noisy variables and structural misspecification, and which structure best uncovers the relevant factors.

In this thesis, we compare two structural constraints under a Bayesian framework, known as the Positive Lower Triangular (PLT) and Unordered Generalized Lower Triangular (UGLT) constraints. We describe the customized Markov Chain Monte Carlo (MCMC) algorithms needed to fit each model, and compare their performance through simulated experiments.

Our results show that certain, non-structural prior specifications do not adapt well to the presence of noisy variables in the data. We also saw that the UGLT structure is better able to determine the number of relevant factors in comparison to the PLT structure.

Abstract

Factor analytic models provide a framework for explaining high-dimensional data through a reduced set of latent variables. A fundamental challenge in these models is structural non-identification: the likelihood is invariant to orthogonal rotations, sign-switching, and column permutations of the factor loadings. To obtain a unique solution, structural constraints must be imposed. Two approaches based on sparsity-induced identification are the positive lower triangular (PLT) constraint and the unordered generalized lower triangular (UGLT) constraint. The PLT constraint resolves the rotation problem by imposing a fixed triangular geometry on the factor loadings, whereas the UGLT constraint generalizes this to a more flexible sparsity pattern.

In this thesis, we provide a comparative study of Bayesian factor models under these two identification strategies. We review the relevant prior specifications for each, including a column wise spike and slab prior that aids in inferring the number of relevant factors, and examine the corresponding Markov chain Monte Carlo (MCMC) algorithms by implementing them in R. Central to both algorithms is parameter expansion (PX), a technique that improves MCMC convergence for latent variable models by sampling from an expanded model that preserves the observed-data likelihood. Specifically, we demonstrate how a PX scheme based on the generalized inverse Gaussian distribution can be used to accelerate MCMC. For the PLT model, we show

that integrating PX into the Gibbs sampler overcomes the slow mixing often inherent in rigid triangular constraints. For the UGLT model, we extend the partially collapsed Gibbs sampler of Frühwirth-Schnatter et al. [2025b] by incorporating PX and adding a post-processing step to determine the number of relevant factors.

We evaluate the performance of both models through a series of simulation studies across varying numbers of observations, variables, and signal-to-noise ratios, under both PLT and UGLT data-generating structures. Model accuracy is assessed by comparing the amount of variance explained by the factor loadings and idiosyncratic variances against their true values, as well as computing the MSE of the posterior estimates of the factor loadings. The quality of the posterior predictive distribution is evaluated using the continuous ranked probability score (CRPS). We also examine the effect of parameter expansion through MCMC diagnostics, including trace plots, ACF plots, and effective sample size.

The results indicate that the UGLT model shows a clear advantage when the data-generating process follows a UGLT structure. Under a PLT structure with noisy variables, the sparse PLT and UGLT models perform worse than the non-sparse PLT model, suggesting that the spike-and-slab column sparsity prior does not adapt well when irrelevant variables are present. Notably, while the sparse models do identify zero loadings, they do not consistently assign them to the correct rows. We designed another simulation setting to test how well each model identifies the relevant factors, we fit both models with more factors than the true number. We find that post-processing under the UGLT model better identifies the irrelevant factors compared to the sparse PLT model.

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Notation, Definitions, and Abbreviations

Notation

x	A real valued constant
X	A real valued random constant
$\mathbf{x}$	A real vector of constants
$\mathbf{x}$	A real valued random vector
$\mathbf{X}$	A real matrix of constants
$\mathbf{X}$	A real valued random matrix
δ	An indicator matrix
$\text{Cov}(\mathbf{x})$	Denotes the covariance matrix of $\mathbf{x}$
$p(x)$	Probability distribution of the random variable X
$P_X(x)$	Cumulative distribution of the random variable X evaluated at x .

$\mathbf{I}_q$	A q dimensional identity matrix
$\mathbf{1}_S(x)$	The indicator function for whether the statement S is true for an element x .
$\mathcal{N}(\mu, \sigma^2)$	A univariate normal distribution with mean μ and variance σ^2
$\mathcal{N}_v(\boldsymbol{\mu}, \Sigma)$	A v variate normal distribution with mean μ and variance σ^2
$\mathcal{U}(c, d)$	A uniform distribution on the interval $[c, d]$
$\mathcal{B}(\alpha, \beta)$	A beta distribution with parameters α, β
$\mathcal{G}^{-1}(a, b)$	An inverse gamma distribution with shape parameter α and scale parameter β
Δ_a	A Dirac (point mass) distribution at a .

Abbreviations

UGLT	Unordered generalized lower triangular
GLT	Generalized Lower Triangular
PLT	Positive lower triangular
i.i.d	Independent and identically distribution

Chapter 1

Introduction

A statistical model is a model that attempts to describe how some observed data has been generated, and the model itself is specified by a set of parameters whose true value is unknown. Inference based on data is performed by finding a description of said parameters that best fits the data. For example, we might consider a psychological study, where we observe responses to a questionnaire on ratings of a subject's level of satisfaction, stress, happiness, and anger.

We can formalize a statistical model for the questionnaire as follows. Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be a list of p -dimensional random vectors representing the data collected from the hypothetical study, so $\mathbf{y}_i$ is a vector that represents subject i 's responses to the questionnaire. We define the likelihood of the data to be a family of distributions $p(\mathbf{y}_1, \dots, \mathbf{y}_n \mid \boldsymbol{\theta})$ where $\boldsymbol{\theta}$ is a d -dimensional vector of parameters which lies in a set Θ , called the parameter space. If we assume that $\boldsymbol{\theta}$ is fixed, one of the ways to perform inference on the parameters for a statistical model is known as maximum likelihood inference. The goal of maximum likelihood inference is to find the value of $\boldsymbol{\theta}^{MLE}$ such

that:

$$\boldsymbol{\theta}^{MLE} = \arg \sup_{\boldsymbol{\theta} \in \Theta} p(\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n \mid \boldsymbol{\theta}). \quad (1.0.1)$$

However, viewing probabilities as a measure of belief, we may consider $\boldsymbol{\theta}$ as random since it is unknown. This is known as the Bayesian paradigm of statistics. In Bayesian statistics, it is required that we impose a prior distribution on the values of the parameters, which we denote as $p(\boldsymbol{\theta})$. The prior distribution reflects our belief on what the parameter values are. If a practitioner places a higher prior probability to a range of values of $\boldsymbol{\theta}$, it should indicate that the practitioner believes that it is more likely that the true value of $\boldsymbol{\theta}$ lies in that range of values. To perform inference in the Bayesian setting, we derive the distribution of $\boldsymbol{\theta}$ given the data $\mathbf{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_n)$ known as the posterior distribution. By Bayes theorem:

$$\pi(\boldsymbol{\theta} \mid \mathbf{Y}) = \frac{p(\mathbf{Y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathbf{Y} \mid \boldsymbol{\theta})dp(\boldsymbol{\theta})}. \quad (1.0.2)$$

Often, the integral in the denominator (the marginal distribution of $\mathbf{y}$) of Eq. (1.0.2) is not computationally tractable. There are many computational algorithms that allow us to sample from the posterior distribution when it is only known up to a proportionality constant. These algorithms will be discussed in Chapter 3.

Recall the data for the psychology study. It's clear that the true level of emotion within a subject is usually unobservable, and thus can be modeled as a continuous latent variable in our model. Latent variable models are a very broad class of statistical models, and an important subclass of these models that will be the focus of this thesis are factor analytic models. Factor analytic models aim to summarize a large set of observed variables in terms of a smaller set of factors. Formally, this is stated

as follows:

For $i = 1, \dots, N$:

$$\mathbf{y}_i = \Lambda \mathbf{f}_i + \mathbf{e}_i, \quad (1.0.3)$$

$\mathbf{y}_i$: is a standardized $v \times 1$ random vector representing the observable variables.

Λ : is a $v \times q$ matrix of constants, called the matrix of factor loadings or the loading matrix.

$\mathbf{f}_i$: is a $q \times 1$ random vector representing the factors.

$\mathbf{e}_i$: is a $v \times 1$ random vector of error terms.

Or, in matrix form:

$$\mathbf{Y} = \Lambda \mathbf{F} + \mathbf{E}, \quad (1.0.4)$$

$\mathbf{Y}$: is a standardized $v \times N$ random matrix representing the observable variables.

The i 'th column of $\mathbf{Y}$ is equal to $\mathbf{y}_i$.

Λ : is a $v \times q$ matrix of factor loadings or the loading matrix.

$\mathbf{F}$: is a $q \times N$ random matrix representing the factors. The i 'th column of $\mathbf{F}$ is equal to $\mathbf{f}_i$.

$\mathbf{E}$: is a $v \times N$ random matrix of error terms. The i 'th column of $\mathbf{E}$ is equal to $\mathbf{e}_i$.

It is most commonly assumed that $\mathbf{f}_i \stackrel{iid}{\sim} \mathcal{N}_q(0, \mathbf{I}_q)$, $\mathbf{e}_i \stackrel{iid}{\sim} \mathcal{N}_v(0, \Sigma)$ and they are independent. As Eq. (1.0.3) suggests, we seek a method to best summarize a large set of variables to a smaller set of latent factors that are linear mixtures of the variables. Note that even in trying to only capture the linear relationship between factors and observed data, the model in Eq. (1.0.3) requires additional constraints for

identification (see Section 2.1). The model assumptions on the error terms and factors outlined above also aid interpretability. Specifically, they allow us to decompose $\text{Cov}(\mathbf{y}_i)$ as follows:

$$\begin{aligned}\text{Cov}(\mathbf{y}_i) &= \Lambda \text{Cov}(\mathbf{f}_i) \Lambda^\top + \Sigma \\ &= \Lambda \Lambda^\top + \Sigma.\end{aligned}\tag{1.0.5}$$

Now, to understand how the variables in $\mathbf{y}_i$ are correlated, one only needs to examine the factor loading matrix Λ , leading to a parsimonious representation of the correlation structure among the variables in $\mathbf{y}_i$. As high-dimensional data has become increasingly common in recent years, the utility of factor analytic methodologies has become increasingly apparent. Furthermore, advances in computational methods such as Markov Chain Monte Carlo (MCMC) have made Bayesian methods more accessible for factor analysis.

A fundamental challenge in factor analysis is structural non-identification: the likelihood is invariant to orthogonal rotations of the factor loadings. As a consequence, the interpretations of these factors are not unique. To obtain a unique solution, structural constraints must be imposed on the loading matrix. Two approaches based on sparsity induced identification are the positive lower triangular (PLT) constraint, which imposes a fixed triangular geometry, and the unordered generalized lower triangular (UGLT) constraint, proposed by Frühwirth-Schnatter et al. [2025b], which generalizes the PLT constraint to a more flexible sparsity pattern.

In this thesis, we provide a comparative study of Bayesian factor models under these two identification strategies. We begin by reviewing the necessary background in Bayesian factor analysis, along with a discussion of model identification constraints, in Chapter 2. In Chapter 3, we provide an overview of Markov chain Monte Carlo

(MCMC) methods and how to diagnose MCMC algorithms. In Chapter 4, we describe customized MCMC algorithms for the Bayesian factor models under the PLT and UGLT constraints. In Chapter 5, we present a series of simulation studies comparing the two structural constraints, assessing model performance through goodness-of-fit measures, the continuous ranked probability score (CRPS), mean squared error (MSE), and proportion of variance explained (Noise Ratio). See Section 5.1.1 for an overview of these performance metrics. We also describe post-processing procedures to evaluate whether the UGLT model can recover the true number of factors where the model is fit with more factors than the true amount.

Chapter 2

Bayesian Factor Models

In this chapter, we introduce the problem of model identification for factor analytic models and discuss identification schemes to address these issues. We then discuss Bayesian factor analytic models and some priors that aid in regularizing parameter estimation and induce structural constraints on the loading matrix.

2.1 Model Identification

In both frequentist and Bayesian statistics, a statistical model is said to be identified if the likelihood function is injective in the parameters — that is, no two distinct parameter values yield the same likelihood. Model identification is essential for reliable estimation and inference. In practice, however, a stronger requirement is needed: identifiability must hold with respect to the observed data likelihood, which we define below.

Suppose we have observed data $\mathbf{y} = (\mathbf{y}_1, \dots, \mathbf{y}_n)$ and latent variables $\mathbf{z} = (\mathbf{z}_1, \dots, \mathbf{z}_n)$, where each $\mathbf{y}_i \in \mathbb{R}^p$ and $\mathbf{z}_i \in \mathbb{R}^k$. Suppose the complete data $(\mathbf{y}, \mathbf{z})$ is governed by a

model with parameters $\boldsymbol{\theta}$. The complete data likelihood is defined as

$$p(\mathbf{y}, \mathbf{z} \mid \boldsymbol{\theta}). \quad (2.1.1)$$

The observed data likelihood is obtained by marginalizing over the latent variables:

$$p(\mathbf{y} \mid \boldsymbol{\theta}) = \int p(\mathbf{y}, \mathbf{z} \mid \boldsymbol{\theta}) d\mathbf{p}(\mathbf{z}). \quad (2.1.2)$$

Identifiability is then required with respect to $p(\mathbf{y} \mid \boldsymbol{\theta})$, since the latent variables $\mathbf{z}$ are unobserved in practice. For example, the factor model in (1.0.3) is not identified. The observed data likelihood for the model is $\mathcal{N}_v(0, \Lambda\Lambda^\top + \Sigma)$. To show non-identifiability, it suffices to show that the conditional covariance matrix of $\mathbf{y}_i$ given (Λ, Σ) does not have a one-to-one relationship with Λ and Σ . Let $\mathbf{R}$ be any $q \times q$ orthogonal matrix. Then, setting $\tilde{\Lambda} = \Lambda\mathbf{R}$ provides us the following:

$$\begin{aligned} \text{Cov}(\mathbf{y}_i \mid \tilde{\Lambda}, \Sigma) &= \Lambda\mathbf{R}\mathbf{R}^\top\Lambda^\top + \Sigma = \Lambda\Lambda^\top + \Sigma \\ &= \text{Cov}(\mathbf{y}_i \mid \Sigma, \Lambda) \end{aligned} \quad (2.1.3)$$

2.1.1 Constraints for Model Identification

The problem of identification in factor analytic models can be solved via model constraints. There have been several model restrictions proposed to handle the problem of identification. The first step is to identify $\Lambda\Lambda^\top$ and Σ from Eq. (1.0.5). Then, we attempt to impose restrictions to identify Λ from $\Lambda\Lambda^\top$.

Definition 2.1.1 (Row Deletion Property [Anderson et al., 1956]). A $v \times q$ factor loading matrix Λ satisfies the row-deletion property if, after deleting any row, two

disjoint sub matrices of rank q remain.

A necessary condition for the row-deletion property is that $\frac{v-1}{2} \geq q$. Further, as noted in Frühwirth-Schnatter et al. [2023a], if there are v_0 zero rows in Λ , we have a more stringent bound of $\frac{v-v_0-1}{2} \geq q$.

Theorem 5.1 in Anderson et al. [1956] states that the row deletion property provides a sufficient condition for identification of $\Lambda\Lambda^\top$ and Σ from $\text{Cov}(\mathbf{y}_i \mid \Lambda, \Sigma)$ in Eq. (1.0.5).

We outline two structural constraints that have proven to be sufficient to identify Λ from $\Lambda\Lambda^\top$. The most common constraint of the two is the positive lower triangular (PLT) structure (see examples in Geweke and Zhou [1996], Lopes and West [2004]).

Definition 2.1.2 (PLT structure). A $v \times q$ matrix Λ has a positive lower triangular (PLT) structure if all the pivot entries lie on the main diagonal, and the entries on the main diagonal are positive (see Eq. (2.1.4) for a 5×3 matrix).

$$\Lambda = \begin{pmatrix} \lambda_{11} & 0 & 0 \\ \lambda_{21} & \lambda_{22} & 0 \\ \lambda_{31} & \lambda_{32} & \lambda_{33} \\ \lambda_{41} & \lambda_{42} & \lambda_{43} \\ \lambda_{51} & \lambda_{52} & \lambda_{53} \end{pmatrix}, \quad \lambda_{11}, \lambda_{22}, \lambda_{33} > 0. \quad (2.1.4)$$

Recently, Frühwirth-Schnatter et al. [2023b] have generalized the PLT constraint so that the pivots of the loading matrix need not lie on the main diagonal. This

constraint is referred to as the Generalized Lower Triangular (GLT) constraint:

Definition 2.1.3 (GLT structure Frühwirth-Schnatter et al. [2023b]). Let l_j denote the row index of the pivot of column j . A matrix Λ has a GLT structure if the pivot rows are in increasing order, i.e. $l_1 < l_2 < \dots < l_q$ (see Eq. (2.1.5) for a 6×3 matrix).

$$\Lambda = \begin{pmatrix} \lambda_{11} & 0 & 0 \\ \lambda_{21} & 0 & 0 \\ \lambda_{31} & 0 & 0 \\ \lambda_{41} & \lambda_{42} & 0 \\ \lambda_{51} & \lambda_{52} & 0 \\ \lambda_{61} & \lambda_{62} & \lambda_{63} \end{pmatrix} \quad (2.1.5)$$

Note that any $v \times q$ matrix Λ with a PLT structure also has a GLT structure, since a PLT matrix has pivots on the main diagonal, corresponding to pivot rows $1, 2, \dots, q$. The GLT constraint on its own is not sufficient for the identification of Λ . Frühwirth-Schnatter et al. [2023b] show that if Λ has a GLT structure and also satisfies the 3579 counting rule, and the pivot entries are positive, then full identification of Λ and Σ is achieved.

Definition 2.1.4 (3579 Counting Rule [Frühwirth-Schnatter et al., 2023b]). A $v \times q$ factor loading matrix Λ satisfies the 3579 counting rule if, for each $r = 1, \dots, q$, every sub-matrix consisting of r columns of Λ contains at least $2r + 1$ non-zero entries.

2.2 Prior Specification

There are several advantages to adopting a Bayesian approach to factor analysis. Priors may be specified in order to impose sparsity in the loading matrix for interpretability. Moreover, model identification can be addressed via post-processing of posterior samples from an MCMC algorithm [Kaufmann and Schumacher, 2017]. More importantly, certain Bayesian factor models allow the number of factors q to remain unknown rather than being fixed as a modeling choice. This is typically accomplished either by post-processing posterior draws or by using reversible-jump MCMC methods, which allow the dimension of the parameter space to change across iterations. Bayesian nonparametric methods have also been proposed to allow the number of factors unconstrained [Bhattacharya and Dunson, 2011].

In what follows, we discuss several priors that may be imposed on the loading matrix Λ . For each prior specification, we assume that the covariance matrix of the error terms in Eq. (1.0.3) takes the form $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_v^2)$, with the following prior placed independently on each variance component. For $k = 1, \dots, v$:

$$\sigma_k^2 \sim \mathcal{G}^{-1}(a_\sigma, b_\sigma). \quad (2.2.1)$$

2.2.1 Priors to Impose the PLT Constraint Model

We begin our discussion of prior selection with a prior that imposes the PLT constraint on the factor loadings. We follow the work of Lopes and West [2004], who impose a PLT constraint on Λ through the following prior:

$$p(\Lambda_k) = \mathcal{N}_{\min(k,q)}(\mathbf{0}, c_0 \mathbf{I}_q) \mathbf{1}_{[\Lambda_{kk} > 0]}(\Lambda_{kk}), \quad (2.2.2)$$

where Λ_k denotes the unconstrained entries in the k 'th row of Λ , and $c_0 \in (0, \infty)$.

In practice, we may replace Eq. (2.2.2) with a prior that only imposes partial identification of the loading matrix, and then apply an appropriate transformation of the parameters to achieve full identification without affecting the model likelihood. To illustrate this approach, suppose that we enforce only the lower trapezoidal part of the PLT constraint through the prior:

$$p(\Lambda_i) = \mathcal{N}_{\min(i,q)}(\mathbf{0}, c_0 \mathbf{I}_q). \quad (2.2.3)$$

While this does not ensure full model identification, it does ensure identification up to a sign change of the columns. That is, if $\Lambda \Lambda^\top + \Sigma = \Lambda^* \Lambda^{*\top} + \Sigma^*$, then $\Lambda^* = \Lambda S$, where S is a $q \times q$ diagonal matrix with entries ± 1 . To see that the model likelihood is invariant under such a transformation, note that

$$\mathbf{y}_i = \Lambda \mathbf{f}_i + \mathbf{e}_i = \Lambda S S^\top \mathbf{f}_i + \mathbf{e}_i. \quad (2.2.4)$$

Since $\mathbf{f}_i \sim \mathcal{N}_q(\mathbf{0}, \mathbf{I}_q)$ and $S^\top S = \mathbf{I}_q$, it follows that $S^\top \mathbf{f}_i \sim \mathcal{N}_q(\mathbf{0}, \mathbf{I}_q)$, and thus the model likelihood is invariant under the transformation

$$(\Lambda, \Sigma, \mathbf{f}_i) \mapsto (\Lambda S, \Sigma, S^\top \mathbf{f}_i). \quad (2.2.5)$$

Since Λ is lower trapezoidal, there exists a sign matrix S^* such that ΛS^* has a PLT structure. This allows us to achieve full model identification via post-processing of the posterior draws.

2.3 Imposing Sparsity on the Loading Matrix

Sparsity in the loading matrix is desirable for two reasons. The first is that sparse loading matrices are easier to interpret. The second is that we are able to determine which factors are relevant based on the number of non-zero entries in the corresponding columns of Λ . To see this, recall the matrix form of the factor model in Eq. (1.0.4):

$$\mathbf{Y} = \Lambda \mathbf{F} + \mathbf{E}. \quad (2.3.1)$$

If it so happens that the r 'th column of Λ is a zero column, then the r 'th row of $\mathbf{F}$ is unrelated to $\mathbf{Y}$. In other words, the r 'th factor for each observation is unrelated to the observed data. The other possible case is when a column of Λ only has one non-zero entry. Suppose the r 'th column of Λ has only 1 non-zero entry λ_{kr} . Note

$$\begin{aligned}
\mathbf{y}_i &= \Lambda \mathbf{f}_i + \mathbf{e}_i \\
&= \begin{pmatrix} \left| \begin{array}{cccc} \cdots & 0 & \cdots \\ \vdots & \vdots & \vdots \\ \Lambda_1 & \cdots & \lambda_{kr} & \cdots & \Lambda_q \\ \vdots & \vdots & \vdots \\ \cdots & 0 & \cdots \end{array} \right| \end{pmatrix} \begin{pmatrix} F_{1i} \\ \vdots \\ F_{qi} \end{pmatrix} + \begin{pmatrix} E_{1i} \\ \vdots \\ E_{vi} \end{pmatrix} \\
&= \Lambda_1 \mathbf{f}_1 + \Lambda_2 \mathbf{f}_2 + \cdots + \begin{pmatrix} 0 \\ \vdots \\ \Lambda_{kr} F_{ri} \\ \vdots \\ 0 \end{pmatrix} + \cdots + \Lambda_q \mathbf{f}_q + \begin{pmatrix} e_{1i} \\ \vdots \\ e_{vi} \end{pmatrix} \tag{2.3.2} \\
&= \Lambda_1 \mathbf{f}_1 + \Lambda_2 \mathbf{f}_2 + \cdots + \Lambda_q \mathbf{f}_q + \begin{pmatrix} E_{1i} \\ \vdots \\ \lambda_{kr} F_{ri} + E_{ki} \\ \vdots \\ E_{vi} \end{pmatrix} \\
&= \Lambda_{-r} \mathbf{f}_{-r,i} + \mathbf{e}'_i,
\end{aligned}$$

where $\mathbf{e}'_i = \begin{pmatrix} E_{1i} & \cdots & \lambda_{kr} F_{ri} & \cdots & E_{vi} \end{pmatrix}^\top$ and $\text{Cov}(\mathbf{e}'_i) = \text{diag}(\sigma_1^2, \dots, \lambda_{kr}^2 + \sigma_k^2, \dots, \sigma_v^2)$.

Thus we may regard the r 'th factor for each observation a spurious factor as its relationship with the observed data could easily be captured by the error terms.

To impose sparsity on the loading matrix, Frühwirth-Schnatter et al. [2025b] employ two strategies. The first is to implement a spike and slab prior, which allows posterior estimates of the loadings to be exactly zero:

$$\lambda_{ij} \mid \tau_j \sim (1 - \tau_j)\Delta_0 + \tau_j P_{\text{slab}}(\lambda_{ij}), \quad (2.3.3)$$

where Δ_0 denotes a point mass at zero (the “spike”), P_{slab} is a continuous distribution such as $\mathcal{N}(0, \zeta^2)$ (the “slab”), and $\tau_j \in [0, 1]$ represents the prior probability that the loadings in column j are exactly zero. Furthermore, a prior is placed on τ_j such as a $\mathcal{B}(\alpha, \beta)$ distribution.

The second strategy is to induce structural zeroes on Λ by implementing a generalization of the GLT structure called the Unordered Generalized Lower Triangular (UGLT) structure.

Definition 2.3.1 (UGLT structure). The matrix A is said to have a UGLT structure if there exists a permutation matrix P such that AP has a GLT structure. Thus, the UGLT structure, together with the 3579 counting rule and constraining the pivots to be positive, only identifies the factor loadings up to a permutation of the columns of Λ .

An equivalent way to define the UGLT structure is to say that a matrix A has a UGLT structure if and only if the pivots of A lie in different rows. To implement the UGLT structure, Frühwirth-Schnatter et al. [2025b] first define a prior on the pivot rows of λ such that the rows are always in different entries.

Let l_j denote the pivot row of column j , and let ℓ_{-j} denote the pivot rows other than l_j . Finally, define $\mathcal{L}(\ell_{-j})$ to be the set of row indexes that are not in the set ℓ_{-j} .

The prior on l_j is defined conditionally on ℓ_{-j} as follows:

$$p(l_j \mid \ell_{-j}) = \begin{cases} \frac{1}{v - q + 1}, & l_j \in \mathcal{L}(\ell_{-j}) \\ 0 & \text{Otherwise.} \end{cases} \quad (2.3.4)$$

Next, a matrix $\delta \in \{0, 1\}^{v \times q}$ is defined to impose the sparsity structure onto Λ . We define that if $\delta_{ij} = 1$ if and only if $\Lambda \neq 0$. The following prior is placed on δ conditionally on the pivots and $\boldsymbol{\tau} = (\tau_1, \dots, \tau_v)$ from the spike and slab distribution:

$$\Pr(\delta_{ij} = 1 \mid l_j, \tau_j) = \begin{cases} 0 & \text{if } i < l_j \\ 1 & \text{if } i = l_j \\ \tau_j & \text{if } i > l_j. \end{cases} \quad (2.3.5)$$

2.4 Shrinkage Priors

For ease of MCMC estimation, Frühwirth-Schnatter et al. [2025b] propose a hierarchical prior $p(\lambda_{ij} \mid \delta_{ij} = 1) = \mathcal{N}(0, \sigma_i^2 B_{ij})$ where a shrinkage prior on B_{ij} may be chosen as to regularize the estimates of λ_{ij} towards zero. One such option is to choose a column shrinkage prior, where B_{ij} is chosen be θ_j , with $\theta_j \sim \mathcal{G}^{-1}(a_\theta, b_\theta)$.

2.5 Models Considered

For the simulation studies presented in Chapter 5, we consider three models. The first follows a UGLT structure, closely following the prior specifications of Frühwirth-Schnatter et al. [2025b]. The remaining two models use a PLT structure. The first PLT model is implemented via Eq. (2.2.3) and Eq. (2.2.1). The second PLT model is

identical to the first, except that sparsity and shrinkage priors are applied in a similar fashion to the UGLT model. The priors used are described in Table 2.1.

Table 2.1: Prior specifications for each model. Here Δ_0 denotes a point mass at zero, l_j is the pivot index for column j .

Parameter	Prior
Non-sparse PLT	
λ_{kj}	$\mathcal{N}(0, c_0)$
σ_k^2	$\mathcal{G}^{-1}(a_\sigma, b_\sigma)$ height
Sparse PLT	
$\lambda_{kj} \mid \delta_{kj}, \sigma_k^2, \theta_j$	$(1 - \delta_{kj}) \Delta_0 + \delta_{kj} \mathcal{N}(0, \sigma_k^2 \theta_j)$
$\delta_{kj} \mid \tau_j$	$\mathbf{1}_{[k=j]}(k, j) + \text{Bernoulli}(\tau_j) \cdot \mathbf{1}_{[i>j]}(k, j)$
τ_j	$\mathcal{B}(\alpha, \beta)$
θ_j	$\mathcal{G}^{-1}(a_\theta, b_\theta)$
σ_k^2	$\mathcal{G}^{-1}(a_\sigma, b_\sigma)$ height
Sparse UGLT	
$\lambda_{kj} \mid \delta_{kj}, \sigma_k^2, \theta_j$	$(1 - \delta_{kj}) \Delta_0 + \delta_{kj} \mathcal{N}(0, \sigma_k^2 \theta_j)$
$\delta_{kj} \mid \tau_j, l_j$	$\mathbf{1}_{[k=j]}(k, j) + \text{Bernoulli}(\tau_j) \cdot \mathbf{1}_{[k>l_j]}(k, j)$
τ_j	$\mathcal{B}(a_\tau, b_\tau)$
θ_j	$\mathcal{G}^{-1}(a_\theta, b_\theta)$
σ_k^2	$\mathcal{G}^{-1}(a_\sigma, b_\sigma)$
$l_j \mid \ell_{-j}$	$\frac{1}{v - q + 1} \cdot \mathbf{1}_{[l_j \in \mathcal{L}(\mathcal{L}_{-j})]}$

Chapter 3

Markov Chain Monte Carlo

The Markov Chain Monte Carlo (MCMC) method works by constructing a Markov chain whose stationary distribution, that is, the distribution the chain converges to after sufficiently many transitions, is the target distribution that we desire to sample from. In many MCMC methods, it is not required for the target distribution to be known fully, but only up to some constant. Referring to Eq. (1.0.2), this means that it is possible to sample from $\pi(\boldsymbol{\theta} \mid y)$ without knowing $\int p(y \mid \boldsymbol{\theta})dp(\boldsymbol{\theta})$.

We will describe two fundamental Markov Chain Monte Carlo methods known as the Metropolis-Hastings algorithm [Metropolis et al., 1953, Hastings, 1970], and the Gibbs sampler [Geman and Geman, 1984, Gelfand and Smith, 1990]. Our aim is to give intuition on how these algorithms work without diving into the technical details of continuous-time Markov chains on general state spaces. For such details, refer to Liu [2001]. We will also provide some discussion for the more specialized techniques used in this thesis, namely parameter expansion and partially collapsed Gibbs sampling, both employed to improve the efficiency of MCMC algorithms.

3.1 The Metropolis-Hastings Algorithm

The Metropolis-Hastings algorithm is a method for generating samples from a target distribution $f(\omega)$ that is known only up to a normalizing constant [Metropolis et al., 1953]. The idea is to construct a Markov chain whose stationary distribution is f by proposing candidates from a proposal distribution $g(\cdot \mid \omega)$ and accepting or rejecting them according to an appropriate criterion. For simplicity, we first present the Metropolis algorithm, which assumes a symmetric proposal distribution, i.e. $g(\omega' \mid \omega_t) = g(\omega_t \mid \omega')$.

Algorithm 3.1.1 Metropolis Hastings

Suppose we would like to generate M samples:

1. Choose a starting value $\omega^{(0)}$.
 2. For $m = 1, \dots, M$:
 - (a) Generate a candidate ω' from $g(\cdot \mid \omega^{(m-1)})$.
 - (b) Compute the acceptance ratio $\alpha = \min\left(\frac{f(\omega')}{f(\omega^{(m-1)})}, 1\right)$.
 - (c) Generate $u \sim \text{Uniform}(0, 1)$.
 - (d) If $u \leq \alpha$, set $\omega^{(m)} = \omega'$. Otherwise, set $\omega^{(m)} = \omega^{(m-1)}$.
-

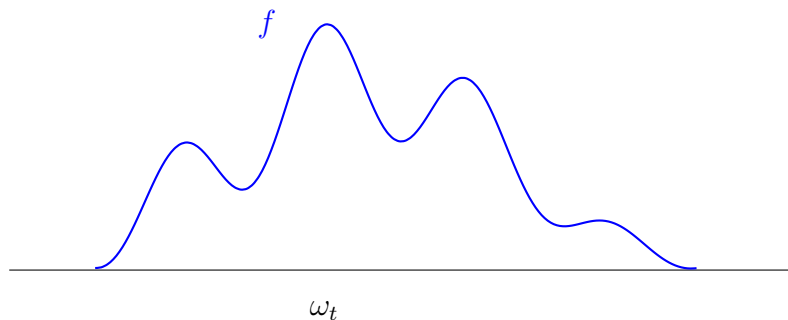
Since α is a ratio of the same density f evaluated at two points, any normalizing constants in f cancel. This is precisely what makes the algorithm useful in Bayesian inference, where the posterior is typically known only up to a constant.

Remark 3.1.1. The Metropolis-Hastings algorithm generalizes the above by accounting for asymmetric proposal distributions. The acceptance ratio becomes

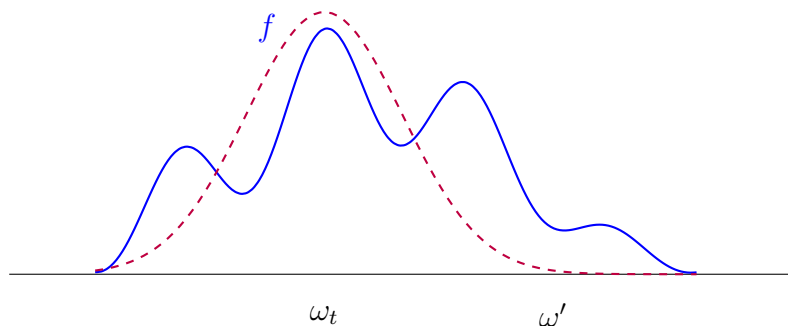
$$\alpha = \min \left(\frac{f(\omega')}{f(\omega^{(c-1)})} \cdot \frac{g(\omega^{(c-1)} | \omega')}{g(\omega' | \omega^{(c-1)})}, 1 \right). \quad (3.1.1)$$

When the proposal is symmetric, the ratio $g(\omega^{(c-1)} | \omega')/g(\omega' | \omega^{(c-1)})$ equals one and we recover the Metropolis algorithm.

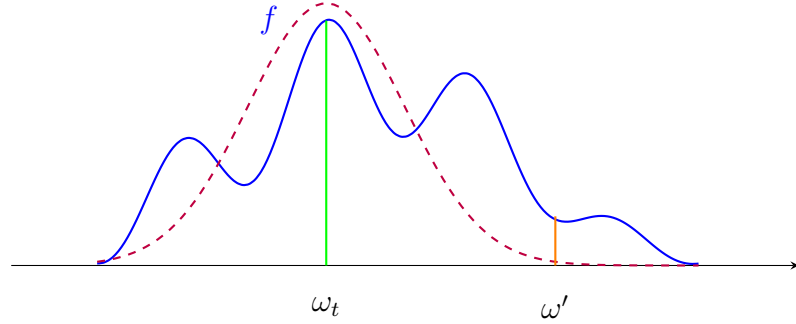
To build intuition for how the algorithm works, consider a visual example. Suppose we wish to sample from the density f depicted below, and that the chain is currently at ω_t .



Let the proposal distribution be $\mathcal{N}(\omega_t, 0.3)$, and suppose we draw a candidate ω' from this distribution.



The next step is to draw from $U \sim \mathcal{U}(0, f(\omega_t))$, and accept the draw if $U < f(\omega')$. This ensures that if candidates are drawn from a lower region of density (i.e., smaller values of f), then we only accept a proportion of α of the draws. This corresponds to the proportion of the amount of density at ω' relative to the density at ω_t . Since the probability of accepting this draw is $\frac{f(\omega')}{f(\omega_t)}$, we can achieve the same goal by sampling $U' \sim \mathcal{U}(0, 1)$, and accepting if $U' < \alpha = \frac{f(\omega')}{f(\omega_t)}$. Conversely, if ω' had landed in a region of higher density, we would have $\alpha \geq 1$ and the candidate would be accepted.



This accept-reject mechanism ensures that the chain spends more time in regions of high density, while still exploring regions of lower density. Over many iterations, the distribution of the samples converges to the target distribution f .

3.2 The Gibbs Sampler

The Metropolis-Hastings algorithm provides a general framework for sampling from complex, high-dimensional distributions. When the full conditional distributions are available in closed form and can be sampled directly, the Gibbs sampler offers a simpler and often more efficient alternative, and is described as follows:

Consider a d -dimensional random vector $\mathbf{w} = (W_1, \dots, W_d)$ with joint density $f(w_1, w_2, \dots, w_d)$, which is the target distribution from which we wish to sample.

Algorithm 3.2.1 Gibbs Sampler

1. Initialize the algorithm at $w^{(0)} = (w_1, w_2, \dots, w_d)$.
 2. Iteratively, for $m = 1, \dots, M$:
 - (a) Generate $w_1^{(m)}$ from $f(w_1 | w_2, \dots, w_d)$.
 - (b) Generate $w_2^{(m)}$ from $f(w_2 | w_1, w_3, \dots, w_d)$.
 - .
 - .
 - .
 - (c) Generate $w_d^{(m)}$ from $f(w_d | w_1, \dots, w_{d-1})$.
 3. Increment m .
-

Note that the Gibbs sampler is a special case of the Metropolis-Hastings algorithm. This can be shown by choosing the proposal distributions as conditional densities of the target distribution, as seen in the following example.

Suppose we are to generate a sample from the joint distribution $f(v, w)$. Assume that $f(v, w)$ is at least known up to a constant multiple, and that the conditional distributions $f(v | w)$ and $f(w | v)$ can be sampled from. Consider Algorithm 3.2.2, where we use the Metropolis Hastings framework to generate a sample for $f(v, w)$:

Algorithm 3.2.2 Gibbs Sampler as a special case of Metropolis Hastings

1. Initialize the algorithm at $(v^{(0)}, w^{(0)})$
2. Iteratively, for $m = 1, \dots, M$:
 - (a) Generate a candidate v' from $f(v | w^{(m-1)})$.
 - (b) Calculate the transition ratio $\alpha_1 = \frac{f(v', w^{(m-1)})}{f(v^{(m-1)}, w^{(m-1)})} \frac{f(v^{(m-1)} | w^{(m-1)})}{f(v' | w^{(m-1)})}$:

$$\begin{aligned}
 \alpha_1 &= \frac{\cancel{f(v' | w^{(m-1)})} f(w^{(m-1)})}{f(v^{(m-1)} | w^{(m-1)}) \cancel{f(w^{(m-1)})}} \frac{f(v^{(m-1)} | w^{(m-1)})}{\cancel{f(v' | w^{(m-1)})}} \\
 &= \frac{\cancel{f(w^{(c-1)})} f(v^{(c-1)} | w^{(c-1)})}{f(v^{(c-1)} | w^{(c-1)}) \cancel{f(w^{(c-1)})}} = \frac{\cancel{f(v^{(c-1)} | w^{(c-1)})}}{\cancel{f(v^{(c-1)} | w^{(c-1)})}} \\
 &= 1.
 \end{aligned}$$

- (c) Generate $u_1 \sim \mathcal{U}(0, 1)$.
 - (d) Since $u_1 \leq \alpha_1$ almost surely, we set $v^{(c)} = v'$.
3. We now generate a sample for w
 - (a) Generate a candidate w' from $f(w | v^{(c)})$. Note that we condition on $v^{(c)}$ rather than $v^{(c-1)}$. This is due to the fact that we have already transitioned from $v^{(c-1)}$ to $v^{(c)}$.
 - (b) Generate u_2 from $\mathcal{U}(0, 1)$.
 - (c) Calculate the second transition ratio $\alpha_2 = \frac{f(v^{(c)}, w')}{f(v^{(c)}, w^{(c-1)})} \frac{f(w^{(c-1)} | v^{(c)})}{f(w' | v^{(c)})}$.

$$\alpha_2 = \frac{\cancel{f(w' | v^{(c)})} f(v^{(c)})}{f(w^{(c-1)} | v^{(c)}) \cancel{f(v^{(c)})}} \frac{f(w^{(c-1)} | v^{(c)})}{\cancel{f(w' | v^{(c)})}} = \frac{\cancel{f(v^{(c)})} f(w^{(c-1)} | v^{(c)})}{\cancel{f(w^{(c-1)} | v^{(c)})} f(v^{(c)})} = 1.$$
 - (d) Since $u_2 \leq \alpha_2$ almost surely, we set $w^{(c)} = w'$.

Since the transition ratios α_1 and α_2 are always equal to 1, every proposed candidate is accepted, and the steps involving the transition ratio may be omitted entirely. This reduces to the standard form of the Gibbs sampler.

In the rest of this paper, we specify Gibbs samplers by specifying only the full conditional sampling steps. For each MCMC iteration, Algorithm 3.2.1 may be written as follows:

Algorithm 3.2.3 Gibbs Sampler

1. Generate $\omega_1^{(c)}$ from $f(\omega_1 \mid \omega_2, \dots, \omega_d)$
 2. Generate $\omega_2^{(c)}$ from $f(\omega_2 \mid \omega_1, \omega_3, \dots, \omega_d)$
 -
 -
 -
 3. Generate $\omega_d^{(c)}$ from $f(\omega_d \mid \omega_1, \dots, \omega_{d-1})$
-

3.3 Partially Collapsed Gibbs Sampling

In some cases, additional structure of the target density $f(\omega_1, \omega_2, \dots, \omega_d)$ may be known, such as certain marginal or conditional distributions in closed form. This additional structure can be exploited by marginalizing variables out of select Gibbs sampling steps to construct a Partially Collapsed Gibbs Sampler [Van Dyk and Park, 2008].

The Partially Collapsed Gibbs Sampler is constructed from a standard Gibbs sampler via three primary operations:

1. Marginalization, which involves moving random variables from being conditioned upon to being integrated over in one or several steps of the Gibbs sampler.
2. Permutation, which refers to reordering the steps in which components of the target distribution are sampled.
3. Trimming, which removes any redundant marginalized components from the sampler.

To illustrate these tools, we consider the example in Van Dyk and Park [2008].

Consider a Gibbs sampler with target distribution $p(w, x, y, z)$:

Algorithm 3.3.1 Gibbs Sampler

1. Sample $w^{(c+1)}$ from $p(w \mid x^{(c)}, y^{(c)}, z^{(c)})$.
 2. Sample $x^{(c+1)}$ from $p(x \mid w^{(c)}, y^{(c)}, z^{(c)})$.
 3. Sample $y^{(c+1)}$ from $p(y \mid w^{(c)}, x^{(c)}, z^{(c)})$.
 4. Sample $z^{(c+1)}$ from $p(z \mid w^{(c)}, x^{(c)}, y^{(c)})$.
-

We apply marginalization. Suppose that we sample from $p(y \mid z)$ and $p(z \mid y)$. We can then replace Steps 3 and 4 by sampling from $p(w, y \mid x^{(c)}, z^{(c)})$ and $p(w, z \mid x^{(c)}, y^{(c)})$ respectively. This does not affect the stationary distribution since we still sample from conditionals of the joint distribution without missing any necessary components. The resulting sampler is given in Algorithm 3.3.2.

Algorithm 3.3.2 Marginalized Gibbs Sampler

1. Sample w^* from $p(w \mid x^{(c)}, y^{(c)}, z^{(c)})$.
 2. Sample $x^{(c+1)}$ from $p(x \mid w^*, y^{(c)}, z^{(c)})$.
 3. Sample $(w^{**}, y^{(c+1)})$ from $p(w, y \mid x^{(c+1)}, z^{(c)})$.
 4. Sample $(w^{(c+1)}, z^{(c+1)})$ from $p(w, z \mid x^{(c+1)}, y^{(c+1)})$.
-

Algorithm 3.3.2 introduces two extra (and unnecessary) intermediate draws of the random variable w . We cannot yet remove any pair of the extra draws of w without affecting the output. This is because Step 2 conditions on w^* , and Step 4 is used in the next iteration of the sampler. However, if we first permute the steps we can simplify the algorithm. In this example, we permute the steps as follows:

Algorithm 3.3.3 Permuted Marginalized Gibbs Sampler

1. Sample $(w^*, y^{(c+1)})$ from $p(w, y \mid x^{(c+1)}, z^{(c)})$
 2. Sample $(w^{**}, z^{(c+1)})$ from $p(w, z \mid x^{(c)}, y^{(c+1)})$
 3. Sample $w^{(c+1)}$ from $p(w \mid x^{(c)}, y^{(c+1)}, z^{(c+1)})$
 4. Sample $x^{(c+1)}$ from $p(x \mid w^{(c+1)}, y^{(c+1)}, z^{(c+1)})$
-

Now, w^* and w^{**} are not conditioned upon in any step, and so we apply trimming to simplify the algorithm. In other words, we integrate out w^* and w^{**} in Steps 1 and 2, as the marginals are available to sample from.

Algorithm 3.3.4 Trimmed, Permuted Marginalized Gibbs Sampler

1. Sample $y^{(c+1)}$ from $p(y \mid x^{(c+1)}, z^{(c)})$
 2. Sample $z^{(c+1)}$ from $p(z \mid x^{(c)}, y^{(c+1)})$
 3. Sample $w^{(c+1)}$ from $p(w \mid x^{(c)}, y^{(c+1)}, z^{(c+1)})$
 4. Sample $x^{(c+1)}$ from $p(x \mid w^{(c+1)}, y^{(c+1)}, z^{(c+1)})$
-

Finally, note that drawing from $p(z \mid y)$ followed by $p(w \mid y, z)$ is equivalent to a single draw from $p(w, z \mid y)$. As a result, we combine Steps 2 and 3 to yield a Partially Collapsed Gibbs Sampler.

Algorithm 3.3.5 Partially Collapsed Gibbs Sampler

1. Sample $y^{(c+1)}$ from $p(y \mid x^{(c+1)}, z^{(c)})$
 2. Sample $(w^{(c+1)}, z^{(c+1)})$ from $p(w, z \mid x^{(c)}, y^{(c+1)})$
 3. Sample $x^{(c+1)}$ from $p(x \mid w^{(c+1)}, y^{(c+1)}, z^{(c+1)})$
-

Van Dyk and Park [2008] show that Partially Collapsed Gibbs Samplers have improved performance over their counterparts.

3.4 Parameter Expansion

Parameter Expansion is a method used to improve the convergence properties of MCMC algorithms, particularly for MCMC algorithms that are used to fit latent

variable models [Liu and Wu, 1999]. The general idea of parameter expansion is to introduce a new parameter to the prior distribution of the latent variables. To illustrate this, we examine the following setting used in Liu and Wu [1999].

Consider a latent variable model where y are the observed variables, and z are the latent variables. Assume that we model the joint distribution of y and z with $f(y, z \mid \theta)$, where θ is a parameter with a prior distribution $f(\theta)$. A data augmentation algorithm is one that iterates between sampling the latent variables z conditionally on (θ, y) , and then sample θ conditionally on (z, y) [Tanner and Wong, 1987]:

Algorithm 3.4.1 Data Augmentation Algorithm

1. Sample $z^{(c+1)}$ from $f(z \mid \theta^{(c)}, y)$.
 2. Sample $\theta^{(c+1)}$ from $f(\theta \mid z^{(c+1)}, y)$.
-

Now, suppose we modify the model where the observed data remains the same, but the latent variables are now w , and $w \sim p(w \mid \theta, \alpha)$. If the joint distribution $p(y, w \mid \alpha, \theta)$ satisfies $\int p(y, w \mid \alpha, \theta) p(\alpha) d\alpha = p(y, z \mid \theta)$, we refer to α as an expansion parameter. Proposition 1 in Liu and Wu [1999] states that if the joint prior $p(\alpha, \theta)$ agrees with $f(\theta)$, and α and θ are independent a priori, then the marginal posterior distribution of θ for both the expanded model and the original model are the same. Thus, an alternative way to generate a sample for the posterior distribution of θ is as follows:

Algorithm 3.4.2 Paramter Expanded Data Augmentation Algorithm

1. Sample $(\alpha^*, w^{(c+1)})$ from $p(w, \alpha \mid \theta^{(c)}, y)$.
 2. Sample $(\theta^{(i+1)}, \alpha^{(c+1)})$ from $p(\theta, \alpha \mid w^{(c+1)}, y)$.
-

Liu and Wu [1999] show that the parameter-expanded data augmentation algorithm is superior to the regular data augmentation algorithm under the following two conditions.

Condition 1: The expansion parameter α indexes a set of one-to-one differentiable maps $\{t_\alpha\}$ such that $z = t_\alpha(w)$. In this case, we have:

$$p(y, w \mid \alpha, \theta) = f(y, t_\alpha(w) \mid \theta) |J_\alpha(w)|, \quad (3.4.1)$$

where $J_\alpha(w)$ is the Jacobian of t_α evaluated at w

Condition 2: The parameters θ and α are a priori independent, meaning $p(\alpha \mid \theta) = p(\alpha)$.

In cases where $p(\alpha \mid w)$ is available, Liu and Wu [1999] propose the following scheme assuming Conditions 1 and 2 are satisfied:

Algorithm 3.4.3 Parameter Expanded Data Augmentation Algorithm

1. Sample z from $f(z \mid \theta, y)$.
 2. Sample α_0 from $p_0(\alpha)$, and compute $w = t_{\alpha_0}^{-1}(z)$. Sample α_1 from $p(\alpha \mid y, w)$.
 3. Compute $z' = t_{\alpha_1}(t_{\alpha_0}^{-1}(z))$.
 4. Sample θ from $f(\theta \mid y, z')$.
-

However, $p(\alpha \mid y, w)$ can be hard to obtain, as was the case for the models we implemented. Since we were able to obtain $p(\alpha \mid y, w, \theta)$, we used the following scheme under, Conditions 1 and 2:

Algorithm 3.4.4 Parameter Expanded Gibbs Sampler

1. Sample θ from $f(\theta \mid y, z')$
 2. Sample z from $f(z \mid \theta, y)$
 3. Sample α_0 from $p_0(\alpha)$, and compute $w = t_{\alpha_0}^{-1}(z)$. Sample α_1 from $p(\alpha \mid y, w, \theta)$.
 4. Compute $z' = t_{\alpha_1}(t_{\alpha_0}^{-1}(z))$.
-

Setting π to $f(z \mid y, \theta)$ in Theorem 1 of Liu and Wu [1999], and then permuting the steps shows that the above is a valid Gibbs sampler for $f(z, \theta \mid y)$. The results regarding the convergence improvements in Liu and Wu [1999] are stated for scheme 1 and scheme 1.1, and do not directly support this scheme. Nevertheless, we saw empirical improvements regarding the convergence rate, which was evaluated using the diagnostic tools discussed in Section 3.5.

3.5 Convergence Diagnostics

MCMC algorithms work by constructing a Markov chain whose stationary distribution is the target distribution of interest. Convergence to the stationary distribution is a limiting property: in practice, the chain is always run for a finite number of iterations, and the stationary distribution is never exactly attained. Nevertheless, as the chain grows longer, the distribution of the samples should approximate the target distribution more closely. More precisely, if π_t denotes the empirical distribution of the chain at iteration t and π denotes the target distribution, then the total variation distance $\|\pi_t - \pi\|_{TV} := \frac{1}{2} \int_{\Omega} |\pi_t(x) - \pi(x)| dx \rightarrow 0$ as $t \rightarrow \infty$ where Ω denotes the

sample space [Roy, 2020]. In practice, the chain must be run for a finite number of iterations, and the rate of this convergence is generally unknown.

This leads to a major drawback — in most cases it is impossible to verify that the chain has converged to within acceptable bounds. However, there are diagnostic methods that can provide evidence of non-convergence. We discuss such diagnostics below: graphical methods such as autocorrelation plots and trace plots, effective sample size (ESS) and Geweke’s diagnostic. All of the diagnostics discussed are for univariate chains. To apply these methods to multivariate settings, it can be useful to apply the diagnostics onto a relevant, real-valued function of the draws, such as the log of the target density [Gelman et al., 1995, Ch. 11]. All of the diagnostics we present are discussed in Roy [2020].

3.5.1 Autocorrelation Plots and Trace Plots

Since samples are generated via a Markov chain, successive draws will tend to be correlated. More importantly, the convergence rate of a Markov chain is directly linked to its correlation structure (see the discussion following Lemma 12.6.4 in Liu [2001]). There are several ways to assess this correlation; we discuss two common graphical methods below.

The first diagnostic we discuss is autocorrelation plots. The lag- k autocorrelation is the correlation between samples that are k steps apart in the chain. The autocorrelation function (ACF) plot displays these values against k . For fast-mixing chains, the lag- k autocorrelation drops to approximately zero quickly as k increases. Conversely, high autocorrelation values at large lags indicate a high degree of dependence between samples and slow mixing of the chain. As a general rule, in order to obtain

precise estimates of the target distribution, an MCMC algorithm should be run for a large multiple of the number of iterations it takes for the autocorrelation to be virtually zero [Roy, 2020]. Figure 3.1 illustrates the difference between desirable and problematic ACF plots.

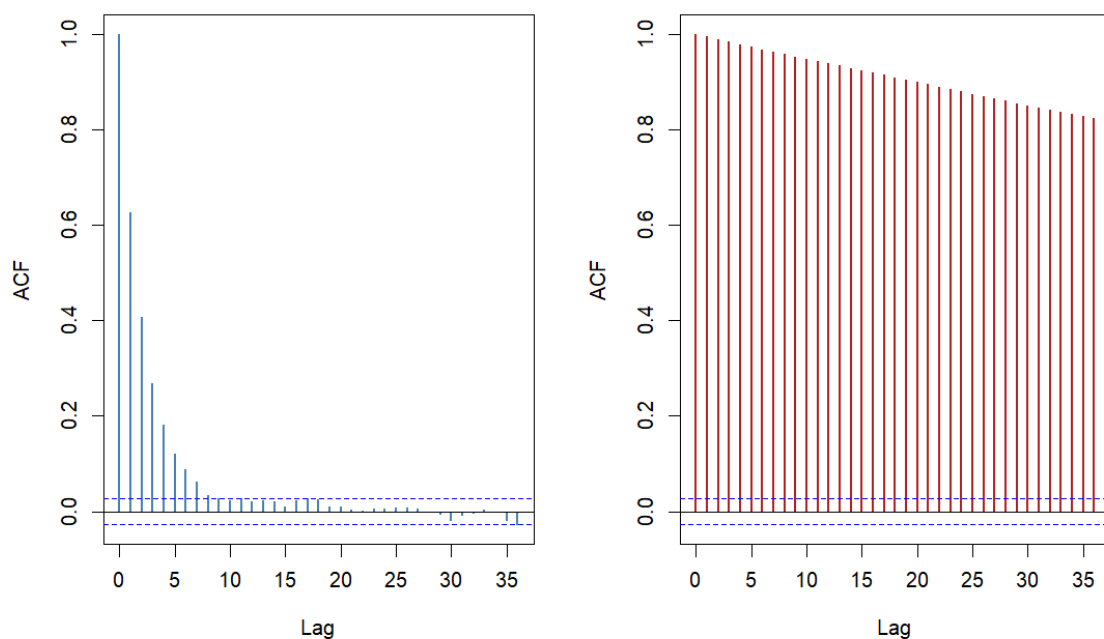


Figure 3.1: Left: An ACF plot of an efficient MCMC algorithm. Right: An ACF of an inefficient MCMC algorithm.

The second graphical diagnostic is the trace plot. Trace plots display the value of the chain on the vertical axis against the iteration number on the horizontal axis. They allow us to visualize how the chain evolves over time, diagnose efficiency issues, and detect cyclical patterns that may indicate a lack of convergence. For instance, if a chain becomes “stuck” in one region of the parameter space, this may indicate that the proposal distribution fails to propose samples in regions of sufficiently high

density under the target distribution. As a result, the chain inefficiently explores the sample space and is unlikely to converge. An example of a trace plot is shown in Figure 3.2.

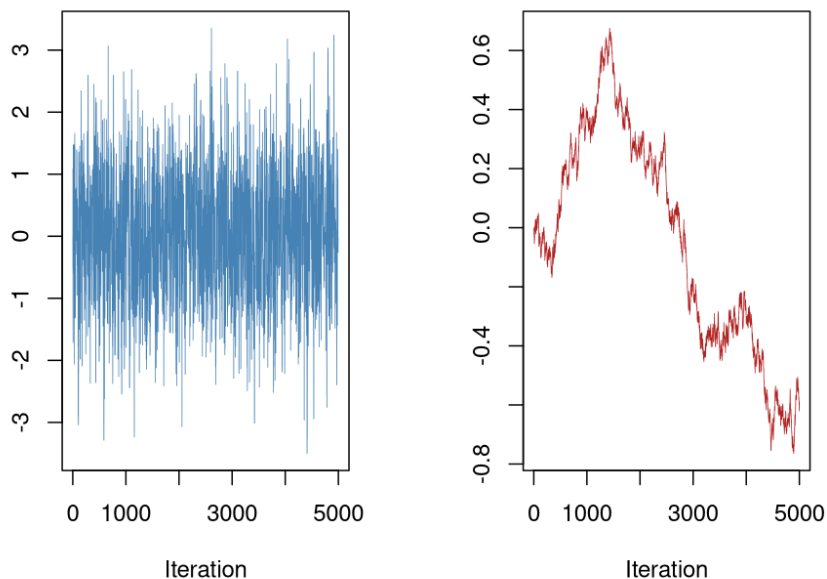


Figure 3.2: Left: A trace plot of an efficient MCMC algorithm. Right: A trace plot of an inefficient MCMC algorithm

3.5.2 Effective Sample Size

The first numeric diagnostic we discuss is the effective sample size (ESS) 3.5. The ESS quantifies the equivalent number of independent draws that the total number of samples from a Markov chain is worth, and is closely related to the autocorrelation. ESS scales the sample size drawn by the estimated correlation of the sample, thus representing how many independent draws the MCMC draws are worth. Thus, the

closer the ESS is to the number of total samples drawn, the better. For this reason, we may report the ratio between the effective sample size and the number of draws (the closer this ratio is to one, the better).

3.5.3 Geweke’s Diagnostic

Geweke’s diagnostic is based on the idea that, for a convergent MCMC chain, accounting for the burn-in period, early parts of the chain should have similar statistical summaries as the latter parts of the chain. Let $\bar{g}_1$ and $\bar{g}_2$ be empirical averages of two separate parts of a chain, where $\bar{g}_1$ is computed using n_1 draws, and $\bar{g}_2$ is computed using n_2 draws. Geweke [1992] develop the following statistic:

$$Z_n = \frac{\bar{g}_1 - \bar{g}_2}{\sqrt{\widehat{S_g(0)}/n_1 + \widehat{S_g(0)}/n_2}}, \quad (3.5.1)$$

where $\widehat{S_g(0)}$ is a non-parametric estimate of the standard deviation of the chain. Geweke [1992] show that Eq. (3.5.1) has a standard normal distribution under the assumption of a Markov chain CLT. Thus, to assess convergence, Eq. (3.5.1) may be treated as a test statistic for the null hypothesis that the chain has converged, and the alternative that it has not.

3.5.4 A note on thinning, and “burn-in”

As discussed, the stationary distribution of a Markov chain is a limiting property, meaning that the initial draws of an MCMC algorithm will likely not be from the target distribution, but will converge to it over time. It is therefore natural to discard the first B samples as a burn-in. A choice for the burn-in B can be determined via

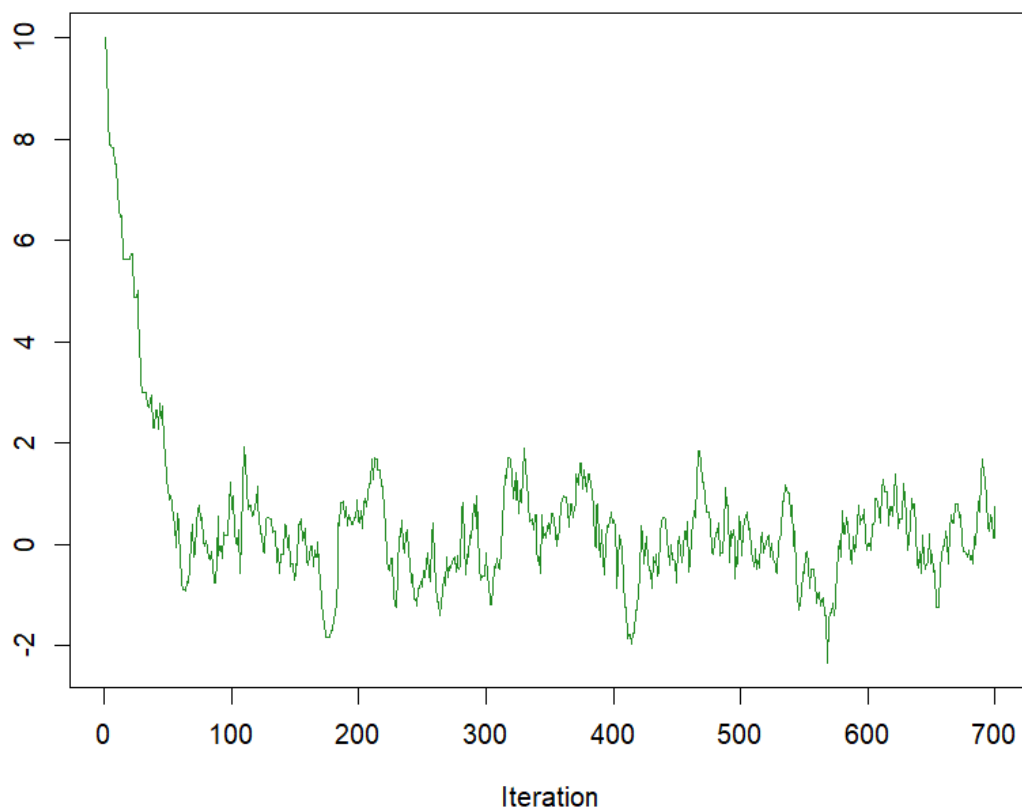


Figure 3.3: Trace plot indicating that a burn-in of at least $B = 90$ iterations is required.

trace plots. As shown in Figure 3.3, the initial portion of the chain behaves noticeably different from the remainder, which is a consequence of the chain’s starting value.

Further, since we use Markov chains to simulate our data, it is natural that the samples are correlated (even if they are being sampled from the target distribution). Thinning is a technique that remedies this by retaining only every p ’th draw from the chain, where p is chosen to be sufficiently large such that the lag- p autocorrelation is negligible, and can be selected using an autocorrelation plot. For example, consider

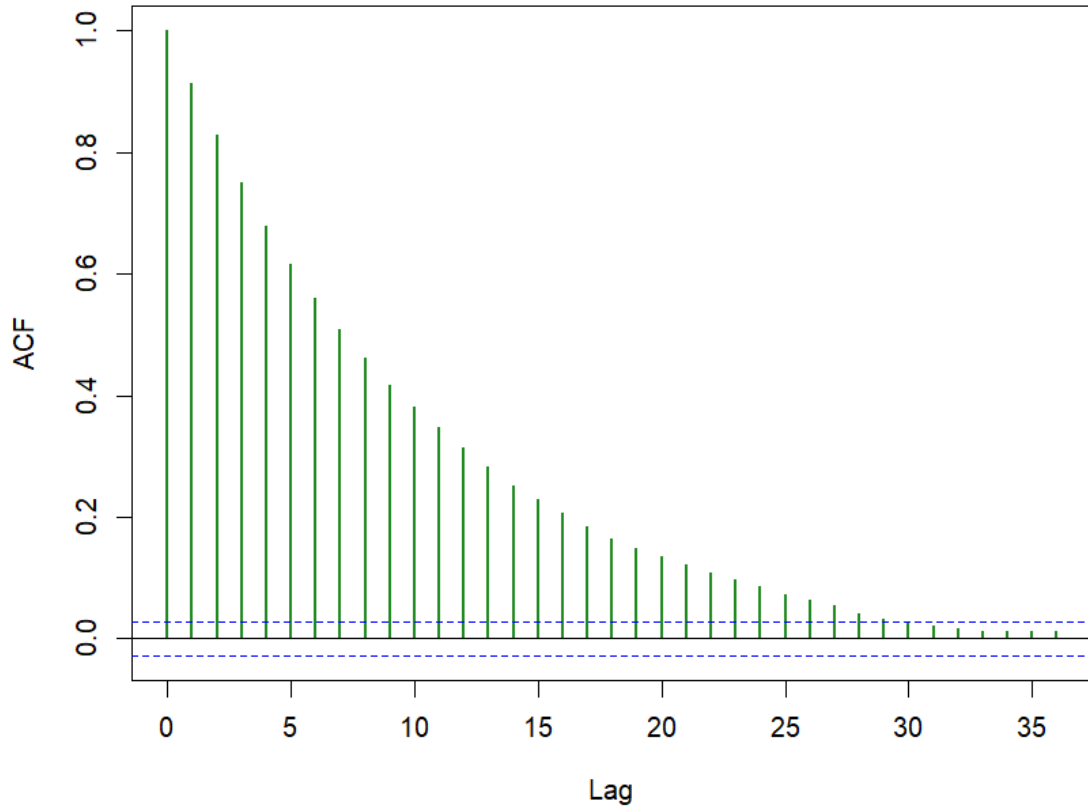


Figure 3.4: ACF plot indicating that a thinning factor of at least 30–35 is required for approximately i.i.d. samples.

Figure 3.4, which suggests that a thinning factor of at least 30–35 is required for samples to be nearly uncorrelated.

Chapter 4

Posterior inference of Bayesian Factor Models via MCMC

In this chapter, we provide details on the MCMC algorithms used for estimation of the PLT and UGLT Bayesian factor models discussed at the end of Section 2.5. We first address the PLT Bayesian factor model, demonstrating how parameter expansion can improve convergence even in this simple case. We then describe the algorithm used to sample the UGLT factor model, with a brief discussion of how an MCMC algorithm for the sparse PLT model can be obtained as a special case. Derivations for each of the full conditional posteriors are given in subsequent sections when available.

4.1 A Gibbs Sampler for the PLT model

We begin with our parameter expanded Gibbs sampler for the PLT model. Recall that the PLT model is as follows:

Independently, for $i = 1, \dots, N$,

$$\mathbf{y}_i \mid \Lambda, \Sigma, \mathbf{f}_i \sim \mathcal{N}_v(\Lambda \mathbf{f}_i, \Sigma), \quad (4.1.1)$$

where $\mathbf{f}_i \sim \mathcal{N}_q(0, \mathbf{I}_q)$, and $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_v^2)$. The prior on Λ is defined row-wise, where each non-zero entry within the rows follows a multivariate normal distribution:

$$\Lambda_k \stackrel{i.i.d.}{\sim} \mathcal{N}_{\min(i,q)}(0, c_0 \mathbf{I}_q). \quad (4.1.2)$$

The prior on Σ is defined element-wise, and each element follows an inverse gamma distribution,

$$\sigma_k^2 \sim \mathcal{G}^{-1}(a_\sigma, b_\sigma). \quad (4.1.3)$$

Note from Eq. (4.1.2) that we only constrain Λ to have a lower triangular structure; the posterior draws are post-processed so that the main diagonal entries of the loading matrix are positive to achieve full identification.

The goal of our MCMC algorithm is to obtain posterior draws of the factors $\mathbf{F}$, the factor loadings Λ , and the idiosyncratic variances $\boldsymbol{\sigma}^2 = (\sigma_1^2, \dots, \sigma_v^2)$. The sampler is outlined below.

Algorithm 4.1.1 Gibbs Sampler for the PLT Model

- Initialize $\Lambda^{(0)}, \sigma^{2(0)}$
 - Iteratively, for $c = 1, \dots, M$:
 1. Sample factors: $(\mathbf{f}_1, \dots, \mathbf{f}_N)^{(c)} \sim p(\mathbf{f}_1, \dots, \mathbf{f}_N \mid \sigma^{2(c-1)}, \Lambda^{(c-1)}, \mathbf{Y})$
 2. Sample loadings: $\Lambda^{(c)} \sim p(\Lambda \mid \mathbf{Y}, \sigma^{2(c-1)}, (\mathbf{f}_1, \dots, \mathbf{f}_N)^{(c)})$
 3. Sample idiosyncratic variances: $\sigma^{2(c)} \sim p(\sigma^2 \mid \mathbf{Y}, \Lambda^{(c)}, (\mathbf{f}_1, \dots, \mathbf{f}_N)^{(c)})$
 4. Apply parameter expansion to enhance convergence
 - Post-process the loadings and factors to ensure model identification
-

Point estimates of the loadings, factors, and variance parameters are obtained by computing the Monte Carlo estimates of the posterior means.

4.1.1 Initializing the algorithm

We initialize the Gibbs sampler for the PLT model using principal component analysis (PCA). PCA may be used to obtain initial values for our model through the reparameterization included below.

Let $\mathbf{y}_i$ be a $v \times 1$ random vector such that $\mathbf{y}_i \sim \mathcal{N}_v(0, \Sigma)$. Since Σ is a covariance matrix, and is thus positive definite, it admits the following spectral decomposition: $\Sigma = \mathbf{C}\mathbf{D}\mathbf{C}^\top$. Define $\mathbf{z}_i := \mathbf{C}^\top \mathbf{y}_i$ to be the principal components of $\mathbf{y}_i$. Note that $\mathbf{z}_i = \mathbf{C}^\top \mathbf{y}_i \Rightarrow \mathbf{y}_i = \mathbf{C}\mathbf{z}_i$ as $\mathbf{C}$ is an orthogonal matrix. This equation leads us to the

following:

$$\begin{aligned}
\mathbf{y}_i &= \mathbf{C}\mathbf{z}_i \\
&= \mathbf{C}\mathbf{D}^{\frac{1}{2}}\mathbf{D}^{-\frac{1}{2}}\mathbf{z}_i \\
&= \underbrace{\mathbf{C}\mathbf{D}^{\frac{1}{2}}}_{v \times v} \underbrace{\tilde{\mathbf{z}}_i}_{v \times 1} \\
&= \left(\underbrace{\mathbf{\Lambda}}_{v \times q} \mid \underbrace{\mathbf{M}}_{v \times (v-q)} \right) \begin{pmatrix} \mathbf{f}_i \\ \mathbf{g}_i \end{pmatrix} \begin{matrix} \leftarrow q \times 1 \\ \leftarrow (v-q) \times 1 \end{matrix} \\
&= \mathbf{\Lambda}\mathbf{f}_i + \mathbf{M}\mathbf{g}_i \\
&= \mathbf{\Lambda}\mathbf{f}_i + \mathbf{e}_i.
\end{aligned} \tag{4.1.4}$$

Since $\tilde{\mathbf{z}}_i = \mathbf{D}^{-\frac{1}{2}}\mathbf{C}^\top\mathbf{y}_i$, and $\mathbf{y}_i \sim \mathcal{N}_v(0, \Sigma = \mathbf{C}\mathbf{D}\mathbf{C}^\top)$, it follows that the first q entries of $\tilde{\mathbf{z}}_i$, denoted above as $\mathbf{f}_i$, follows a $\mathcal{N}_q(0, \mathbf{I}_q)$ distribution. As a result, principal components themselves may be treated as factors, and the loadings of the principal components are considered as the factor loadings. The variances can then be derived by computing the variance of the residuals since,

$$\mathbf{y}_i - \mathbf{\Lambda}\mathbf{f}_i = \mathbf{e}_i. \tag{4.1.5}$$

Note that if $N < v$, we can also initialize the algorithm using singular value decomposition.

4.1.2 Derivation for $p(\mathbf{F}|\boldsymbol{\sigma}^{2(c-1)}, \mathbf{\Lambda}^{(c-1)}, \mathbf{Y})$

Since $\mathbf{f}_1, \dots, \mathbf{f}_N$ are assumed to be independent, we draw each $\mathbf{f}_i$ from

$$\mathbf{f}_i \sim p(\mathbf{f}_i \mid \boldsymbol{\sigma}^{2(c-1)}, \mathbf{\Lambda}^{(c-1)}, \mathbf{Y}). \tag{4.1.6}$$

We derive $p(\mathbf{f}_i \mid \boldsymbol{\sigma}^{2(c-1)}, \Lambda^{(c-1)}, \mathbf{Y})$ as follows:

$$\begin{aligned}
p(\mathbf{f}_i \mid \boldsymbol{\sigma}^2, \Lambda, \mathbf{Y}) &\propto p(\mathbf{Y} \mid \boldsymbol{\sigma}^2, \mathbf{f}_i, \Lambda) p(\mathbf{f}_i) \quad \text{by Bayes' Theorem} \\
&= p(\mathbf{y}_i \mid \boldsymbol{\sigma}^2, \mathbf{f}_i, \Lambda) p(\mathbf{f}_i) \quad \text{by independence} \\
&\propto \exp \left\{ -\frac{1}{2} (\mathbf{y}_i - \Lambda \mathbf{f}_i)^\top \Sigma^{-1} (\mathbf{y}_i - \Lambda \mathbf{f}_i) \right\} \exp \left\{ -\frac{1}{2} \mathbf{f}_i^\top \mathbf{f}_i \right\} \\
\log p(\mathbf{f}_i \mid \boldsymbol{\sigma}^2, \mathbf{f}_i, \Lambda) &= -\frac{1}{2} (\mathbf{y}_i - \Lambda \mathbf{f}_i)^\top \Sigma^{-1} (\mathbf{y}_i - \Lambda \mathbf{f}_i) - \frac{1}{2} \mathbf{f}_i^\top \mathbf{f}_i + \text{const.} \\
&= -\frac{1}{2} \mathbf{y}_i^\top \Sigma^{-1} \mathbf{y}_i + \mathbf{f}_i^\top \Lambda^\top \Sigma^{-1} \mathbf{y}_i - \frac{1}{2} (\Lambda \mathbf{f}_i)^\top \Sigma^{-1} \Lambda \mathbf{f}_i - \frac{1}{2} \mathbf{f}_i^\top \mathbf{f}_i + \text{const.} \\
&= -\frac{1}{2} \mathbf{f}_i^\top (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q) \mathbf{f}_i + \mathbf{f}_i^\top \Lambda^\top \Sigma^{-1} \mathbf{y}_i + \text{const.} \\
&= -\frac{1}{2} \mathbf{f}_i^\top (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q) \mathbf{f}_i + \frac{1}{2} \mathbf{f}_i^\top (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q) (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q)^{-1} \\
&\quad \Lambda^\top \Sigma^{-1} \mathbf{y}_i + \frac{1}{2} \mathbf{y}_i^\top \Sigma^{-1} \Lambda (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q)^{-1} (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q) \mathbf{f}_i + \text{const.} \\
&= \frac{1}{2} \left(\mathbf{f}_i - (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q)^{-1} \Lambda^\top \Sigma^{-1} \mathbf{y}_i \right)^\top (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q) \\
&\quad \left(\mathbf{f}_i - (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q)^{-1} \Lambda^\top \Sigma^{-1} \mathbf{y}_i \right) \\
&\implies p(\mathbf{f}_i \mid \Lambda, \boldsymbol{\sigma}^2, \mathbf{Y}) \propto \exp \left\{ -\frac{1}{2} \left(\mathbf{f}_i - (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q)^{-1} \Lambda^\top \Sigma^{-1} \mathbf{y}_i \right)^\top \right. \\
&\quad \left. \times (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q) \left(\mathbf{f}_i - (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q)^{-1} \Lambda^\top \Sigma^{-1} \mathbf{y}_i \right) \right\}. \tag{4.1.7}
\end{aligned}$$

Recognizing the last line in Eq. (4.1.7) is the kernel of the density of a multivariate normal distribution, we find that

$$\mathbf{f}_i \mid \sigma^2, \Lambda, \mathbf{Y} \sim \mathcal{N}_q \left((\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q)^{-1} \Lambda^\top \Sigma^{-1} \mathbf{y}_i, (\Lambda^\top \Sigma^{-1} \Lambda + \mathbf{I}_q)^{-1} \right). \tag{4.1.8}$$

4.1.3 Derivation for $p(\Lambda \mid \mathbf{Y}, \mathbf{F}, \sigma^2)$

Recall from Eq. (1.0.4) that

$$\mathbf{Y} = \Lambda \mathbf{F} + \mathbf{E}. \quad (4.1.9)$$

Writing this row-wise yields

$$\tilde{\mathbf{y}}_k = \mathbf{F}^\top \Lambda_k + \tilde{\mathbf{e}}_k, \quad (4.1.10)$$

where

- $\tilde{\mathbf{y}}_k$ is a $N \times 1$ column vector containing row k of $\mathbf{Y}$,
- Λ_k is a $1 \times q$ column vector containing the entries of the k th row of Λ ,
- $\tilde{\mathbf{e}}_k$ is a $N \times 1$ containing the k th row of $\mathbf{E}$. Since $\mathbf{e}_i \sim \mathcal{N}_v(0, \text{diag}(\sigma_1^2, \dots, \sigma_v^2))$ independently for $i = 1, \dots, N$, it follows that $\tilde{\mathbf{e}}_k \sim \mathcal{N}_N(0, \sigma_k^2 \mathbf{I}_N)$.

This row-wise representation is useful because it allows us to express the full conditional posterior of Λ row-wise. Note that, since Λ has a PLT structure, Eq. (4.1.10) can be reduced to

$$\tilde{\mathbf{y}}_k = \mathbf{F}^{k\top} \tilde{\Lambda}_k + \tilde{\mathbf{e}}_k, \quad (4.1.11)$$

where $\tilde{\Lambda}_k$ is a column vector containing the non-zero entries of the k 'th row of Λ . Under the PLT structure, these are the first $\min(k, q)$ entries of that row. Correspondingly, $\mathbf{F}^k$ is a $N \times \min(k, q)$ matrix formed by the first $\min(k, q)$ columns of $\mathbf{F}^\top$. As a result,

$$\tilde{\mathbf{y}}_k \mid \mathbf{F}, \Lambda_k \sim \mathcal{N}_N \left(\mathbf{F}^{k\top} \tilde{\Lambda}_k, \sigma_k^2 \mathbf{I}_N \right). \quad (4.1.12)$$

Moreover, since $\Lambda_{kj} \stackrel{iid}{\sim} \mathcal{N}(0, c_0)$,

$$\tilde{\Lambda}_k \stackrel{i.i.d.}{\sim} \mathcal{N}_{\min(k, q)}(0, c_0 \mathbf{I}_{\min(k, q)}). \quad (4.1.13)$$

Thus, the conditional posterior for the non-zero loadings can be obtained as follows:

$$\begin{aligned}
p\left(\tilde{\Lambda}_k \mid \mathbf{Y}, \mathbf{F}, \boldsymbol{\sigma}^2\right) &= p\left(\tilde{\Lambda}_k \mid \tilde{\mathbf{y}}_k, \mathbf{F}^k\right) \\
&\propto p\left(\tilde{\Lambda}_k\right) p\left(\tilde{\mathbf{y}}_k \mid \mathbf{F}^k, \sigma_k^2, \tilde{\Lambda}_k\right) \quad \text{by Bayes theorem} \\
&\propto \exp \left\{-\frac{1}{2}\left(\tilde{\mathbf{y}}_k - \mathbf{F}^{k\top} \tilde{\Lambda}_k\right)^\top \left(\sigma_k^2\right)^{-1}\left(\tilde{\mathbf{y}}_k - \mathbf{F}^{k\top} \tilde{\Lambda}_k\right)\right\} \\
&\quad \cdot \exp \left\{-\frac{1}{2} \tilde{\Lambda}_k^\top c_0^{-1} \tilde{\Lambda}_k\right\}
\end{aligned} \tag{4.1.14}$$

Then,

$$\begin{aligned}
\log p\left(\tilde{\Lambda}_k \mid \mathbf{Y}, \mathbf{F}, \boldsymbol{\sigma}^2\right) &= -\frac{1}{2}\left(\tilde{\mathbf{y}}_k - \mathbf{F}^{k\top} \tilde{\Lambda}_k\right)^\top \left(\sigma_k^2\right)^{-1}\left(\tilde{\mathbf{y}}_k - \mathbf{F}^{k\top} \tilde{\Lambda}_k\right) \\
&\quad - \frac{1}{2} \tilde{\Lambda}_k^\top c_0^{-1} \tilde{\Lambda}_k + \text{const.} \\
&= -\frac{1}{2} \tilde{\mathbf{y}}_k^\top \left(\sigma_k^2\right)^{-1} \tilde{\mathbf{y}}_k + \tilde{\Lambda}_k^\top \mathbf{F}^k \left(\sigma_k^2\right)^{-1} \tilde{\mathbf{y}}_k - \frac{1}{2} \tilde{\Lambda}_k^\top \mathbf{F}^k \left(\sigma_k^2\right)^{-1} \mathbf{F}^{k\top} \tilde{\Lambda}_k + \text{const.} \\
&= -\frac{1}{2} \tilde{\Lambda}_k^\top \left(\mathbf{F}^k \left(\sigma_k^2\right)^{-1} \mathbf{F}^{k\top} + \mathbf{I}_{\min(q,k)} c_0^{-1}\right) \tilde{\Lambda}_k + \tilde{\Lambda}_k^\top \mathbf{F}^k \tilde{\mathbf{y}}_k \left(\sigma_k^2\right)^{-1} + \text{const.}
\end{aligned} \tag{4.1.15}$$

Now let

$$\Xi := \left(\sigma_k^2\right)^{-1} \mathbf{F}^k \mathbf{F}^{k\top} + c_0^{-1} \mathbf{I}_{\min(q,k)}. \tag{4.1.16}$$

Then, the log posterior of lambda is

$$\begin{aligned}
\log p\left(\tilde{\Lambda}_k \mid \tilde{\mathbf{y}}_k, \boldsymbol{\sigma}^2, \mathbf{F}^k\right) &= -\frac{1}{2} \tilde{\Lambda}_k^\top \Xi \tilde{\Lambda}_k + \frac{1}{2} \left(\sigma_k^2\right)^{-1} \Lambda_k^\top \Xi \Xi^{-1} \mathbf{F}^k \tilde{\mathbf{y}}_k + \\
&\quad \frac{1}{2} \left(\sigma_k^2\right)^{-1} \tilde{\mathbf{y}}_k^\top \mathbf{F}^{k\top} + \Xi^{-1} \Xi \Lambda_k + \text{const.} \\
&= -\frac{1}{2} \tilde{\Lambda}_k^\top \Xi \tilde{\Lambda}_k + \frac{1}{2} \left(\sigma_k^2\right)^{-1} \Lambda_k^\top \Xi \Xi^{-1} \mathbf{F}^k \tilde{\mathbf{y}}_k + \frac{1}{2} \left(\sigma_k^2\right)^{-1} \tilde{\mathbf{y}}_k^\top \mathbf{F}^{k\top} \Xi^{-1} \Xi \Lambda_k \\
&\quad - \frac{1}{2} \left(\sigma_k^2\right)^{-1} \tilde{\mathbf{y}}_k^\top \mathbf{F}^{k\top} \Xi^{-1} \Xi \Xi^{-1} \mathbf{F}^k \tilde{\mathbf{y}}_k \left(\sigma_k^2\right)^{-1} + \text{const.}
\end{aligned} \tag{4.1.17}$$

Define

$$\zeta := \Xi^{-1} (\sigma_k^2)^{-1} \mathbf{F}^k \tilde{\mathbf{y}}_k. \quad (4.1.18)$$

Substituting the last line in Eq. (4.1.18) into Eq. (4.1.17) gives

$$\log p \left(\tilde{\Lambda}_k \mid \tilde{\mathbf{y}}_k, \boldsymbol{\sigma}^2, \mathbf{F}^k \right) = -\frac{1}{2} \tilde{\Lambda}_k^\top \Xi \tilde{\Lambda}_k + \frac{1}{2} \Lambda_k^\top \Xi \zeta + \frac{1}{2} \zeta^\top \Xi \Lambda_k - \frac{1}{2} \zeta^\top \Xi \zeta + \text{const.} \quad (4.1.19)$$

Finally,

$$p \left(\tilde{\Lambda}_k \mid \tilde{\mathbf{y}}_k, \boldsymbol{\sigma}^2, \mathbf{F}^k \right) \propto \exp \left\{ -\frac{1}{2} \left(\tilde{\Lambda}_k - \zeta \right)^\top \Xi \left(\tilde{\Lambda}_k - \zeta \right) \right\}. \quad (4.1.20)$$

We conclude that $\tilde{\Lambda}_k \mid \tilde{\mathbf{y}}_k, \boldsymbol{\sigma}^2, \mathbf{F} \sim \mathcal{N}(\zeta, \Xi^{-1})$.

4.1.4 Derivation for $p(\boldsymbol{\sigma}^2 \mid \mathbf{Y}, \mathbf{F}, \Lambda)$

Recall from Eq. (4.1.12):

$$\tilde{\mathbf{y}}_k \mid \boldsymbol{\sigma}^2, \mathbf{F}, \Lambda \sim \mathcal{N}_N(\mathbf{F}^\top \Lambda_k, \sigma_k^2 \mathbf{I}_N). \quad (4.1.21)$$

Note that $\sigma_1^2, \dots, \sigma_v^2$ are assumed to be independent a priori, and conditionally on $\mathbf{F}$ and Λ , $\tilde{\mathbf{y}}_k$ only depends on σ_k^2 . Thus, to sample $\boldsymbol{\sigma}^2$ from the conditional posterior $p(\boldsymbol{\sigma}^2 \mid \mathbf{Y}, \mathbf{F}, \Lambda)$, we sample each σ_k^2 from $p(\sigma_k^2 \mid \mathbf{Y}, \mathbf{F}, \Lambda)$. Recall the prior on $\boldsymbol{\sigma}^2$:

$$\boldsymbol{\sigma}_k^2 \sim \mathcal{G}^{-1}(a_\sigma, b_\sigma). \quad (4.1.22)$$

We derive $p(\sigma_k^2 \mid \Lambda, \mathbf{F}, \tilde{\mathbf{y}}_k)$ as follows:

$$\begin{aligned}
p(\sigma_k^2 \mid \Lambda, \mathbf{F}, \mathbf{Y}) &\propto p(\tilde{\mathbf{y}}_k \mid \mathbf{F}, \Lambda, \sigma_k^2) p(\sigma_k^2) \quad \text{by Bayes Theorem} \\
&\propto (\sigma_k^2)^{-\frac{N}{2}} \exp \left\{ -\frac{1}{2\sigma_k^2} (\tilde{\mathbf{y}}_k - \mathbf{F}^\top \Lambda_k)^\top (\tilde{\mathbf{y}}_k - \mathbf{F}^\top \Lambda_k) \right\} (\sigma_k^2)^{-a_\sigma - 1} \\
&\quad \cdot \exp \left\{ -\frac{b_\sigma}{\sigma_k^2} \right\} \\
&= (\sigma_k^2)^{-(\frac{N}{2} + a_\sigma) - 1} \exp \left\{ -\frac{\left(b_\sigma + \frac{1}{2} (\tilde{\mathbf{y}}_k - \mathbf{F}^\top \Lambda_k)^\top (\tilde{\mathbf{y}}_k - \mathbf{F}^\top \Lambda_k) \right)}{\sigma_k^2} \right\}.
\end{aligned} \tag{4.1.23}$$

Recognizing Eq. (4.1.23) as the kernel of an inverse gamma distribution, we conclude that

$$\sigma_k^2 \mid \tilde{\mathbf{y}}_k, \Lambda_k, \mathbf{F} \sim \mathcal{G}^{-1} \left(a_\sigma + \frac{N}{2}, \quad b_\sigma + \frac{1}{2} (\tilde{\mathbf{y}}_k - \mathbf{F}^\top \Lambda_k)^\top (\tilde{\mathbf{y}}_k - \mathbf{F}^\top \Lambda_k) \right). \tag{4.1.24}$$

4.1.5 Parameter Expansion

We implement scheme 3.4.4 by regarding the latent variables to be $\Lambda, \mathbf{F}$. This allows us to define a family of transformations that leaves $p(\mathbf{Y} \mid \Lambda, \boldsymbol{\sigma}^2, \mathbf{F})$ invariant. The scheme is implemented as follows:

1. Sample $\psi_j \sim \mathcal{G}^{-1}(a_\psi, \beta_\psi)$ independently for $j = 1, \dots, q$ and let

$$\Psi = \text{diag}(\psi_1, \dots, \psi_q), \quad \boldsymbol{\psi} = (\psi_1, \dots, \psi_q).$$

2. Transform the original model to an expanded one:

For $i = 1, \dots, N$,

$$\begin{aligned}
\mathbf{f}_i^* &= (\Psi)^{\frac{1}{2}} \mathbf{f}_i \\
\Lambda^* &= \Lambda (\Psi)^{-\frac{1}{2}}.
\end{aligned} \tag{4.1.25}$$

3. Draw $\boldsymbol{\psi}_\dagger$ from $p(\boldsymbol{\psi} \mid \mathbf{Y}, \mathbf{F}^*, \Lambda^*)$ and define $\Psi_\dagger = \text{diag}(\boldsymbol{\psi}_\dagger)$

4. Transform back:

$$\begin{aligned}\mathbf{f}_i^\dagger &= \Psi_\dagger^{-\frac{1}{2}} \mathbf{f}_i^* \\ \Lambda^* &= \Lambda \Psi_\dagger^{\frac{1}{2}}.\end{aligned}\tag{4.1.26}$$

We now derive $p(\boldsymbol{\psi} \mid \mathbf{Y}, \mathbf{F}^*, \Lambda^*)$ in order to implement Step 3 in 3.4.4. First, note that under the transformed model:

$$\begin{aligned}\Lambda^* \mathbf{f}_i^* + \mathbf{e}_i &= \Lambda \Psi^{-\frac{1}{2}} \Psi^{\frac{1}{2}} \mathbf{f}_i + \mathbf{e}_i \\ &= \Lambda \mathbf{f}_i + \mathbf{e}_i \\ &= \mathbf{y}_i.\end{aligned}\tag{4.1.27}$$

Thus, the conditional likelihood $\mathbf{y}_i \mid \mathbf{f}_i, \Lambda, \Sigma$ the likelihood is as the same as the conditional likelihood of $\mathbf{y}_i \mid \mathbf{f}_i^*, \Lambda^*, \Sigma$ under the expanded model. That is, $\mathbf{y}_i$ is independent of Ψ given $\mathbf{F}^*, \Lambda^*$. As a result

$$\begin{aligned}p(\boldsymbol{\psi} \mid \mathbf{Y}, \mathbf{F}^*, \Lambda^*, \boldsymbol{\sigma}^2) &\propto p(\boldsymbol{\psi} \mid \mathbf{F}^*, \Lambda^*, \boldsymbol{\sigma}^2) \\ &\propto p(\Lambda^*, \mathbf{F}^* \mid \boldsymbol{\psi}, \boldsymbol{\sigma}^2) p(\boldsymbol{\psi} \mid \boldsymbol{\sigma}^2) \text{ By Bayes' Theorem.}\end{aligned}\tag{4.1.28}$$

Now, since $\Lambda, \boldsymbol{\sigma}^2, \mathbf{F}$ are considered to be a priori independent, $\boldsymbol{\sigma}^2, \Lambda^* = \Lambda \Psi^{-\frac{1}{2}}$ and $\mathbf{F}^* = \Psi^{\frac{1}{2}} \mathbf{F}$ are independent conditionally on Ψ . Thus,

$$\begin{aligned}p(\Lambda^*, \mathbf{F}^* \mid \boldsymbol{\psi}, \boldsymbol{\sigma}^2) p(\boldsymbol{\psi} \mid \boldsymbol{\sigma}^2) &= p(\Lambda^* \mid \boldsymbol{\psi}, \boldsymbol{\sigma}^2) p(\mathbf{F}^* \mid \boldsymbol{\psi}, \boldsymbol{\sigma}^2) p(\boldsymbol{\psi} \mid \boldsymbol{\sigma}^2) \\ &= p(\Lambda^* \mid \boldsymbol{\psi}) p(\mathbf{F}^* \mid \boldsymbol{\psi}) p(\boldsymbol{\psi}).\end{aligned}\tag{4.1.29}$$

Let $\tilde{\mathbf{f}}_k$ be a $N \times 1$ column vector containing the entries of the j 'th row of $\mathbf{F}$. Since $\mathbf{f}_i \stackrel{i.i.d.}{\sim} \mathcal{N}_q(0, \mathbf{I}_q)$, we have that $\tilde{\mathbf{f}}_j \stackrel{iid}{\sim} \mathcal{N}_N(0, \mathbf{I}_N)$. Letting $\tilde{\mathbf{f}}_j^*$ be a $N \times 1$ column vector

containing the entries of the j 'th row of $\mathbf{F}^*$, we note

$$\tilde{\mathbf{f}}_j^* = \psi_j^{\frac{1}{2}} \tilde{\mathbf{f}}_j \mid \psi_j \sim \mathcal{N}_N(0, \psi_j \mathbf{I}_N). \quad (4.1.30)$$

Moreover, since $\Lambda^* = \Lambda \Psi^{-\frac{1}{2}}$, we have that $\Lambda_{\cdot,j}^* = \psi_j^{-1} \Lambda_{\cdot,j}$ where $\Lambda_{\cdot,j}$ denotes the j 'th column of Λ . Let $\tilde{\Lambda}_{\cdot,j}$ be the non zero entries of $\Lambda_{\cdot,j}$. Since Λ has a PLT structure, $\tilde{\Lambda}_{\cdot,j}$ is equal to the $\Lambda_{\cdot,j}$ with the first $j-1$ entries removed. Therefore, $\tilde{\Lambda}_{\cdot,j}$ is a $v-j+1$ dimensional vector and

$$\tilde{\Lambda}_{\cdot,j} \sim \mathcal{N}_q(0, c_0 \mathbf{I}_q) \implies \tilde{\Lambda}_{\cdot,j}^* \sim \mathcal{N}_{v-j+1}(0, c_0 \psi_j^{-1} \mathbf{I}_q). \quad (4.1.31)$$

Let $s_j = v - j + 1$. We now derive the conditional distribution as follows:

$$\begin{aligned} p(\Lambda^* \mid \boldsymbol{\psi}) p(\mathbf{F}^* \mid \boldsymbol{\psi}) p(\boldsymbol{\psi}) &= \prod_{j=1}^q p(\tilde{\Lambda}_{\cdot,j}^* \mid \boldsymbol{\psi}) p(\tilde{\mathbf{f}}_j \mid \psi_j) p(\psi_j) \\ &\propto \prod_{j=1}^q c_0^{-\frac{s_j}{2}} \psi_j^{\frac{s_j}{2}} \exp \left\{ -\frac{1}{2} \frac{\psi_j}{c_0} \tilde{\Lambda}_{\cdot,j}^\top \tilde{\Lambda}_{\cdot,j} \right\} \psi_j^{-\frac{N}{2}} \exp \left\{ -\frac{1}{2\psi_j} \tilde{\mathbf{f}}_j^\top \tilde{\mathbf{f}}_j \right\} \\ &\quad \psi_j^{-\alpha_\psi - 1} \exp \left(\frac{\beta_\psi}{\psi_j} \right) \\ &= \prod_{j=1}^q \psi_j^{-\left(\alpha_\psi + \frac{N}{2} \right) + \frac{s_j}{2} - 1} \exp \left\{ -\frac{1}{2} \left[\frac{2\beta_\psi + \tilde{\mathbf{f}}_j^\top \tilde{\mathbf{f}}_j}{\psi_j} + \frac{\psi_j}{c_0} \tilde{\Lambda}_{\cdot,j}^\top \tilde{\Lambda}_{\cdot,j} \right] \right\}. \end{aligned} \quad (4.1.32)$$

So,

$$\psi_j \mid \mathbf{F}^*, \Lambda^* \sim \mathcal{GIG} \left(-\left(\alpha_\psi + \frac{N}{2} \right) + \frac{s_j}{2}, 2\beta_\psi + \tilde{\mathbf{f}}_j^\top \tilde{\mathbf{f}}_j, \frac{1}{c_0} \tilde{\Lambda}_{\cdot,j}^\top \tilde{\Lambda}_{\cdot,j} \right). \quad (4.1.33)$$

The parameterization used for the GIG in this thesis is the one used in Leydold

and Hormann [2023].

4.2 A sampler for the UGLT model

We follow Frühwirth-Schnatter et al. [2025b] and describe their partially collapsed Gibbs sampler, with some modifications. The first is our use of the generalized inverse Gaussian distribution for the parameter expansion scheme, akin to what was done for the PLT sampler 4.1. Additionally, we propose to use post-processing to determine the number of relevant factors, whereas Frühwirth-Schnatter et al. [2025b] implemented a reversible Jump MCMC. Our empirical studies show that in some cases, we can recover the number of factors with post-processing, and this corroborates with one of the proposals in the discussion of Frühwirth-Schnatter et al. [2025b]. We outline the sampler in Algorithm 4.2.1, and then present the derivations for most steps in the order of the algorithm.

Algorithm 4.2.1 Partially Collapsed Gibbs Sampler for the UGLT Model

1. For $j = 1, \dots, q$, sample $\tau_j \mid l_j, \delta \sim p(\tau_j \mid l_j, \delta)$.
 2. Loop over all columns of the sparsity matrix δ in a random order and sample δ_{kj} below the pivot l_j from $p(\delta_{kj} \mid l_j, \delta_{\cdot, -j}, \mathbf{F}, \tau_j, \mathbf{Y})$, where $\delta_{\cdot, -j}$ denotes the remaining columns. Note that we do not condition on Λ or $\boldsymbol{\sigma}^2$, as they have been collapsed out.
 3. Loop over all columns j of δ in a random order and sample a new pivot row l_j from $p(l_j \mid \delta_{\cdot, -j}, \mathbf{F}, \mathbf{Y})$, without conditioning on Λ , $\boldsymbol{\sigma}^2$, or the slab probabilities $\boldsymbol{\tau}$.
 4. Sample the model parameters Λ and $\sigma_1^2, \dots, \sigma_v^2$ jointly from $p(\Lambda, \boldsymbol{\sigma}^2 \mid \delta, \mathbf{F}, \mathbf{Y})$.
 5. Sample the latent factors $\mathbf{F}$, from $p(\mathbf{F} \mid \Lambda, \sigma_1^2, \dots, \sigma_v^2, \mathbf{Y})$.
 6. Update the column shrinkage parameters $\theta_1, \dots, \theta_q$.
 7. Apply parameter expansion to improve convergence.
-

The algorithm above is an example of a partially collapsed Gibbs sampler, as Steps 1, 2, and 3 do not condition on the loadings and the idiosyncratic variances.

Posterior estimates of Λ are obtained by first computing the Monte Carlo estimate of the posterior mode δ_{mode} of δ , and then taking the mean of the posterior draws of Λ whose structure conforms to δ_{mode} . For the remaining parameters, Monte Carlo estimates of the posterior means are taken.

4.2.1 Sampling the Sparsity probabilities

Note that, conditionally on δ , $\mathbf{Y}$ is independent of $\boldsymbol{\tau}$. That is,

$$p(\mathbf{Y} \mid \mathbf{F}, \boldsymbol{\sigma}^2, \Lambda, \delta, \boldsymbol{\tau}, \boldsymbol{\theta}) = p(\mathbf{Y} \mid \mathbf{F}, \boldsymbol{\sigma}^2, \Lambda, \delta, \boldsymbol{\theta}). \quad (4.2.1)$$

Thus,

$$p(\boldsymbol{\tau} \mid \mathbf{Y}, \mathbf{F}, \boldsymbol{\sigma}^2, \Lambda, \delta, \boldsymbol{\theta}) = p(\boldsymbol{\tau} \mid \mathbf{F}, \boldsymbol{\sigma}^2, \Lambda, \delta, \boldsymbol{\theta}). \quad (4.2.2)$$

Since $\boldsymbol{\tau}$ and $(\boldsymbol{\theta}, \Lambda, \boldsymbol{\sigma}^2)$ are conditionally independent given δ , we conclude that

$$p(\boldsymbol{\tau} \mid \mathbf{F}, \boldsymbol{\sigma}^2, \Lambda, \delta, \boldsymbol{\theta}) = p(\boldsymbol{\tau} \mid \delta). \quad (4.2.3)$$

It remains to find the closed form of $p(\boldsymbol{\tau} \mid \delta)$. Note that the components of $\boldsymbol{\tau} = (\tau_1, \tau_2, \dots, \tau_q)$ are independent, and since τ_j governs the sparsity of the unconstrained entries in column j of δ , it depends only on $\delta_{:,j}$. Specifically,

$$p(\boldsymbol{\tau} \mid \delta) = \prod_{j=1}^q p(\tau_j \mid \delta_{:,j}). \quad (4.2.4)$$

Moreover, for the unconstrained entries of column j , i.e. for row indices $i > l_j$ since δ follows a UGLT structure, we have $\mathbf{P}(\delta_{kj} = 0 \mid \tau_j) = \tau_j$ and $\mathbf{P}(\delta_{kj} = 1 \mid \tau_j) = 1 - \tau_j$. Conditionally on τ_j , the entries $\delta_{l_j+1,j}, \dots, \delta_{q,j}$ are therefore independent Bernoulli(τ_j). Note that, a priori, $\tau \sim \mathcal{B}(\alpha, \beta)$. It follows by standard conjugacy that

$$\tau_j \mid \delta_{:,j} \sim \mathcal{B}(\alpha + d_j, \beta + v - l_j - d_j), \quad (4.2.5)$$

where $d_j = \sum_{k=l_j+1}^q \delta_{kj}$ and $\delta_{:,j}$ denotes the j 'th column of δ .

4.2.2 Sampling the Sparsity Matrix

To sample each indicator δ_{kj} , we can compute the conditional posterior odds of $\delta_{kj} = 1$ versus $\delta_{kj} = 0$ given all other indicators and sample accordingly. Referring to (4.1.10), note that $\tilde{\mathbf{y}}_k$ only depends on the k 'th row of Λ . This implies that $\tilde{\mathbf{y}}_k$ only depends on the k 'th row of δ for the UGLT model. As a result, we write:

$$p(\delta_{kj} \mid \delta_{k,-j}, \mathbf{F}, \tau_j, \mathbf{Y}) \propto \int \int p(\tilde{\mathbf{y}}_k \mid \mathbf{F}, \delta_{k,\cdot}, \Lambda_k, \boldsymbol{\sigma}^2) p(\Lambda_k \mid \boldsymbol{\theta}, \delta_k, \boldsymbol{\sigma}^2) d\Lambda_k d\boldsymbol{\sigma}^2 \cdot p(\delta_{kj} \mid \tau_j), \quad (4.2.6)$$

where Eq. (4.2.6) is due to the decomposition shown in Eq.(4.1.10). Let Λ_k^δ denote the unconstrained entries in the k 'th row of Λ . Let X_k^δ be the indices o of $\mathbf{F}^\top$ such that $\delta_{ko} \neq 0$. Then,

$$\tilde{\mathbf{y}}_k \mid X_k^\delta, \Lambda_k^\delta, \boldsymbol{\sigma}^2 \sim \mathcal{N}_N(X_k^\delta \Lambda_k^\delta, \mathbf{I}_N \sigma_k^2). \quad (4.2.7)$$

Since $\lambda_{kj} \mid \delta_{kj} = 1 \sim \mathcal{N}(0, \sigma_k^2 \theta_j)$, the prior distribution of the unconstrained loadings of row k is

$$\Lambda_k^\delta \sim \mathcal{N}_{d_k}(0, \sigma_k^2 L_{k0}), \quad (4.2.8)$$

where L_{k0} is a $d_k \times d_k$ diagonal matrix consisting of θ_{k_j} for all k_j where $\delta_{kk_j} = 1$, and $d_k = \sum_{j=1}^q \delta_{kj}$. We proceed first by marginalizing out Λ_k from $p(\tilde{\mathbf{y}}_k \mid X_k^\delta, \Lambda_k^\delta, \boldsymbol{\sigma}^2, \boldsymbol{\theta})$:

$$\begin{aligned}
p(\tilde{\mathbf{y}}_k | \mathbf{X}_k^\delta, \boldsymbol{\sigma}^2) &= \int (2\pi)^{-\frac{N}{2}} \sigma_k^{-N} \exp \left\{ -\frac{1}{2} [\tilde{\mathbf{y}}_k - \mathbf{X}_k^\delta \Lambda_k^\delta]^\top \mathbf{I}_N \sigma_k^{-2} [\tilde{\mathbf{y}}_k - \mathbf{X}_k^\delta \Lambda_k^\delta] \right\} \\
&\quad \times (2\pi)^{-\frac{d_k}{2}} (\sigma_k^2)^{-\frac{d_k}{2}} |\mathbf{L}_{k0}|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} (\Lambda_k^\delta)^\top (\mathbf{L}_{k0} \sigma_k^2)^{-1} \Lambda_k^\delta \right\} d\Lambda_k^\delta \\
&= (2\pi)^{-\frac{N+d_k}{2}} (\sigma_k^2)^{-\frac{N+d_k}{2}} |\mathbf{L}_{k0}|^{-\frac{1}{2}} \int \exp \left\{ -\frac{1}{2} [\tilde{\mathbf{y}}_k - \mathbf{X}_k^\delta \Lambda_k^\delta]^\top \mathbf{I}_N \sigma_k^{-2} \right. \\
&\quad \left. [\tilde{\mathbf{y}}_k - \mathbf{X}_k^\delta \Lambda_k^\delta] \right\} \cdot \exp \left\{ -\frac{1}{2} (\Lambda_k^\delta)^\top (\mathbf{L}_{k0} \sigma_k^2)^{-1} \Lambda_k^\delta \right\} d\Lambda_k^\delta \\
&= (2\pi)^{-\frac{N+d_k}{2}} (\sigma_k^2)^{-\frac{N+d_k}{2}} |\mathbf{L}_{k0}|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} \tilde{\mathbf{y}}_k^\top (\mathbf{L}_{k0} \sigma_k^2)^{-1} \tilde{\mathbf{y}}_k \right\} \\
&\quad \int \exp \left\{ -\frac{1}{2} \left(\Lambda_k^{\delta\top} \mathbf{X}_k^{\delta\top} \mathbf{I}_N \sigma_k^{-2} \mathbf{X}_k^\delta \Lambda_k^\delta - 2 \Lambda_k^{\delta\top} \mathbf{X}_k^{\delta\top} \sigma_k^{-2} \tilde{\mathbf{y}}_k \right) \right. \\
&\quad \left. - \frac{1}{2} \Lambda_k^{\delta\top} (\mathbf{L}_{k0} \sigma_k^2)^{-1} \Lambda_k^\delta \right\} d\Lambda_k^\delta \\
&= (2\pi)^{-\frac{N+d_k}{2}} (\sigma_k^2)^{-\frac{N+d_k}{2}} |\mathbf{L}_{k0}|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} \tilde{\mathbf{y}}_k^\top (\mathbf{L}_{k0} \sigma_k^2)^{-1} \tilde{\mathbf{y}}_k \right\} \\
&\quad \cdot \int \exp \left\{ -\frac{1}{2} \left(\Lambda_k^{\delta\top} \left(\mathbf{X}_k^{\delta\top} \mathbf{I}_N \sigma_k^{-2} \mathbf{X}_k^\delta + (\mathbf{L}_{k0} \sigma_k^2)^{-1} \right) \Lambda_k^\delta - 2 \Lambda_k^{\delta\top} \mathbf{X}_k^{\delta\top} \sigma_k^{-2} \tilde{\mathbf{y}}_k \right) \right\} d\Lambda_k^\delta.
\end{aligned} \tag{4.2.9}$$

Define,

$$\mathbf{Q}_k^{-1} = \left(\mathbf{X}_k^{\delta\top} \mathbf{I}_N \sigma_k^{-2} \mathbf{X}_k^\delta + (\mathbf{L}_{k0} \sigma_k^2)^{-1} \right). \tag{4.2.10}$$

Plugging in Eq. (4.2.10) to Eq. (4.2.9):

$$\begin{aligned}
&(2\pi)^{-\frac{N+d_k}{2}} (\sigma_k^2)^{-\frac{N+d_k}{2}} |\mathbf{L}_{k0}|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} \tilde{\mathbf{y}}_k^\top (\mathbf{L}_{k0} \sigma_k^2)^{-1} \tilde{\mathbf{y}}_k \right\} \\
&\quad \times \int \exp \left\{ -\frac{1}{2} \left(\Lambda_k^{\delta\top} \mathbf{Q}_k^{-1} \Lambda_k^\delta - 2 \Lambda_k^{\delta\top} \mathbf{Q}_k^{-1} \mathbf{Q}_k \mathbf{X}_k^{\delta\top} \sigma_k^{-2} \tilde{\mathbf{y}}_k \right) \right\} d\Lambda_k^\delta.
\end{aligned} \tag{4.2.11}$$

Let

$$\mathbf{M}_k = \mathbf{Q}_k \mathbf{X}_k^{\delta\top} \sigma_k^{-2} \tilde{\mathbf{y}}_k. \tag{4.2.12}$$

Plugging in M_k into Eq. (4.2.11), we get:

$$\begin{aligned}
& (2\pi)^{-\frac{N+d_k}{2}} (\sigma_k^2)^{-\frac{N+d_k}{2}} |L_{k0}|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} \tilde{\mathbf{y}}_k^\top (L_{k0} \sigma_k^2)^{-1} \tilde{\mathbf{y}}_k + \frac{1}{2} M_k^\top Q_k^{-1} M_k \right\} \\
& \times \int \exp \left\{ -\frac{1}{2} (\Lambda_k^\delta - M_k)^\top Q_k^{-1} (\Lambda_k^\delta - M_k) \right\} d\Lambda_k^\delta \\
& = (\sigma_k^2)^{-\frac{N+d_k}{2}} |L_{k0}|^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} \tilde{\mathbf{y}}_k^\top (L_{k0} \sigma_k^2)^{-1} \tilde{\mathbf{y}}_k + \frac{1}{2} M_k^\top Q_k^{-1} M_k \right\} |Q_k|^{\frac{1}{2}}.
\end{aligned} \tag{4.2.13}$$

Note:

$$\begin{aligned}
Q_k &= \left(X_k^{\delta\top} X_k^\delta \sigma_k^{-2} + (L_{k0} \sigma_k^2)^{-1} \right)^{-1} \\
&= \left(X_k^\delta X_k^\delta \sigma_k^{-2} + L_{k0}^{-1} \sigma_k^{-2} \right)^{-1} \\
&= \underbrace{\left(X_k^\delta X_k^\delta + L_{k0}^{-1} \right)^{-1}}_{:=L_{kN}} \sigma_k^2 \\
&= L_{kN} \sigma_k^2.
\end{aligned} \tag{4.2.14}$$

And so:

$$|Q_k| = \sigma_k^{\frac{d_k}{2}} |L_{kN}|. \tag{4.2.15}$$

Thus,

$$\begin{aligned}
 p(\tilde{\mathbf{y}}_k \mid \mathbf{F}, \delta, \sigma_k^2) &= (2\pi)^{-\frac{N+d_k}{2}} (\sigma_k^2)^{-\frac{N+d_k}{2}} |\mathbf{L}_{k0}|^{-\frac{1}{2}} \\
 &\exp\left(-\frac{1}{2} \left(\tilde{\mathbf{y}}_k^\top (\mathbf{L}_{k0} \sigma_k^2)^{-1} \tilde{\mathbf{y}}_k\right) + \frac{1}{2} (\mathbf{M}_k^\top \mathbf{Q}_k^{-1} \mathbf{M}_k)\right) (2\pi)^{\frac{d_k}{2}} \sigma_k^{\frac{d_k}{2}} |\mathbf{L}_{kN}|^{\frac{1}{2}} \\
 &= (2\pi)^{-\frac{N}{2}} (\sigma_k^2)^{-\frac{N}{2}} \left(\frac{|\mathbf{L}_{kN}|}{|\mathbf{L}_{k0}|}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2} \left(\tilde{\mathbf{y}}_k^\top (\mathbf{L}_{k0} \sigma_k^2)^{-1} \tilde{\mathbf{y}}_k\right) + \frac{1}{2} (\mathbf{M}_k^\top \mathbf{Q}_k^{-1} \mathbf{M}_k)\right) \\
 &= (2\pi)^{-\frac{N}{2}} (\sigma_k^2)^{-\frac{N}{2}} \left(\frac{|\mathbf{L}_{kN}|}{|\mathbf{L}_{k0}|}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2} \left(\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} (\sigma_k^2)^{-1} \tilde{\mathbf{y}}_k\right) + \right. \\
 &\quad \left. \frac{1}{2} (\mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} (\sigma_k^2)^{-1} \mathbf{M}_k)\right) \\
 &= (2\pi)^{-\frac{N}{2}} (\sigma_k^2)^{-\frac{N}{2}} \left(\frac{|\mathbf{L}_{kN}|}{|\mathbf{L}_{k0}|}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2\sigma_k^2} (\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} \tilde{\mathbf{y}}_k - \mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} \mathbf{M}_k)\right).
 \end{aligned} \tag{4.2.16}$$

We now integrate out σ_k^2 in Eq. 4.2.16:

$$\begin{aligned}
 p(\tilde{\mathbf{y}}_k \mid \boldsymbol{\theta}, \mathbf{F}, \delta) &= \int (2\pi)^{-\frac{N}{2}} (\sigma_k^2)^{-\frac{N}{2}} \left(\frac{|\mathbf{L}_{kN}|}{|\mathbf{L}_{k0}|}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2\sigma_k^2} \left(\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} \tilde{\mathbf{y}}_k - \right. \right. \\
 &\quad \left. \left. \mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} \mathbf{M}_k\right)\right) \frac{b_\sigma^{a_\sigma}}{\Gamma(a_\sigma)} (\sigma_k^2)^{-a_\sigma-1} \exp\left(\frac{-b_\sigma}{\sigma_k^2}\right) d\sigma_k^2 \\
 &= (2\pi)^{-\frac{N}{2}} \left(\frac{|\mathbf{L}_{kN}|}{|\mathbf{L}_{k0}|}\right)^{\frac{1}{2}} \frac{b_\sigma^{a_\sigma}}{\Gamma(a_\sigma)} \int (\sigma_k^2)^{-\frac{N}{2}} \exp\left(-\frac{1}{2\sigma_k^2} \left(\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} \tilde{\mathbf{y}}_k \right. \right. \\
 &\quad \left. \left. - \mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} \mathbf{M}_k\right)\right) (\sigma_k^2)^{-a_\sigma-1} \exp\left(\frac{-b_\sigma}{\sigma_k^2}\right) d\sigma_k^2 \\
 &= (2\pi)^{-\frac{N}{2}} \left(\frac{|\mathbf{L}_{kN}|}{|\mathbf{L}_{k0}|}\right)^{\frac{1}{2}} \frac{b_\sigma^{a_\sigma}}{\Gamma(a_\sigma)} \times \int (\sigma_k^2)^{-(\frac{N}{2}+a_\sigma)} \\
 &\quad \cdot \exp\left(-\frac{\frac{1}{2} (\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} \tilde{\mathbf{y}}_k - \mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} \mathbf{M}_k) + b_\sigma}{\sigma_k^2}\right) d\sigma_k^2
 \end{aligned} \tag{4.2.17}$$

Note that the integrand in the last line of Eq. (4.2.17) is the kernel of $\mathcal{G}^{-1}(a_\sigma +$

$\frac{N}{2}$, $b_\sigma + \frac{1}{2}(\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} \tilde{\mathbf{y}}_k - \mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} \mathbf{M}_k)$. Evaluating this integral gives:

$$\left(\frac{|\mathbf{L}_{kN}|}{|\mathbf{L}_{k0}|} \right)^{\frac{1}{2}} (2\pi)^{-\frac{N}{2}} \frac{b_\sigma^{a_\sigma}}{\Gamma(a_\sigma)} \frac{\Gamma\left(\frac{N}{2} + a_\sigma\right)}{\left(b_\sigma + \frac{1}{2}(\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} \tilde{\mathbf{y}}_k - \mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} \mathbf{M}_k)\right)^{\frac{N}{2} + a_\sigma}}. \quad (4.2.18)$$

If $\delta_{kj} = 0$ for all j , we have:

$$p(\tilde{\mathbf{y}}_k \mid \delta_{kj} = 0 \text{ for all } j) = (2\pi)^{-\frac{N}{2}} \frac{b_\sigma^{a_\sigma}}{\Gamma(a_\sigma)} \frac{\Gamma\left(\frac{N}{2} + a_\sigma\right)}{\left(b_\sigma + \frac{1}{2}(\tilde{\mathbf{y}}_k^\top \tilde{\mathbf{y}}_k)\right)^{\frac{N}{2} + a_\sigma}}. \quad (4.2.19)$$

4.2.3 Sampling the Pivots

To sample the pivots, Frühwirth-Schnatter et al. [2025b] use a Metropolis Hastings algorithm with a proposal distribution $g(\cdot \mid \delta)$ that ensures proposals correspond to a UGLT structure. The proposal distribution randomly chooses from three possible move (i.e., proposal) types, being a shift move, a switch move, or an add/delete move. Equivalently, proposal distribution can be expressed as a mixture:

$$g(\cdot \mid \delta) = p_{\text{shift}} g_{\text{shift}}(\cdot \mid \delta) + p_{\text{switch}} g_{\text{switch}}(\cdot \mid \delta) + (1 - p_{\text{shift}} - p_{\text{switch}}) g_{\text{add/delete}}(\cdot \mid \delta) \quad (4.2.20)$$

As explained noted in Algorithm 4.2.1, the columns in δ are randomly chosen in order, and for each column a move is made. Details for proposal distributions are provided in the supplementary materials for Frühwirth-Schnatter et al. [2025b].

4.2.4 Sampling from $p(\Lambda, \sigma^2 \mid \delta, \mathbf{F}, \mathbf{Y})$

Note that $p(\Lambda, \sigma^2 \mid \mathbf{Y}, \mathbf{F}, \delta, \tau, \boldsymbol{\theta}) = p(\Lambda \mid \sigma^2, \mathbf{Y}, \mathbf{F}, \delta, \tau, \boldsymbol{\theta})p(\sigma^2 \mid \mathbf{Y}, \mathbf{F}, \delta, \tau, \boldsymbol{\theta})$. Thus to draw a sample from the joint distribution, we may first sample σ^2 and then Λ from the corresponding marginals. We first derive $p(\sigma^2 \mid \mathbf{Y}, \mathbf{F}, \delta, \tau, \boldsymbol{\theta})$.

Conditionally on δ , $\mathbf{Y}$ is independent of τ . Moreover, note from Eq. (4.2.7) that $\tilde{\mathbf{y}}_k \mid \sigma^2, \mathbf{F}, \Lambda \sim \mathcal{N}_N(\mathbf{F}^\top \Lambda_k, \sigma_k^2 \mathbf{I}_N)$. Thus $\tilde{\mathbf{y}}_k$ is independent of σ_j for $j \neq k$. Further, $\sigma_1^2, \sigma_2^2, \dots, \sigma_v^2, \mathbf{F}, \tau, \boldsymbol{\theta}$ are all a priori independent. Thus,

$$p(\sigma^2 \mid \mathbf{Y}, \mathbf{F}, \delta, \tau, \boldsymbol{\theta}) = \prod_{k=1}^v p(\sigma_k^2 \mid \tilde{\mathbf{y}}_k, \mathbf{F}, \delta). \quad (4.2.21)$$

Thus, we derive $p(\sigma_k^2 \mid \tilde{\mathbf{y}}_k, \mathbf{F}, \delta, \boldsymbol{\theta})$. First note that by Bayes' Theorem:

$$p(\sigma_k^2 \mid \tilde{\mathbf{y}}_k, \mathbf{F}, \delta, \boldsymbol{\theta}) \propto p(\tilde{\mathbf{y}}_k \mid \sigma_k^2, \mathbf{F}, \delta, \boldsymbol{\theta}) p(\sigma_k^2) \quad (4.2.22)$$

Substituting Eq. (4.2.16) (i.e the marginal likelihood), we have that $p(\sigma_k^2 \mid \tilde{\mathbf{y}}_k, \mathbf{F}, \delta, \boldsymbol{\theta})$ is proportional to:

$$(2\pi)^{-\frac{N}{2}} (\sigma_k^2)^{-\frac{N}{2}} \left(\frac{|\mathbf{L}_{kN}|}{|\mathbf{L}_{k0}|} \right)^{\frac{1}{2}} \exp \left(-\frac{1}{2\sigma_k^2} (\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} \tilde{\mathbf{y}}_k - \mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} \mathbf{M}_k) \right) \quad (4.2.23)$$

$$\propto (\sigma_k^2)^{-(\frac{N}{2} + a_\sigma)} \exp \left(-\frac{\frac{1}{2} (\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} \tilde{\mathbf{y}}_k - \mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} \mathbf{M}_k) + b_\sigma}{\sigma_k^2} \right). \quad (4.2.24)$$

Which is the kernel of a

$$\mathcal{G}^{-1} \left(a_\sigma + \frac{N}{2}, \quad b_\sigma + \frac{1}{2} (\tilde{\mathbf{y}}_k^\top \mathbf{L}_{k0}^{-1} \tilde{\mathbf{y}}_k - \mathbf{M}_k^\top \mathbf{L}_{kN}^{-1} \mathbf{M}_k) \right). \quad (4.2.25)$$

To derive $p(\Lambda \mid \sigma^2, \mathbf{Y}, \mathbf{F}, \boldsymbol{\theta}, \delta)$ we can use very similar steps that were used in

finding the conditional posterior for the loadings in the PLT model. To see this, recall from (4.2.7) that

$$\tilde{\mathbf{y}}_k \mid \mathbf{F}, \Lambda_k^\delta, \boldsymbol{\sigma}^2 \sim \mathcal{N}_N(\mathbf{X}_k^\delta \Lambda_k^\delta, \sigma_k^2 \mathbf{I}_N). \quad (4.2.26)$$

And,

$$\Lambda_k^\delta \mid \boldsymbol{\sigma}^2, \mathbf{F}, \boldsymbol{\theta} \sim \mathcal{N}_{d_k}(0, \sigma_k^2 \mathbf{L}_{k0}). \quad (4.2.27)$$

Using Bayes' Theorem, $p(\Lambda_k^\delta \mid \boldsymbol{\sigma}^2, \boldsymbol{\theta}, \mathbf{F}, \mathbf{Y})$, is proportional to the product of Eq. (4.2.26) and Eq. (4.2.27). Applying similar steps to the ones in Section 4.1.3, we arrive at:

$$\Lambda_k^\delta \mid \boldsymbol{\sigma}^2, \boldsymbol{\theta}, \mathbf{F}, \mathbf{Y} \sim \mathcal{N}_{d_k}(\mathbf{M}_k, \mathbf{L}_{kN}), \quad (4.2.28)$$

where

$$\mathbf{L}_{kN} = \sigma_k^2 \left(\mathbf{X}_k^\delta \mathbf{X}_k^{\delta\top} + \mathbf{L}_{0k}^{-1} \right)^{-1}, \quad (4.2.29)$$

and

$$\mathbf{M}_k = \mathbf{X}_k^\delta \tilde{\mathbf{y}}_k. \quad (4.2.30)$$

4.2.5 Column Shrinkage

Since $\mathbf{Y}$ is conditionally independent from $\boldsymbol{\theta}$ given Λ ,

$$p(\boldsymbol{\theta} \mid \mathbf{Y}, \mathbf{F}, \Lambda, \delta, \boldsymbol{\sigma}^2, \boldsymbol{\tau}) = p(\boldsymbol{\theta} \mid \mathbf{F}, \Lambda, \delta, \boldsymbol{\sigma}^2, \boldsymbol{\tau}). \quad (4.2.31)$$

Moreover, the unconstrained entries of the columns of lambda $\Lambda_{:,j}^\delta \mid \delta, \boldsymbol{\sigma}^2 \sim \mathcal{N}_{d^j}(0, \sigma_k^2 \theta_j)$, where d^j is the number of unconstrained entries of column j of Λ . Each

$\Lambda_{:,j}^\delta$ is conditionally independent of θ_k for $k \neq j$. Thus,

$$p(\boldsymbol{\theta} \mid \mathbf{F}, \Lambda, \delta, \boldsymbol{\sigma}^2, \boldsymbol{\tau}) \propto p(\Lambda \mid \delta, \boldsymbol{\sigma}^2, \boldsymbol{\tau}, \boldsymbol{\theta})p(\boldsymbol{\theta}), \quad (4.2.32)$$

by Bayes' theorem, and prior independence of $\boldsymbol{\tau}, \mathbf{F}, \boldsymbol{\theta}, \boldsymbol{\sigma}^2, \delta$. Note,

$$p(\Lambda \mid \delta, \boldsymbol{\sigma}^2, \boldsymbol{\tau}, \boldsymbol{\theta})p(\boldsymbol{\theta}) = p(\Lambda \mid \delta, \boldsymbol{\sigma}^2, \boldsymbol{\theta})p(\boldsymbol{\theta}). \quad (4.2.33)$$

Since, given δ , Λ does not depend on $\boldsymbol{\theta}$, and further,

$$p(\Lambda \mid \delta, \boldsymbol{\sigma}^2, \boldsymbol{\theta})p(\boldsymbol{\theta}) = p(\Lambda^\delta \mid \delta, \boldsymbol{\sigma}^2, \boldsymbol{\theta})p(\boldsymbol{\theta}), \quad (4.2.34)$$

as $\boldsymbol{\theta}$ is independent of the constrained entries. Finally,

$$p(\Lambda^\delta \mid \delta, \boldsymbol{\sigma}^2, \boldsymbol{\theta}) = \prod_{j=1}^q p(\Lambda_{:,j}^\delta \mid \delta, \boldsymbol{\sigma}^2, \boldsymbol{\theta})p(\boldsymbol{\theta}). \quad (4.2.35)$$

As $\Lambda_{:,j}^\delta \mid \delta, \boldsymbol{\sigma}^2 \sim \mathcal{N}_{dj}(0, \theta_j B_j)$, and $\theta_j \sim \mathcal{G}^{-1}(a_\theta, b_\theta)$, it follows from Normal-Inverse Gamma conjugacy for the variance parameter that:

$$\theta_j \mid \mathbf{Y}, \mathbf{F}, \Lambda, \delta, \boldsymbol{\sigma}^2, \boldsymbol{\tau} \sim \mathcal{G}^{-1} \left(a_\theta + \frac{d^j}{2}, b_\theta + \frac{1}{2} \Lambda_{:,j}^{\delta \top} B_j^{-1} \Lambda_{:,j}^\delta \right) \quad (4.2.36)$$

4.2.6 Parameter Expansion for the UGLT Model

Parameter expansion for the UGLT model is similar to the PLT model. We use a similar scheme as Section 4.1.5. The only difference is in the conditional posterior of

ψ , though this is similar, as shown below.

Note

$$p(\psi \mid \mathbf{Y}, \mathbf{F}^*, \Lambda^*, \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta) = p(\psi \mid \mathbf{Y}, \mathbf{F}^*, \Lambda^*, \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta), \quad (4.2.37)$$

by (4.1.27). By Bayes' theorem, we have:

$$p(\psi \mid \mathbf{F}^*, \Lambda^*, \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta) \propto p(\Lambda^*, \mathbf{F}^* \mid \psi, \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta) p(\psi \mid \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta) \quad (4.2.38)$$

$$= p(\Lambda^*, \mathbf{F}^* \mid \psi, \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta) p(\psi), \quad (4.2.39)$$

where the last line is due to a priori independence. Similarly, since $\mathbf{F}$, and Λ are considered to be independent, and $\mathbf{F}$ is also independent of $\boldsymbol{\theta}$, $\boldsymbol{\tau}$, σ^2 , it follows that (4.2.39) is equal to

$$p(\Lambda^* \mid \psi, \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta) p(\mathbf{F} \mid \psi, \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta) p(\psi). \quad (4.2.40)$$

And,

$$\begin{aligned} p(\Lambda^* \mid \psi, \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta) p(\mathbf{F} \mid \psi, \sigma^2, \boldsymbol{\theta}, \boldsymbol{\tau}, \delta) p(\psi) &= p(\Lambda^* \mid \psi, \sigma^2, \boldsymbol{\theta}, \delta) \\ &\cdot p(\mathbf{F}^* \mid \psi) p(\psi). \end{aligned} \quad (4.2.41)$$

Denote the unconstrained entries of the columns of Λ , as $\Lambda_{\cdot,j}^\delta$ which is a $d_j = \sum_{i=1}^v \delta_{ij}$ dimensional vector by the definition of δ . Then,

$$\Lambda_{\cdot,j}^\delta \mid \delta, \theta_j, \sigma^2 \sim \mathcal{N}_{d_j}(0, \theta_j \Sigma). \quad (4.2.42)$$

Thus, considering the same transformations as before where $\Lambda^* = \Lambda(\Psi)^{-\frac{1}{2}}$, we have:

$$\tilde{\Lambda}_{:,j}^\delta | \delta, \theta_j, \sigma^2 \sim \mathcal{N}_{d_j}(0, \psi_j^{-1} \theta_j \Sigma). \quad (4.2.43)$$

From (4.1.30) we have:

$$\tilde{\mathbf{f}}_j^* = \psi_j^{\frac{1}{2}} \tilde{\mathbf{f}}_j \sim \mathcal{N}_N(0, \psi_j \mathbf{I}_N) \text{ conditionally on } \Psi_j. \quad (4.2.44)$$

Thus, we finally derive the conditional distribution:

$$\begin{aligned} p(\Lambda | \psi, \sigma^2, \theta, \delta) p(\mathbf{F}^* | \psi) p(\psi) &= \prod_{j=1}^q p(\tilde{\Lambda}_{:,j}^\delta | \psi, \sigma^2, \theta, \delta) \\ p(\tilde{\mathbf{f}}_j | \psi_j) p(\psi_j) &\propto \prod_{j=1}^q |\Sigma|^{-\frac{1}{2} d_j} \theta_j^{-\frac{1}{2} d_j} \psi_j^{\frac{d_j}{2}} \exp \left\{ -\frac{1}{2\theta_j} \psi_j \tilde{\Lambda}_{:,j}^\delta \top \Sigma^{-1} \tilde{\Lambda}_{:,j}^\delta \right\} \psi_j^{-\frac{N}{2}} \exp \left\{ -\frac{1}{2\psi_j} \tilde{\mathbf{f}}_j^\top \tilde{\mathbf{f}}_j \right\} \quad (4.2.45) \\ \psi_j^{-\alpha_\psi - 1} \exp \left(\frac{\beta_\psi}{\psi_j} \right) &= \prod_{j=1}^q \psi_j^{-(\alpha_\psi + \frac{N}{2}) + \frac{d_j}{2} - 1} \exp \left\{ -\frac{1}{2} \left[\frac{2\beta_\psi + \tilde{\mathbf{f}}_j^\top \tilde{\mathbf{f}}_j}{\psi_j} + \frac{\psi_j}{\theta_j} \tilde{\Lambda}_{:,j}^\delta \top \Sigma^{-1} \tilde{\Lambda}_{:,j}^\delta \right] \right\}. \end{aligned}$$

So,

$$\psi_j | \sigma^2, \theta, \Lambda, \mathbf{F}, \delta \sim \mathcal{GIG} \left(-\left(\alpha_\psi + \frac{N}{2} \right) + \frac{d_j}{2}, 2\beta_\psi + \tilde{\mathbf{f}}_j^\top \tilde{\mathbf{f}}_j, \frac{1}{\theta_j} \tilde{\Lambda}_{:,j}^\delta \top \Sigma^{-1} \tilde{\Lambda}_{:,j}^\delta \right) \quad (4.2.46)$$

Chapter 5

Simulations

5.1 General Setup

We conduct three simulation studies to compare models under PLT and UGLT identification constraints. In the first study, we generate data under a PLT structure with two factors across eight settings, formed by changing the number of observations ($N \in \{200, 400\}$), the number of variables ($v \in \{10, 20\}$), and the presence or absence of noisy variables for a total of eight settings. We compare a non-sparse PLT model against sparse PLT and UGLT models equipped with column sparsity priors. In the second study, we generate data under a sparse UGLT structure with three factors and $N = 1200$ observations, varying the number of variables ($v \in \{20, 30, 40\}$), and compare the two sparse models to assess whether the sparse PLT model can adequately capture signal generated under a UGLT constraint. In the third study, we evaluate the effectiveness of post-processing for determining the number of relevant factors. We generate data under a UGLT structure with four factors, $N = 2000$ observations, and $v = 30$ variables, and fit both sparse models with seven factors, while

the true number of factors was set to be four.

We opt for weakly informative priors for each study, they are given in Table 5.1.

Table 5.1: Prior specifications for each model.

Parameter	PLT	Sparse UGLT	Sparse PLT
Λ	$\Lambda_{kj} \stackrel{iid}{\sim} \mathcal{N}(0, 1)$	$\Lambda_{kj} \mid \delta_{kj} \sim (1 - \delta_{kj})\Delta_0 + \delta_{kj}\mathcal{N}(0, \sigma_k^2\theta_j)$	Same as UGLT
δ	—	$\delta_{kj} \mid \tau_j, l_j \sim \mathbf{1}_{k>l_j} \cdot \text{Bern}(\tau_j) + \mathbf{1}_{k=l_j} \cdot \Delta_1$	Same as UGLT
τ	—	$\tau_j \stackrel{iid}{\sim} \mathcal{B}(1, 1)$	Same as UGLT
ℓ	—	$l_j \mid \ell_{-j} \sim \mathcal{U}\{[V] \setminus \ell_{-j}\}$	—
σ^2	$\sigma_k \stackrel{iid}{\sim} \mathcal{G}^{-1}(1.5, 1.5)$	$\sigma_k \stackrel{iid}{\sim} \mathcal{G}^{-1}(1.5, 1.5)$	Same as UGLT
θ	—	$\theta_j \stackrel{i.i.d.}{\sim} \mathcal{G}^{-1}(1.5, 1.5)$	Same as UGLT
Ψ	$\Psi_j \sim \mathcal{G}^{-1}(1.5, 1.5)$	Same as PLT	Same as PLT

The models were assessed using the continuous ranked probability score (CRPS), goodness-of-fit, and Noise Ratio. We provide a brief discussion of these metrics in Section 5.1.1.

5.1.1 Performance Metrics

We provide a brief discussion of the metrics used to assess model performance.

CRPS [Gneiting and Raftery, 2007] is a scoring rule that evaluates probabilistic forecasts by measuring the integrated squared difference between the forecast cumulative distribution function (CDF) F and the empirical CDF of an observation y :

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(x) - \mathbf{1}\{x \geq y\})^2 dx. \quad (5.1.1)$$

A lower CRPS indicates a better forecast, with a score of zero achieved only by a perfect point forecast at the true observation. We compute the average CRPS between the true value of Λ and the posterior draws Λ element-wise.

The signal-to-noise ratio measures the proportion of total data variance attributable to the factor structure. Specifically, the variance explained by the factors is given by $\text{tr}(\Lambda\Lambda^\top)$, while the total variance is computed using the trace of the estimate of $\text{Cov}(\mathbf{y}_i)$ which we denote with $\hat{\Omega}$:

$$\text{Noise Ratio}(\Lambda, \Sigma) = \frac{\text{tr}(\Lambda\Lambda^\top)}{\text{tr}(\hat{\Omega})}. \quad (5.1.2)$$

To assess estimation accuracy, we report the absolute error between the true and estimated noise ratios:

$$\left| \frac{\text{tr}(\Lambda\Lambda^\top)}{\text{tr}(\hat{\Omega})} - \frac{\text{tr}(\Lambda_{\text{est}}\Lambda_{\text{est}}^\top)}{\text{tr}(\hat{\Omega})} \right|. \quad (5.1.3)$$

Goodness-of-fit is assessed by simulating $S = 20$ datasets under both the fitted and true models, and computing the MSE between data simulated under the full model. That is, for $S = 1, \dots, 20$, two data sets $\mathbf{y}_{1\text{est}}^{(s)}, \dots, \mathbf{y}_{N\text{est}}^{(s)}, \mathbf{y}_1^{(s)}, \dots, \mathbf{y}_N^{(s)}$ were simulated under the distributions $\mathbf{y}_{i\text{est}}^{(s)} \sim \mathcal{N}_v(\Lambda_{\text{est}}\mathbf{f}_{i\text{est}}, \Sigma_{\text{est}})$ and $\mathbf{y}_i^{(s)} \sim \mathcal{N}_v(\Lambda\mathbf{f}_i, \Sigma)$ and the MSE is computed as follows:

$$\text{MSE} = \frac{1}{N \times S} \sum_{s=1}^S \sum_{i=1}^N \frac{1}{v} \left\| \mathbf{y}_{i\text{est}}^{(s)} - \mathbf{y}_i^{(s)} \right\|_{L_2}^2, \quad (5.1.4)$$

where $\Lambda_{\text{est}}, \mathbf{y}_{i\text{est}}^{(s)}, \mathbf{f}_{i\text{est}}, \Sigma_{\text{est}}$ are Monte Carlo estimates from a model fitted on data generated based on.

The mean squared error is computed between the true observed data covariance $\Lambda\Lambda^\top + \Sigma$ and its posterior estimate $\Lambda_{\text{est}}\Lambda_{\text{est}}^\top + \Sigma_{\text{est}}$:

$$\text{MSE} = \frac{1}{v^2} \sum_{i=1}^v \sum_{j=1}^v \left([\Lambda\Lambda^\top + \Sigma]_{ij} - [\Lambda_{\text{est}}\Lambda_{\text{est}}^\top + \Sigma_{\text{est}}]_{ij} \right)^2. \quad (5.1.5)$$

5.2 Simulation 1: Adding noisy variables to the PLT Bayesian factor model

In this section, we provide a simulation study to assess how noisy variables affect model performance across three Bayesian factor models: the non-sparse PLT model, the sparse PLT model, and the sparse UGLT model. We define a variable to be noisy or irrelevant if none of the factors are related to it. That is, if the k 'th variable is noisy, then the k 'th row of the loading matrix Λ in Eq. (1.0.3) is entirely zero.

We consider eight simulation settings, where the datasets are generated according to Eq. (1.0.3) with $q = 2$ factors. The number of observations N , the number of variables v , and the presence of irrelevant variables are varied across settings as summarized in Table 5.2. When irrelevant variables are present, the last 20% of rows

Table 5.2: Summary of simulation settings for Study 1. Each row represents one simulated dataset configuration.

Setting	N	v	Number of Irrelevant Variables
1	200	10	0
2	200	10	2
3	200	20	0
4	200	20	4
5	400	10	0
6	400	10	2
7	400	20	0
8	400	20	4

in Λ (see Figure 5.1 for an example).

The non-zero entries of the loading matrix were drawn from a normal distribution with mean zero and variance 0.5, and the absolute values for the diagonal entries were taken. The data-generating distributions are summarized in Table 5.3.

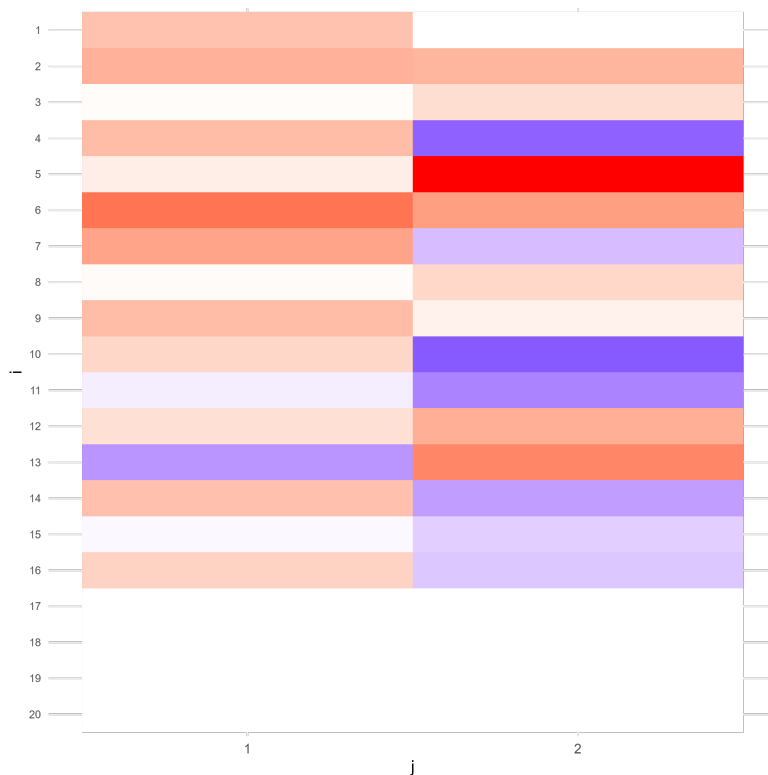


Figure 5.1: An example heatmap of a loading matrix with irrelevant variables (the last 4 rows are zero).

We fit three Bayesian models to each dataset: a non-sparse PLT model, a sparse PLT model, and a sparse UGLT model. Posterior inference was carried out via MCMC using the algorithms discussed in Chapter 4. The prior specifications for each model are summarized in Table 5.1. For the non-sparse PLT model, an MCMC chain of length 5000 was generated, with the first 2000 used as a burn in, and the chain was thinned by a factor of 10. For the other two models, chains of length 5000 was generated, the first 500 were discarded for a burn in, and a thinned by a factor of 2. We post-process the draws to ensure draws of Λ had a PLT structure. For both the sparse PLT and sparse UGLT model, none of the draws of Λ were identifiable for the setting with noisy variables included, where $v = 10$, and $N = 200$.

Table 5.3: Simulation Study 1: Data-generating distributions under a PLT structure with noisy variables.

Parameter	Value
Identification structure	PLT
Number of factors (q)	2
Number of observations (N)	$\{200, 400\}$
Number of variables (v)	$\{10, 20\}$
Proportion of noisy variables	$\{0\%, 20\%\}$
Total settings	8
Factor distribution	$\mathbf{f}_i \sim \mathcal{N}_q(0, I_q)$
Pivot rows	ℓ is sampled uniformly from $\{1, \dots, v-3\}$ without replacement.
Loading distribution	$\Lambda_{kj} \sim \mathcal{N}_q(1, \sigma = 0.5)$ if $j > k$, otherwise $\Lambda_{kj} = X $ where $X \sim \mathcal{N}_q(1, \sigma = 0.5)$.
Idiosyncratic variance	$\sigma_k^2 = 1$ for each $k \in \{1, \dots, v\}$
Noise distribution	$e_i \sim \mathcal{N}_v(0, \text{diag}(\boldsymbol{\sigma}^2))$
Data generation	$\mathbf{y}_i \mid \Lambda, \boldsymbol{\sigma}^2 \sim \mathcal{N}_v(\Lambda \mathbf{f}_i, \text{diag} \boldsymbol{\sigma}^2)$
Models compared	Non-sparse PLT, Sparse PLT, Sparse UGLT
Evaluation metrics	CRPS, MSE, Signal to noise.

Convergence diagnostics are reported in Table 5.4. The corresponding trace and autocorrelation plots are in Appendix B.1.

The model performance metrics are reported in Table 5.5 and Figure 5.2. Referring to Figure 5.2, we observe that the non-sparse PLT model outperformed all other models when all variables were relevant. There were mixed results for the setting with noisy variables. First, for the setting with $N = 200, v = 10$, neither sparse models were able to fit the data, and the non-sparse PLT model tended to out perform the other two. Moreover, apart from the setting where $N = 200$ and $v = 10$, the non-sparse PLT model had the best goodness-of-fit and lowest MSE in settings where noisy variables were included. The UGLT model tended to perform similarly to the sparse PLT model in terms of goodness-of-fit and MSE. These results indicate that a column-wise spike and slab prior may adapt poorly to the presence of noisy variables.

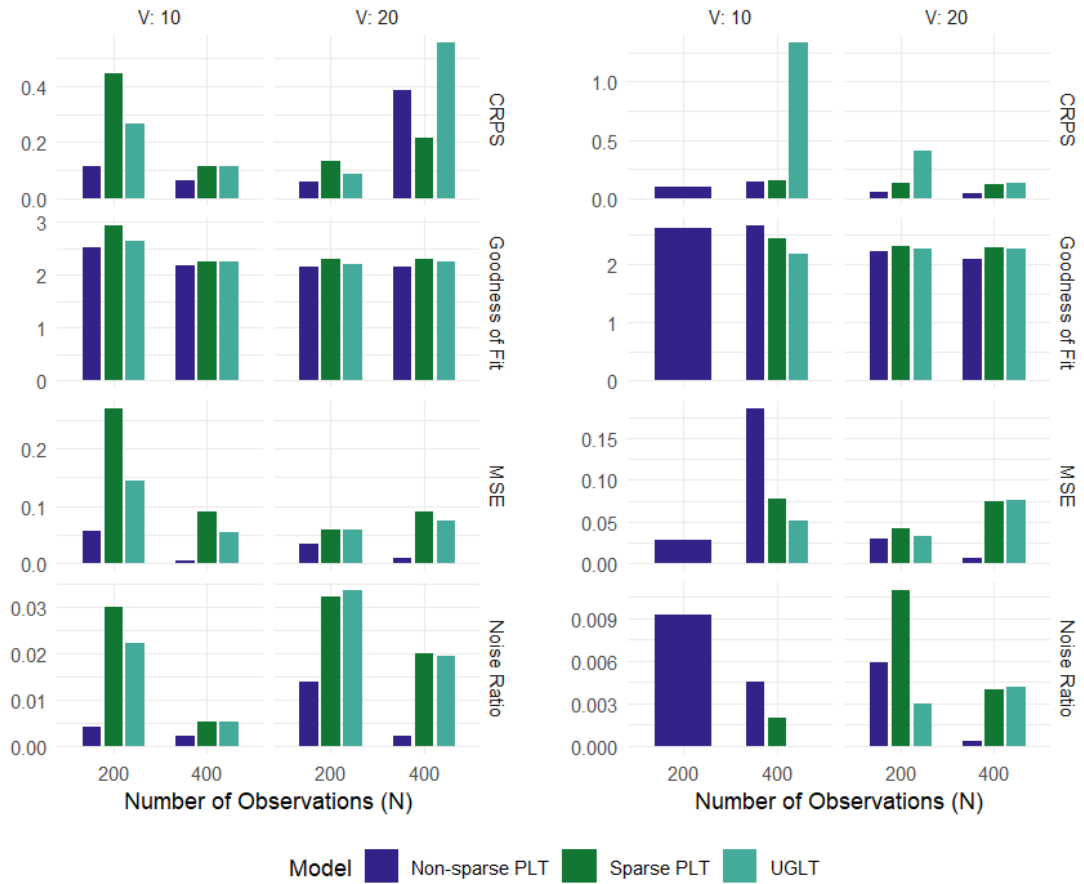


Figure 5.2: Performance metrics for simulation study 1. (Left) Metrics of each model where all variables are relevant. (Right) Metrics of each model where some of the variables are irrelevant.

Table 5.4: MCMC convergence diagnostics for Simulation Study 1. Effective sample size (ESS) ratio and Geweke’s diagnostic are reported for the trace of $\Lambda\Lambda^\top + \Sigma$. Not enough samples were gathered to compute Geweke’s diagnostic for the UGLT model in the setting without noisy variables, with $N = 200$ and $v = 10$.

Setting	Model	ESS Ratio	Geweke
With noisy variables			
$v = 10, N = 200$	Non-sparse PLT	0.696	−1.890
	Sparse PLT	—	—
	UGLT	—	—
$v = 10, N = 400$	Non-sparse PLT	0.452	−0.158
	Sparse PLT	1.000	0.335
	UGLT	0.830	0.529
$v = 20, N = 200$	Non-sparse PLT	0.764	−1.180
	Sparse PLT	1.000	0.214
	UGLT	1.000	−0.093
$v = 20, N = 400$	Non-sparse PLT	1.000	1.680
	Sparse PLT	1.000	−0.005
	UGLT	1.000	1.230
Without noisy variables			
$v = 10, N = 200$	Non-sparse PLT	0.886	−0.686
	Sparse PLT	0.647	1.460
	UGLT	1.000	NA
$v = 10, N = 400$	Non-sparse PLT	1.000	0.709
	Sparse PLT	1.000	1.210
	UGLT	1.000	1.240
$v = 20, N = 200$	Non-sparse PLT	0.852	−2.600
	Sparse PLT	1.000	−0.740
	UGLT	0.864	−0.896
$v = 20, N = 400$	Non-sparse PLT	0.897	−0.929
	Sparse PLT	0.939	−0.702
	UGLT	1.000	0.914

This conclusion is further supported in Figure 5.3, where we see that while the UGLT model correctly identifies variables 16–20 as noisy, it incorrectly classifies some other variables as noisy as well.

Table 5.5: Model performance metrics for Simulation Study 1 (grouped by setting).
The Noise Column represents the absolute difference between the true noise ratio
and the estimated noise ratio.

Setting	Model	GoF	MSE	CRPS	Noise	True Noise
With noisy variables						
$v = 10, N = 200$	Non-sparse PLT	2.640	0.028	0.101	0.009	0.019
	Sparse PLT	—	—	—	—	—
	UGLT	—	—	—	—	—
$v = 10, N = 400$	Non-sparse PLT	2.680	0.186	0.146	0.005	0.017
	Sparse PLT	2.470	0.149	0.078	0.011	
	UGLT	2.170	0.052	1.330	0.003	
$v = 20, N = 200$	Non-sparse PLT	2.210	0.029	0.050	0.006	0.041
	Sparse PLT	2.330	0.127	0.042	0.002	
	UGLT	2.260	0.412	0.032	0.000	
$v = 20, N = 400$	Non-sparse PLT	2.100	0.007	0.038	0.000	0.019
	Sparse PLT	2.290	0.123	0.075	0.004	
	UGLT	2.280	0.128	0.077	0.004	
Without noisy variables						
$v = 10, N = 200$	Non-sparse PLT	2.530	0.058	0.112	0.009	0.024
	Sparse PLT	2.930	0.446	0.271	0.013	
	UGLT	2.610	0.265	0.143	0.006	
$v = 10, N = 400$	Non-sparse PLT	2.200	0.004	0.061	0.000	0.014
	Sparse PLT	2.230	0.115	0.091	0.003	
	UGLT	2.240	0.112	0.055	0.005	
$v = 20, N = 200$	Non-sparse PLT	2.170	0.034	0.057	0.009	0.045
	Sparse PLT	2.320	0.133	0.060	0.000	
	UGLT	2.200	0.086	0.060	0.000	
$v = 20, N = 400$	Non-sparse PLT	2.150	0.010	0.385	0.002	0.024
	Sparse PLT	2.290	0.214	0.090	0.003	
	UGLT	2.260	0.556	0.074	0.003	

5.3 Simulation 2: Comparing the UGLT and PLT Models

In this simulation, we assess the performance of the sparse UGLT model in comparison to the sparse PLT model when the data-generating process follows a UGLT structure.

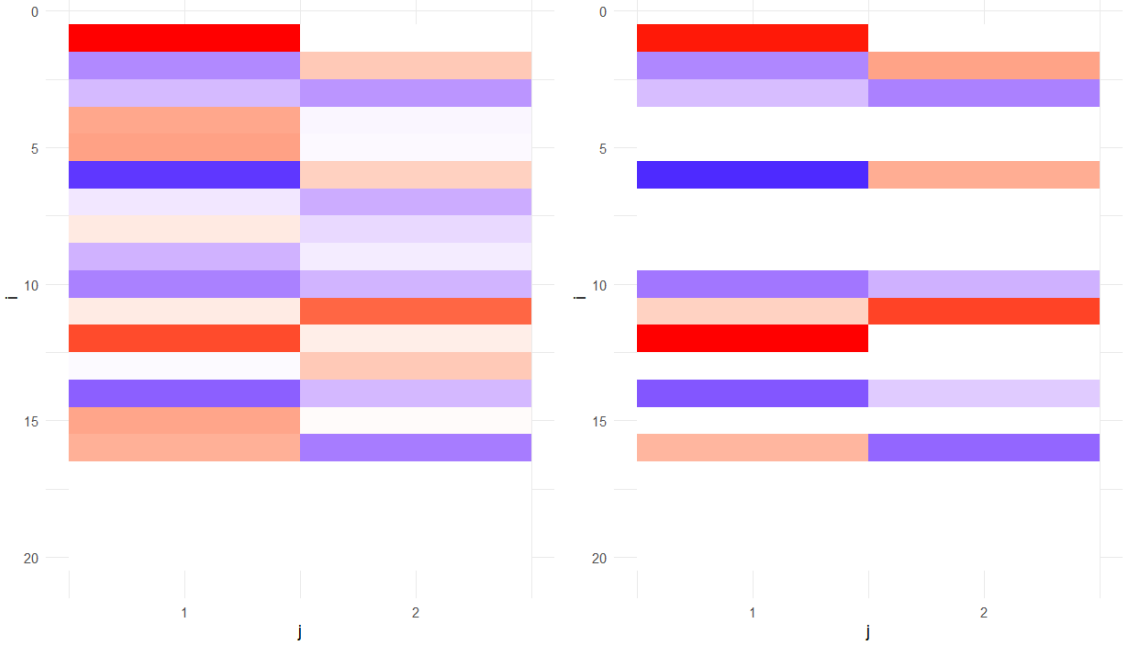


Figure 5.3: Left: Heatmap of the true loading matrix for the setting with noisy variables ($N = 400$, $v = 20$). Right: Heatmap of the estimated loading matrix for the UGLT model, with noisy variables ($N = 400$, $v = 20$). In both heatmaps, red indicates higher positive values and blue indicates lower negative values.

We consider four settings with $q = 3$ factors and $N = 1200$ observations. The only quantity that varies across settings is the number of variables, $v \in \{20, 30, 40\}$.

The data were simulated according to the UGLT factor model. The idiosyncratic variances and the column shrinkage parameters $\boldsymbol{\theta}$ were all set to 1, and the loading matrix was simulated as follows. First, pivots $l_1, \dots, l_q$ were sampled uniformly from $\{1, 2, \dots, v - 3\}$. Note, we only sample to $v - 3$ for the sake of ensuring that the 3579 counting rule is satisfied. Then, for each index (k, j) in the loading matrix with $k > l_j$:

$$\lambda_{kj} \sim \frac{j}{2q+1} \Delta_0 + \left(1 - \frac{j}{2q+1}\right) \mathcal{N}(0, 1), \quad (5.3.1)$$

so that the probability of a zero loading increases with the column index, producing a

sparser structure in later columns. The loading matrix is then multiplied by a signed matrix $\mathbf{P}$ so that the pivots are all positive. The data generating distributions are summarized in Table 5.6.

Table 5.6: Data-generating settings under a sparse UGLT structure for simulation study 2.

Parameter	Value
Identification structure	Sparse UGLT
Number of factors (q)	3
Number of observations (N)	1200
Number of variables (v)	$\{20, 30, 40\}$
Total settings	4
Factor distribution	$\mathbf{f}_i \sim \mathcal{N}_q(0, \mathbf{I}_q)$
Loading distribution	See (5.3.1) and the discussion that follows.
Idiosyncratic variance	$\sigma_k^2 = 1$
Noise distribution	$\mathbf{e}_i \sim \mathcal{N}_v(0, \mathbf{I}_v)$

Like simulation study 1, posterior inference was carried out via MCMC using the algorithms discussed in Chapter 4. For both the sparse PLT and UGLT models, 7000 draws were generated, the first 1000 were discarded for a burn-in, and then the chains were thinned by a factor of 2. Post-processing for model identification was then applied. We report the ESS and Geweke’s diagnostics for each posterior simulation in Table 5.7. Corresponding Trace and ACF plots are in Appendix B.2.

Table 5.7: MCMC convergence diagnostics for Simulation Study 2. Effective sample size (ESS) ratio and Geweke’s diagnostic are reported for $\Lambda\Lambda^\top + \Sigma$.

Setting	Model	ESS Ratio	Geweke
$v = 20, N = 1200$	Sparse PLT	1.000	2.130
	UGLT	1.000	4.000
$v = 30, N = 1200$	Sparse PLT	1.000	-1.710
	UGLT	1.000	-1.420
$v = 40, N = 1200$	Sparse PLT	0.710	1.050
	UGLT	0.788	-0.292

The model performance metrics are shown in Table 5.8 and Figure 5.4. We can

Table 5.8: Model performance metrics for Simulation Study 2 (grouped by setting).

Setting	Model	GoF	MSE	CRPS	Noise
$v = 20, N = 1200$	Sparse PLT	3.560	0.361	0.190	0.001
	UGLT	2.490	0.022	0.158	0.002
$v = 30, N = 1200$	Sparse PLT	3.350	1.190	0.330	0.012
	UGLT	2.150	0.076	0.042	0.001
$v = 40, N = 1200$	Sparse PLT	2.870	0.451	0.278	0.012
	UGLT	2.820	0.080	0.318	0.013

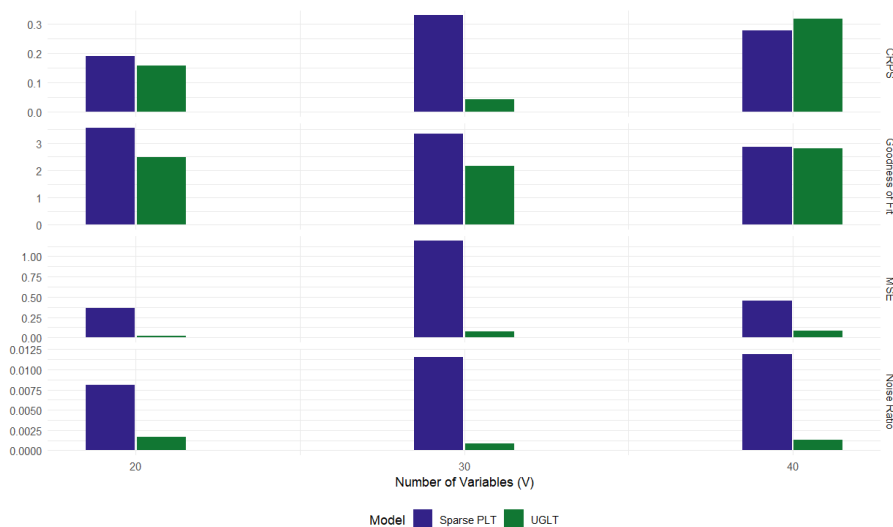


Figure 5.4: Performance metrics for simulation study 2.

see from the disparity CRPS, MSE, and Noise Ratio in Figure 5.4 that the UGLT model outperforms the PLT model across all metrics in the settings with $v = 20$ and 30 variables. In the $v = 40$ case, we observe that the Sparse PLT model performs similarly to the UGLT model. This indicates that in settings with fewer variables, the Sparse PLT model may be sensitive to structural misspecification.

5.3.1 Simulation Study 3: Recovering the Number of Factors

The goal of this study is to evaluate how well post-processing can recover the true number of latent factors when the model is overspecified. We generate data under both sparse PLT and UGLT structures with $q = 4, 10$ true factors, $N = 2000$ observations, and $v = 30$ variables. Both the sparse PLT and sparse UGLT models are then fitted with three more than the true number. The data-generating settings are summarized in Table 5.9.

Table 5.9: Simulation Study 3: Overfitting study under a UGLT structure.

Parameter	Value
Identification structure	UGLT & Sparse PLT
True number of factors (q)	4, 10
Fitted number of factors	7, 13
Number of observations (N)	2000
Number of variables (v)	30
Total settings	1
Factor distribution	$f_i \sim \mathcal{N}_q(0, \mathbf{I}_q)$
Pivot rows	For the UGLT model, ℓ is sampled uniformly from $\{1, \dots, v - 3\}$ without replacement. For the PLT model, it is fixed to the diagonal entries.
Loading distribution	See (5.3.1) and the discussion that follows.
Noise distribution	$\mathbf{e}_i \sim \mathcal{N}_q(0, \mathbf{I}_v)$

Like the other simulations, posterior inference was carried out via MCMC using the algorithms discussed in 4. For both the sparse PLT and UGLT models, 6000 draws were generated, the first 1000 were discarded for a burn-in, and a thinned by a factor of 2. Post-processing for model identification was then applied. The ESS and Geweke’s diagnostics for each posterior simulation are provided in table 5.10. Corresponding Trace and ACF plots are in Appendix B.3.

We then applied post-processing to identify and remove spurious columns from

Table 5.10: MCMC convergence diagnostics for Simulation Study 3. Effective sample size (ESS) ratio and Geweke’s diagnostic are reported for $\Lambda\Lambda^\top + \Sigma$.

Setting	True q	Model	ESS Ratio	Geweke
UGLT structure	4	Sparse PLT	0.692	−0.163
		UGLT	0.500	−1.350
PLT structure	4	Sparse PLT	0.673	0.994
		UGLT	0.793	1.300
UGLT structure	10	Sparse PLT	0.470	−0.264
		UGLT	0.732	−0.105
PLT structure	10	Sparse PLT	0.354	−0.737
		UGLT	0.306	−2.580

the estimated factor loadings matrix via the transformation shown in Section 2.3.

The estimated number of factors for each model are reported in Tables 5.11 and 5.12.

Table 5.11: Frequency of the number of factors selected by post-processing in Simulation Study 3 ($q = 4$). Models were fit with 7 factors.

Data structure	Model	Number of factors selected			
		4	5	6	7
UGLT	UGLT	807	276	72	11
	Sparse PLT	1797	611	89	4
PLT	UGLT	236	315	237	2
	Sparse PLT	147	1020	905	429

Table 5.12: Frequency of the number of factors selected by post-processing in Simulation Study 3 ($q = 10$). Models were fit with 13 factors.

Data structure	Model	Number of factors selected			
		10	11	12	13
UGLT	UGLT	926	398	133	1
	Sparse PLT	997	1188	299	17
PLT	UGLT	348	726	194	3
	Sparse PLT	0	39	450	2012

Referring to Tables 5.11 and 5.12, we see that the UGLT model tends to better estimates the factors than the PLT model, though in most cases the number of factors

is still overestimated. The sparse PLT model, tends to overestimate the number of factors by a larger degree, sometimes not even identifying a single irrelevant factor. This suggests that the UGLT structure is better suited to distinguishing active factors from redundant ones when we overspecify the number of factors.

Chapter 6

Conclusion

In this thesis, we conducted a comparative study of three Bayesian factor models under two sparsity-induced identification strategies: the sparse and non-sparse positive lower triangular (PLT) models, and the unordered generalized lower triangular (UGLT) model, similar to the one introduced in Frühwirth-Schnatter et al. [2025b]. We described Bayesian factor models under these constraints, including prior specifications and model identification, provided an overview of Markov Chain Monte Carlo methods, derived the conditional posterior distributions, and developed a parameter expansion scheme to accelerate posterior sampling for each model.

We derived the conditional posterior distributions for each model, developed the corresponding MCMC sampling algorithms in R, and introduced parameter expansion via the generalized inverse Gaussian distribution to accelerate posterior simulation in all three models.

Three simulation studies were conducted to assess robustness under noisy variables, how well the PLT constraint performs under structural misspecification, and how accurately each model estimates the number of relevant factors.

In the first study, we investigated the robustness of each model to the presence of noisy variables. We generated data under a PLT structure with two factors across eight settings, formed by crossing the number of observations ($N \in \{200, 400\}$), the number of variables ($v \in \{10, 20\}$), and the presence or absence of noisy variables. The non-sparse PLT model generally outperformed the sparse PLT and UGLT models across most settings.

In the second study, we examined how well the PLT model performs when the true data-generating structure follows a UGLT constraint. We generated data under a UGLT structure with three factors and $N = 1200$ observations, where the number of variables was varied ($v \in \{20, 30, 40\}$). The UGLT model consistently performed better than the sparse PLT model across all settings, though goodness of fit scores and continuous ranked probability score (CRPS) were similar when the number of variables was 40.

In the third study, we evaluated how accurately each model estimates the true number of factors by generating data under both a UGLT and PLT structure with $q = 4, 10$ factors, and fitting the model with 3 extra factors. The number of observations was held at 2000, and the number of variables 30. In most cases, both the PLT and UGLT model tended to over estimate the number of factors, though the estimates from the UGLT model were closer.

These simulation studies support four main findings. From our first study, we observed that the non-sparse PLT model outperformed the sparse UGLT and sparse PLT model under the presence of noisy variables. This indicates that the column-wise spike and slab prior may not adapt well when irrelevant variables are present. This was supported by the fact that while the sparse models did produce zero loadings, there

were some relevant variables that were deemed irrelevant. Second, we observed that the sparse PLT model had worse goodness of fit scores than the sparse UGLT model in settings where the number of variables was less than 40. This suggests that models that use the PLT structure are not robust to structural misspecification in situations with few variables. Third, recovering the true number of factors remains a challenge for both sparse models, though the UGLT model tended to better estimate the number of factors compared to the PLT. Thus, the additional flexibility of the UGLT structure was beneficial for determining relevant factors. Finally, our parameter expansion scheme via the $\mathcal{GIG}$ distribution improved the efficiency of the MCMC algorithms.

6.1 Further Research

There are several avenues for further research based on our work. First, our simulation in the first study revealed that the spike-and-slab prior performed poorly under scenarios with irrelevant variables. A potential solution is to implement a row-wise sparsity prior such as the one used in Kaufmann and Schumacher [2017].

In our second simulation study, we observed that the PLT model performs poorly in the UGLT setting, we have yet to see how both the PLT and UGLT models compare in settings where the true loading matrix takes neither a PLT nor a UGLT structure. This direction is supported by Theorem 1 in Frühwirth-Schnatter et al. [2023a] which states that for a full rank matrix Λ , there exists a rotation matrix G such that ΛG has a GLT structure. Based on the theorem and results of simulation study 1, UGLT model may be robust to structural misclassifications, unlike the PLT model.

Initial investigation was conducted into applying the PLT and UGLT constraints

within Bayesian dynamic factor models, though we have yet to achieve model estimation via MCMC.

Appendix A

Code

The code for this thesis is available on github: https://github.com/simb2/Thesis_Code

Appendix B

Convergence Plots from Simulations

B.1 Convergence Plots for Simulation Study 1

B.1.1 Non-Sparse PLT model

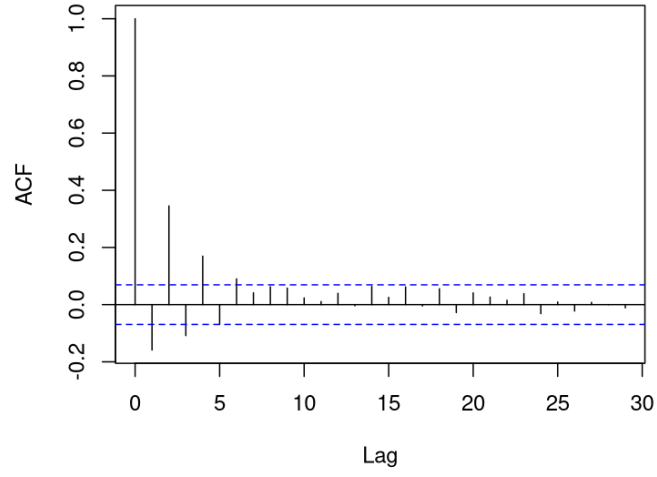


Figure B.1: ACF plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, noise variables included, $N = 200$, $v = 10$.

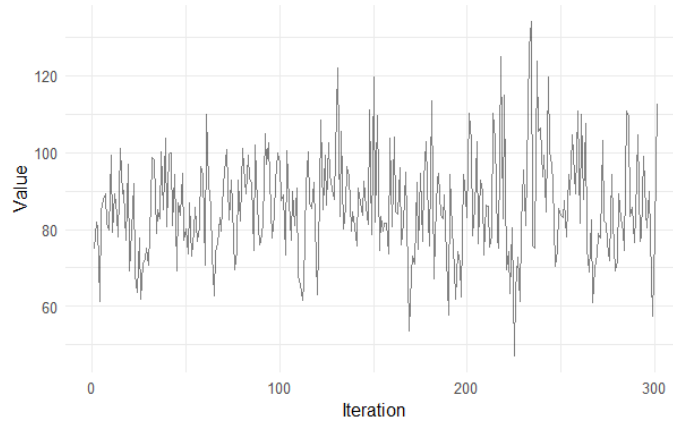


Figure B.2: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, noise variables included, $N = 200$, $v = 10$.

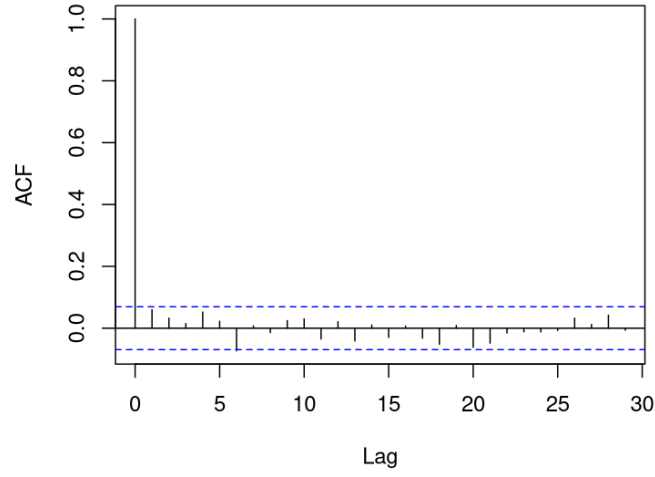


Figure B.3: ACF plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, noise variables included, $N = 200$, $v = 10$.

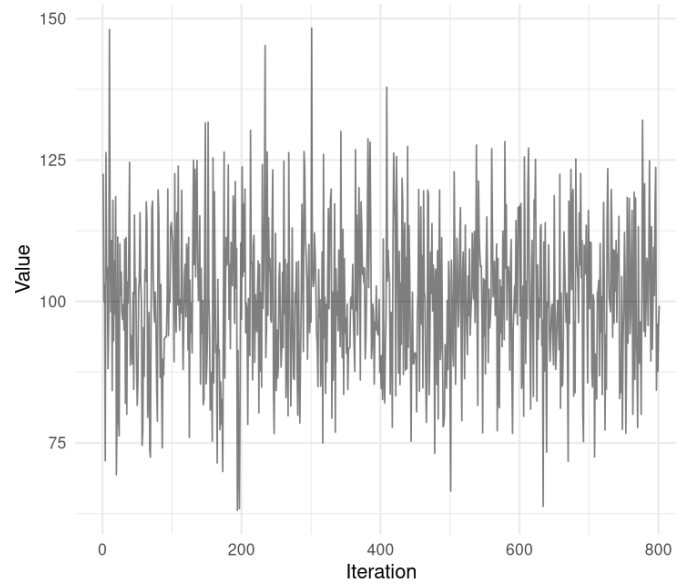


Figure B.4: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, no noise variables included, $N = 200$, $v = 10$.

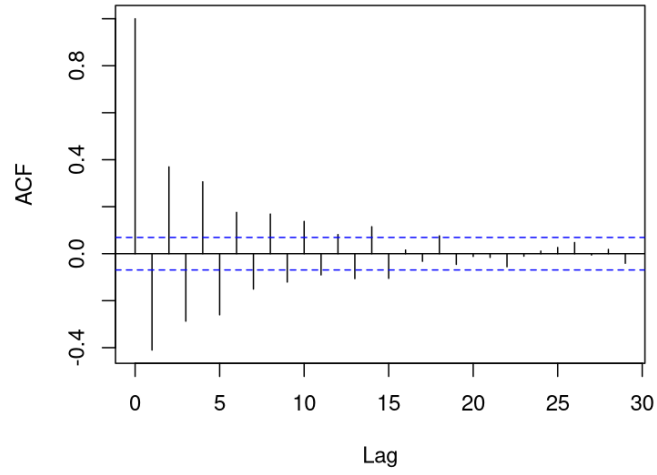


Figure B.5: ACF plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, noise variables included, $N = 200$, $v = 20$.

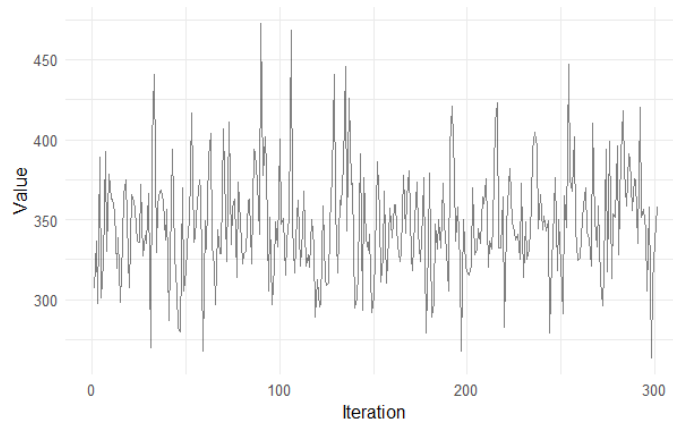


Figure B.6: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, noise variables included, $N = 200$, $v = 20$.

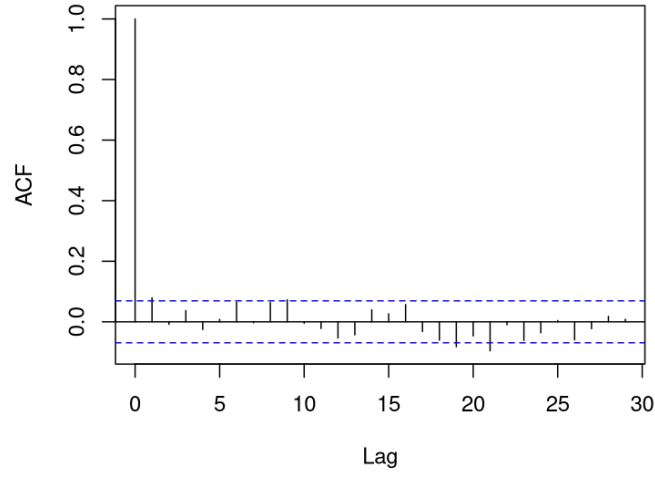


Figure B.7: ACF plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, no noise variables included, $N = 200$, $v = 20$.

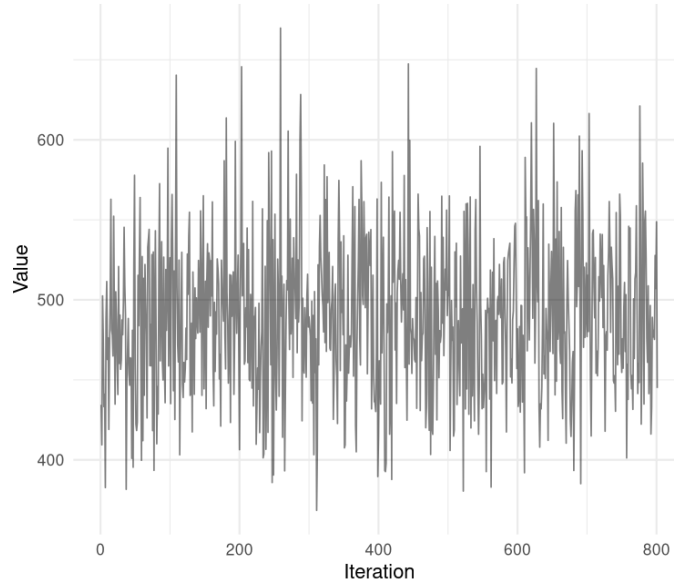


Figure B.8: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, no noise variables included, $N = 200$, $v = 20$.

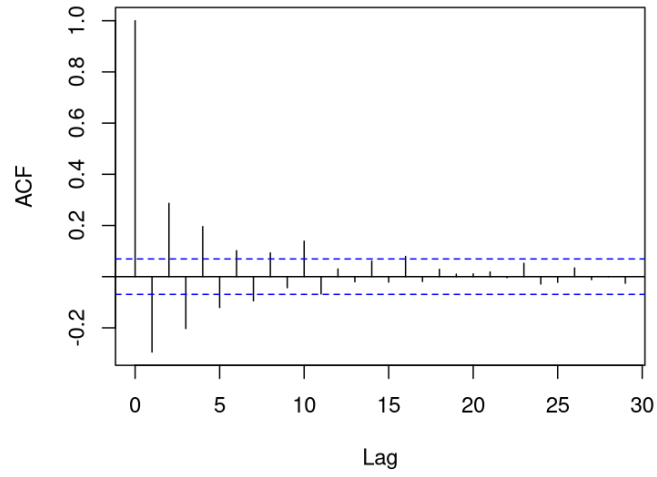


Figure B.9: ACF plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, noise variables included, $N = 400$, $v = 10$.

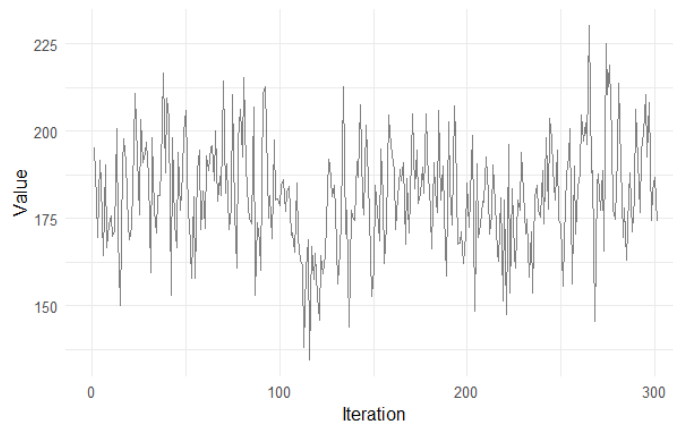


Figure B.10: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, noise variables included, $N = 400$, $v = 10$.

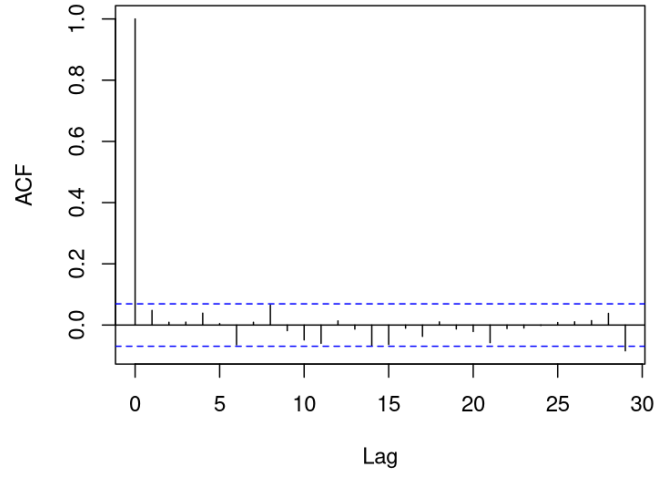


Figure B.11: ACF plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, no noise variables included, $N = 400$, $v = 10$.

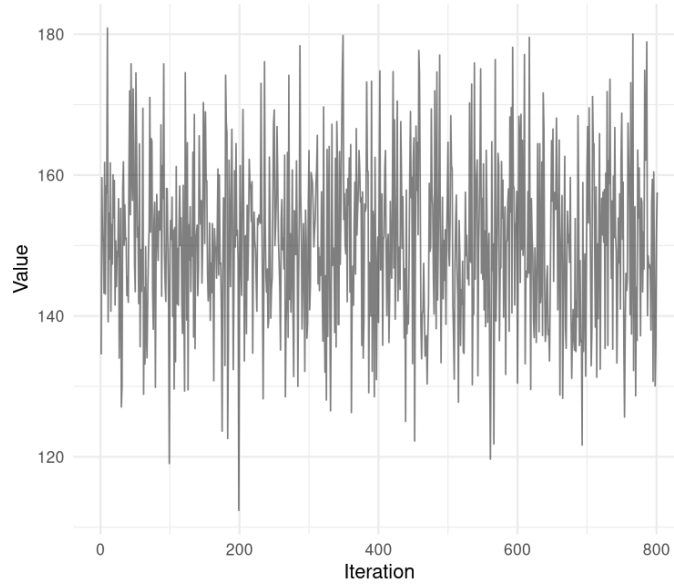


Figure B.12: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, no noise variables included, $N = 400$, $v = 10$.

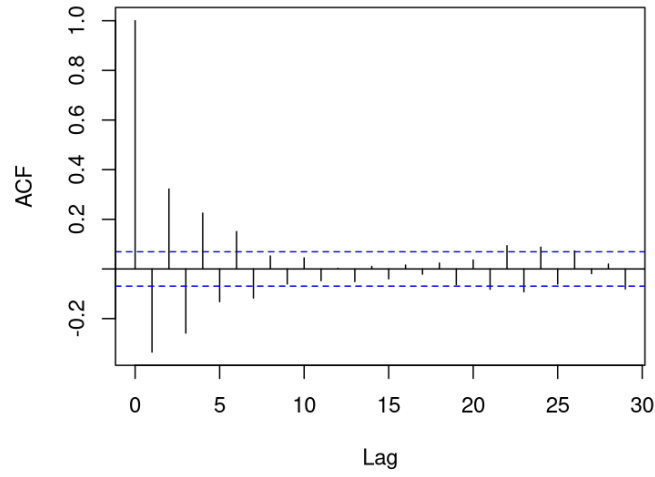


Figure B.13: ACF plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, noise variables included, $N = 400$, $v = 20$.

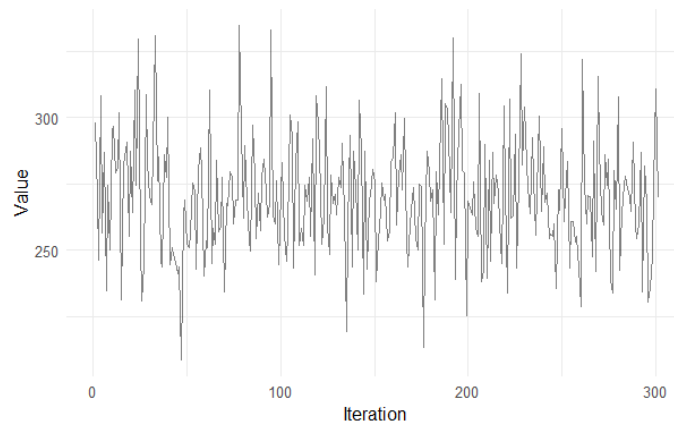


Figure B.14: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, noise variables included, $N = 400$, $v = 20$.

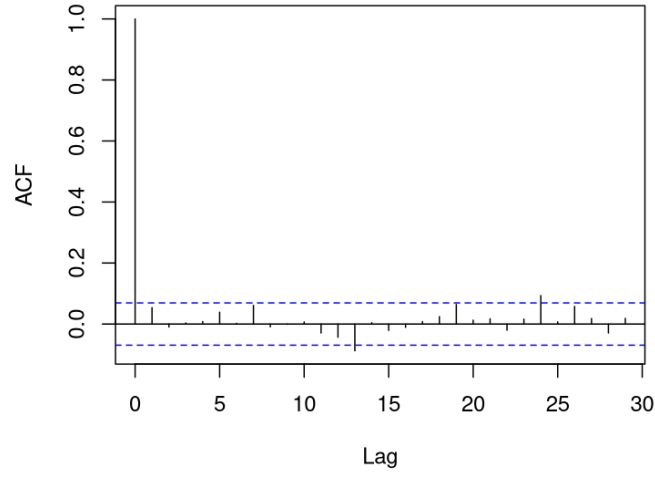


Figure B.15: ACF plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, no noise variables included, $N = 400$, $v = 20$.

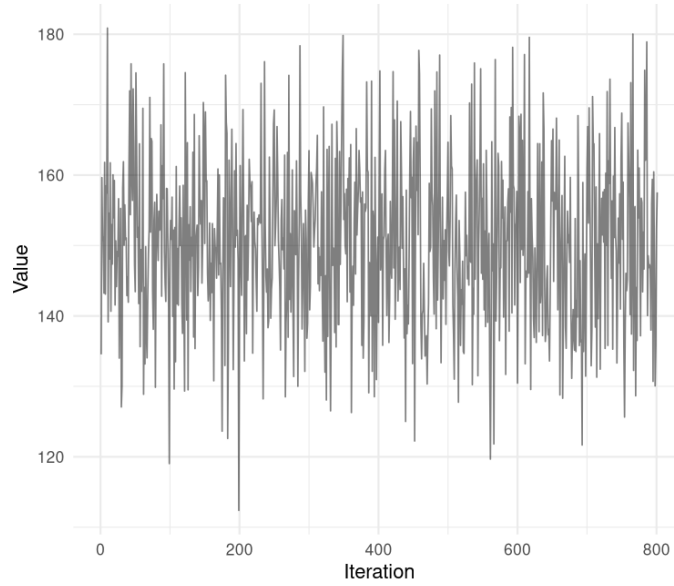


Figure B.16: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for non-sparse PLT model, no noise variables included, $N = 400$, $v = 20$.

B.1.2 Sparse PLT Model

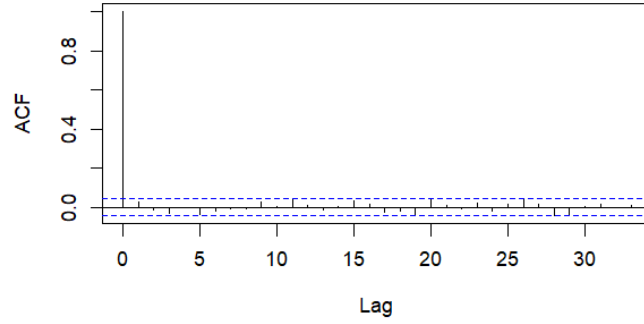


Figure B.17: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for Sparse PLT model, noise variables included, $N = 200$, $v = 20$.

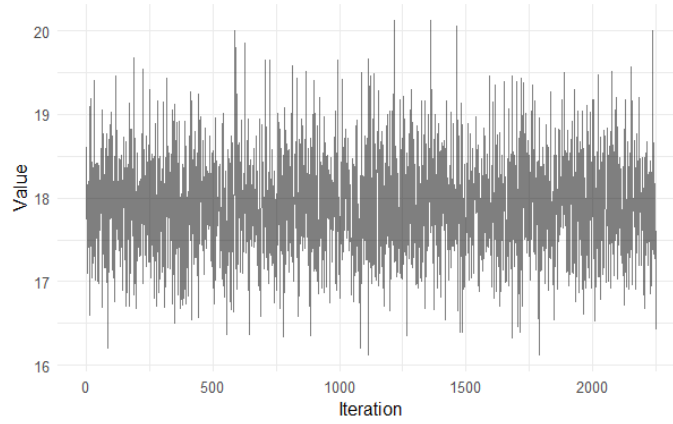


Figure B.18: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for Sparse PLT model, noise variables included, $N = 200$, $v = 20$.

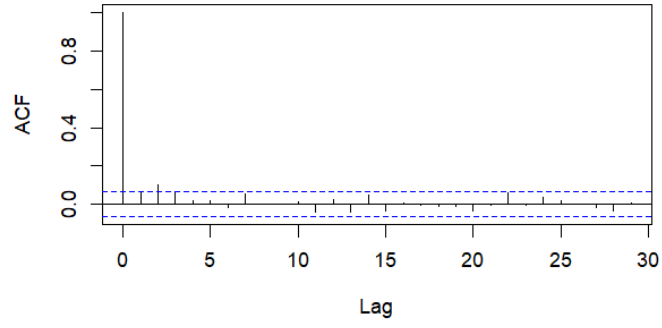


Figure B.19: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, no noise variables included, $N = 200$, $v = 20$.

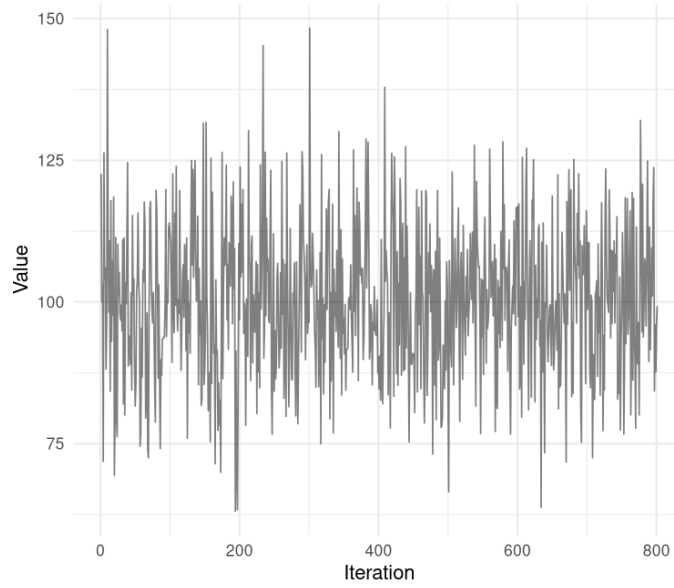


Figure B.20: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, no noise variables included, $N = 200$, $v = 20$.

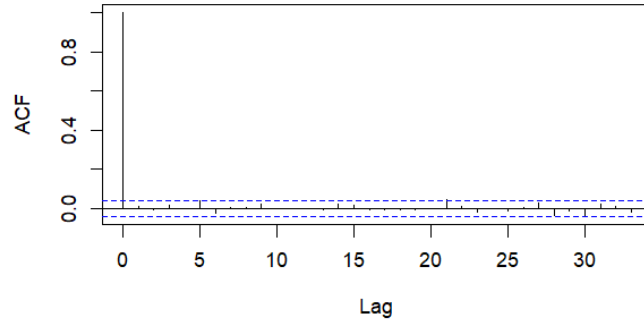


Figure B.21: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for Sparse PLT model, noise variables included, $N = 400$, $v = 10$.

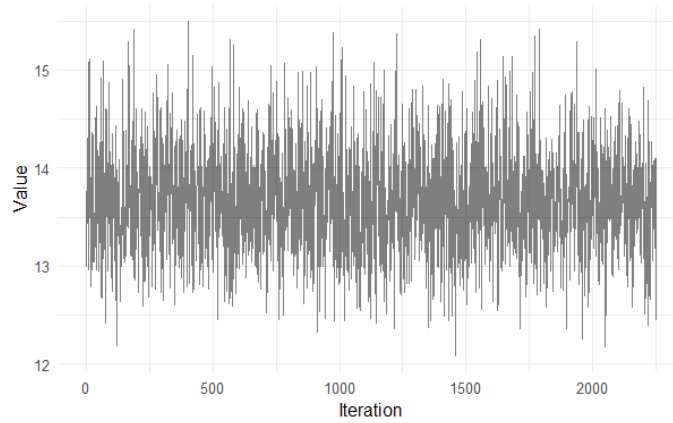


Figure B.22: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for Sparse PLT model, noise variables included, $N = 400$, $v = 10$.

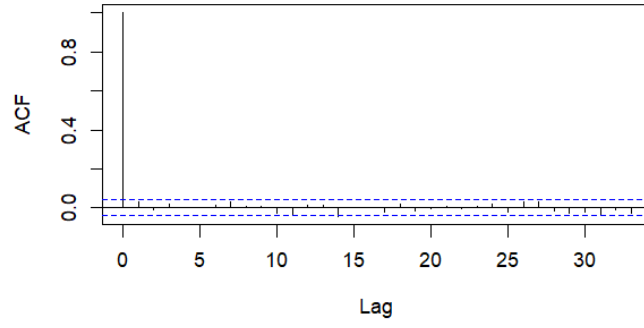


Figure B.23: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, no noise variables included, $N = 400$, $v = 10$.

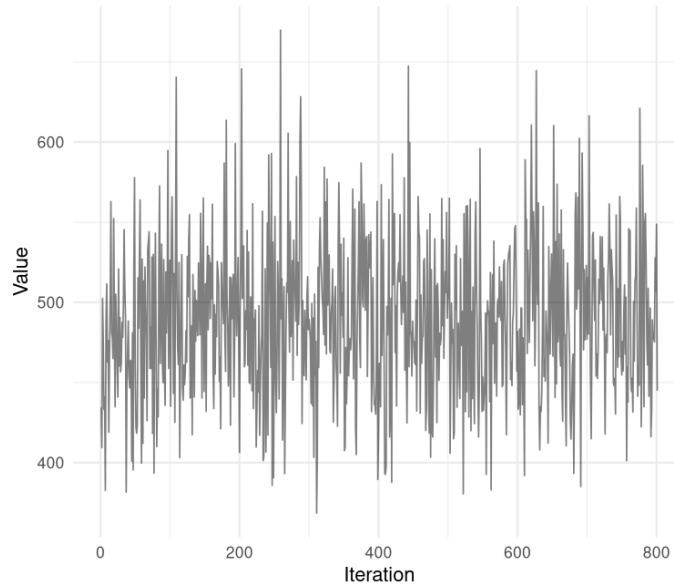


Figure B.24: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, no noise variables included, $N = 400$, $v = 10$.

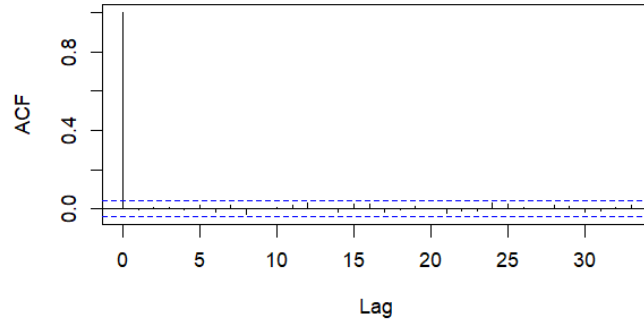


Figure B.25: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for Sparse PLT model, noise variables included, $N = 400$, $v = 20$.

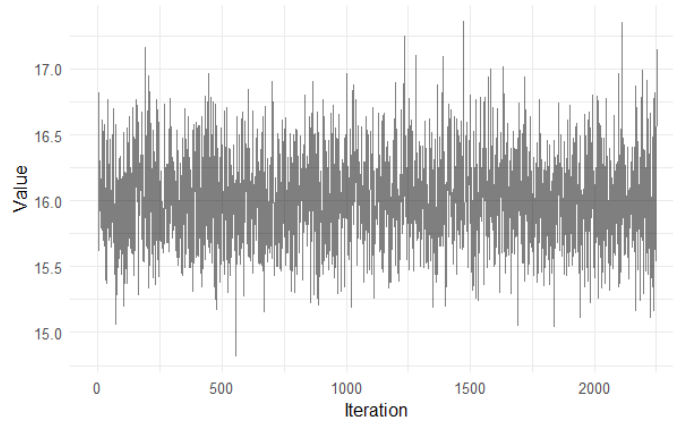


Figure B.26: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for Sparse PLT model, noise variables included, $N = 400$, $v = 20$.

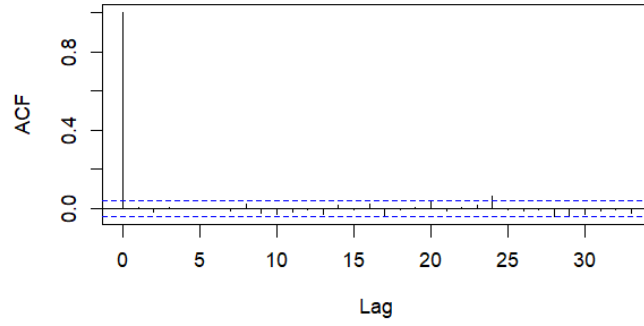


Figure B.27: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, no noise variables included, $N = 400$, $v = 20$.

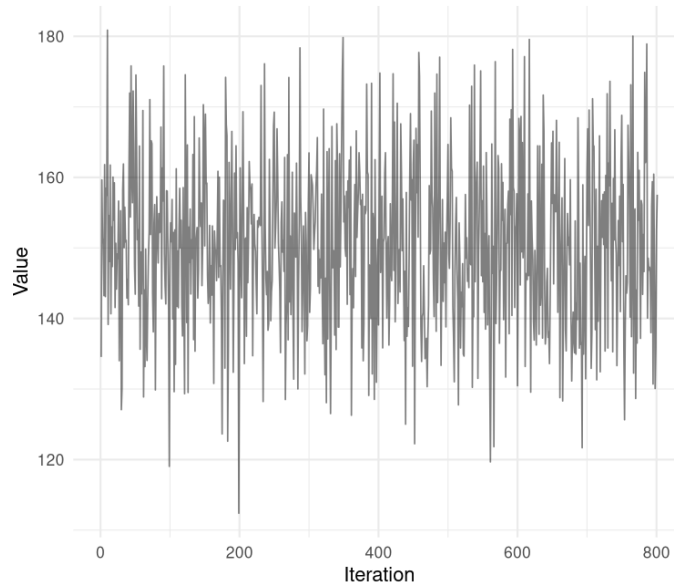


Figure B.28: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, no noise variables included, $N = 400$, $v = 20$.

B.1.3 UGLT Model

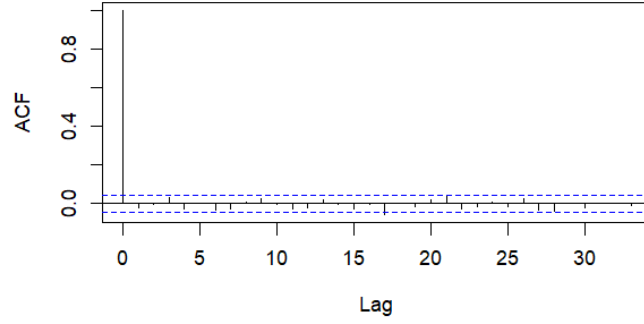


Figure B.29: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, noise variables included, $N = 200$, $v = 20$.

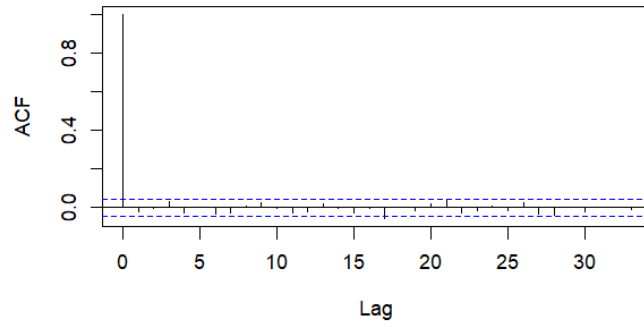


Figure B.30: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, noise variables included, $N = 200$, $v = 20$.

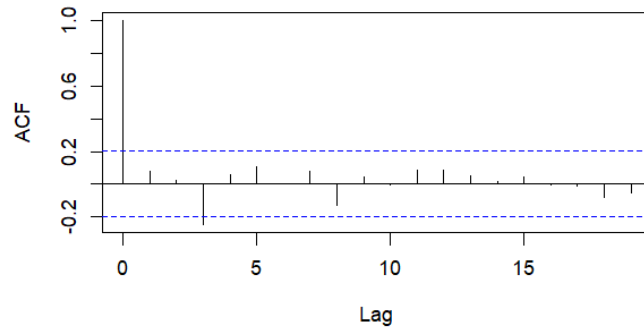


Figure B.31: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, no noise variables included, $N = 200$, $v = 20$.

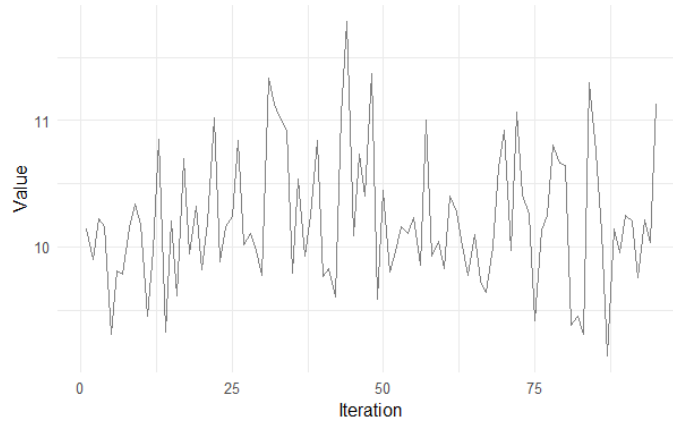


Figure B.32: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT, no noise variables included, $N = 200$, $v = 20$.

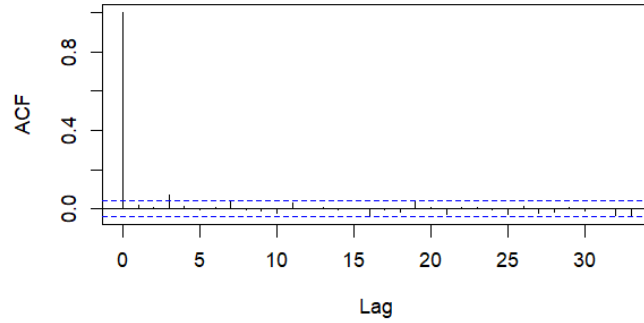


Figure B.33: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, noise variables included, $N = 400$, $v = 10$.

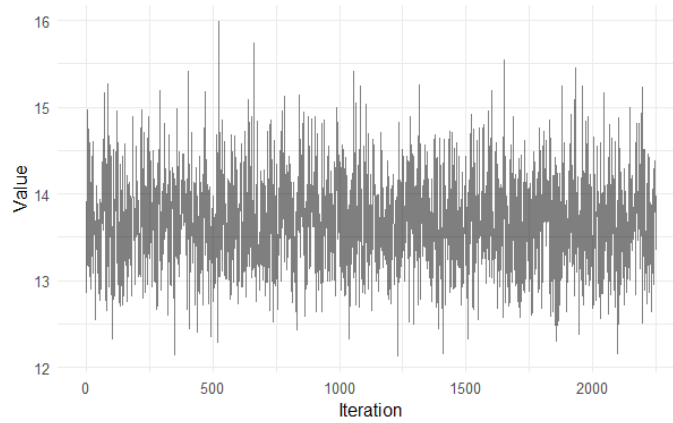


Figure B.34: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, noise variables included, $N = 400$, $v = 10$.

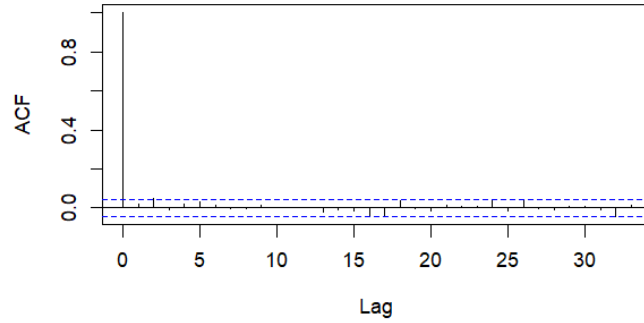


Figure B.35: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, no noise variables included, $N = 400$, $v = 10$.

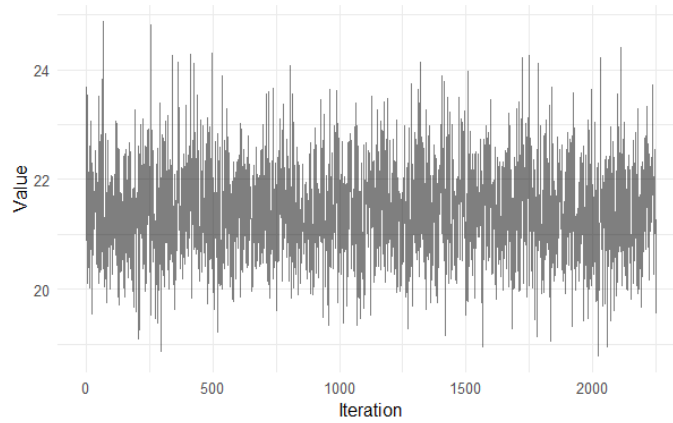


Figure B.36: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, no noise variables included, $N = 400$, $v = 10$.

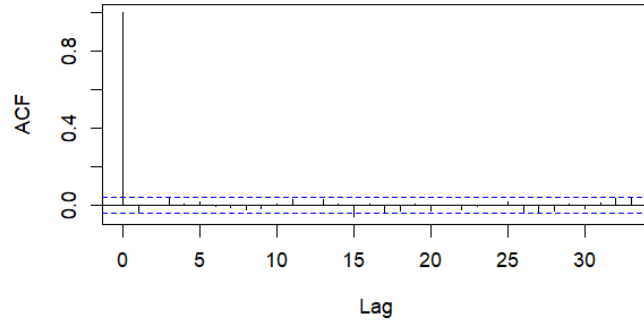


Figure B.37: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for UGLT model, noise variables included, $N = 400$, $v = 20$.

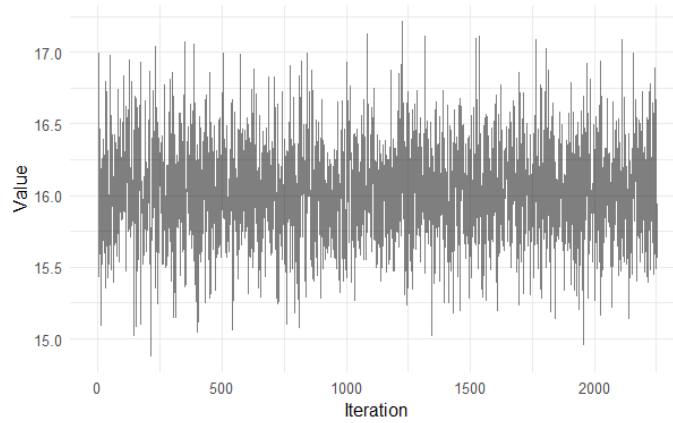


Figure B.38: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for UGLT model, noise variables included, $N = 400$, $v = 20$.

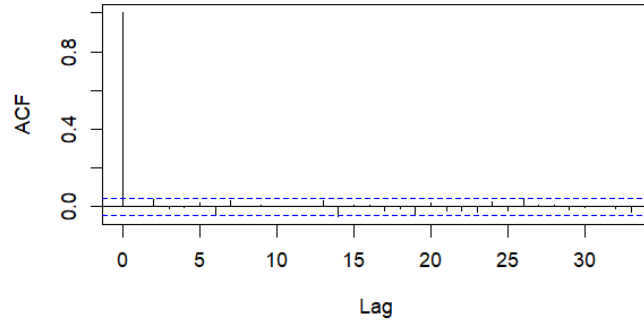


Figure B.39: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, no noise variables included, $N = 400$, $v = 20$.

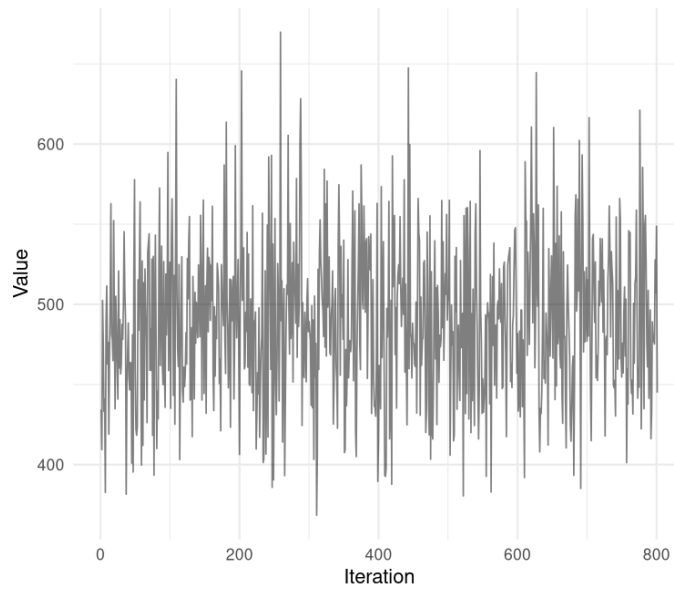


Figure B.40: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, no noise variables included, $N = 400$, $v = 20$.

B.2 Convergence Plots for Simulation Study 2

B.2.1 Sparse PLT Model

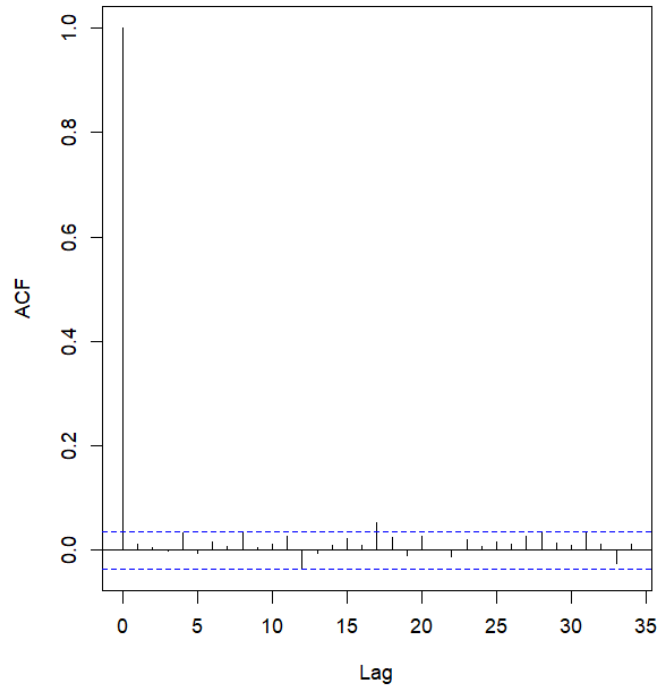


Figure B.41: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $v = 20$.

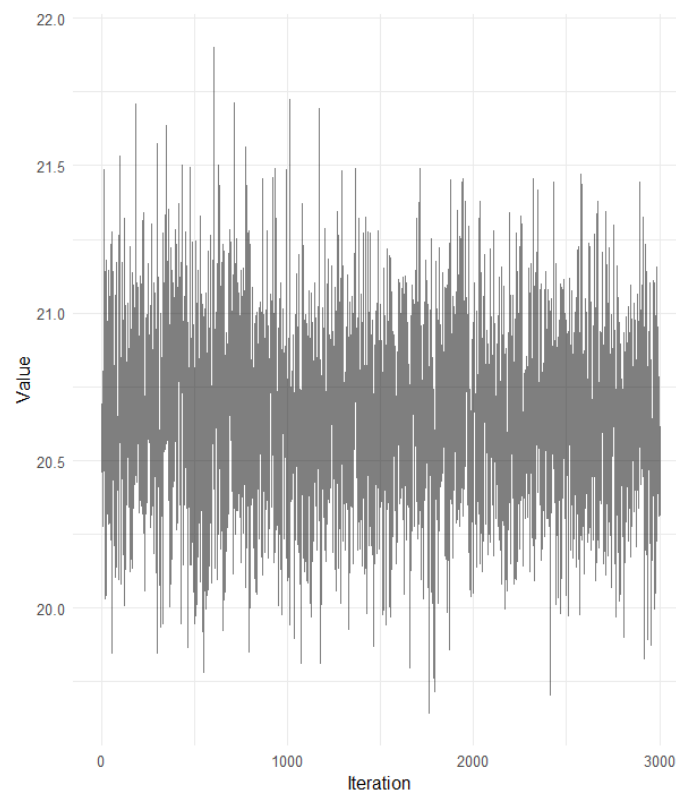


Figure B.42: Trace plot of $\text{tr}(\Lambda\Lambda^T + \Sigma)$ for the Sparse PLT model, $v = 20$.

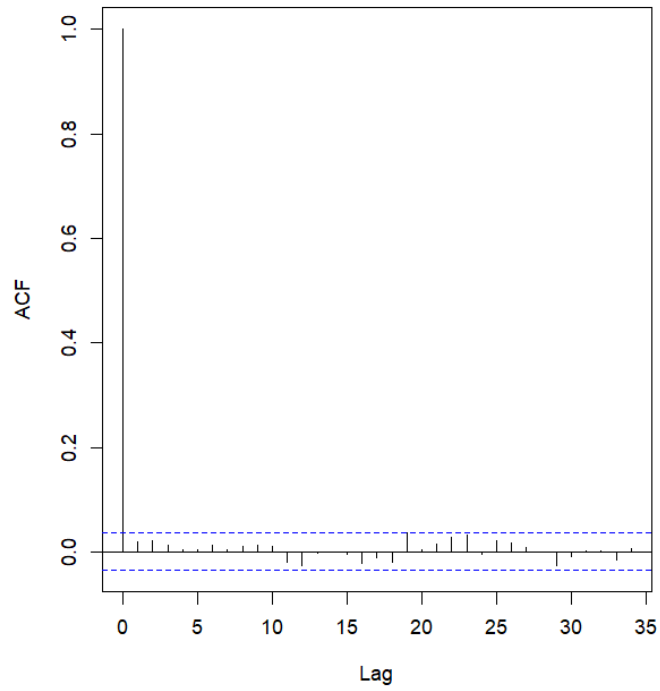


Figure B.43: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $v = 30$.

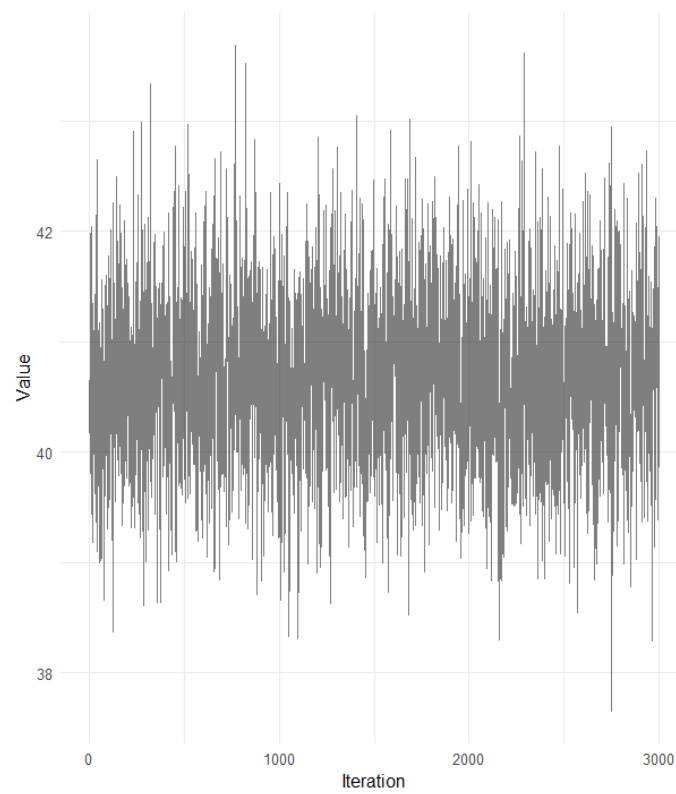


Figure B.44: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $v = 30$.

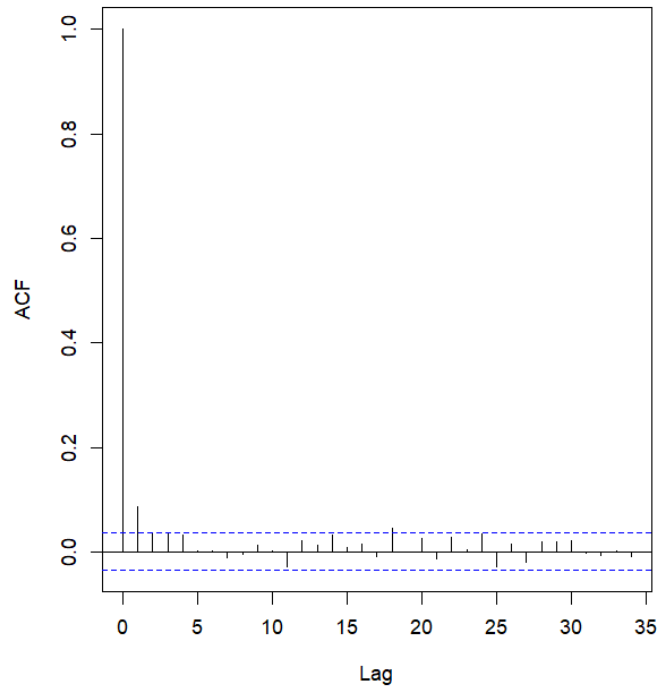


Figure B.45: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $v = 40$.

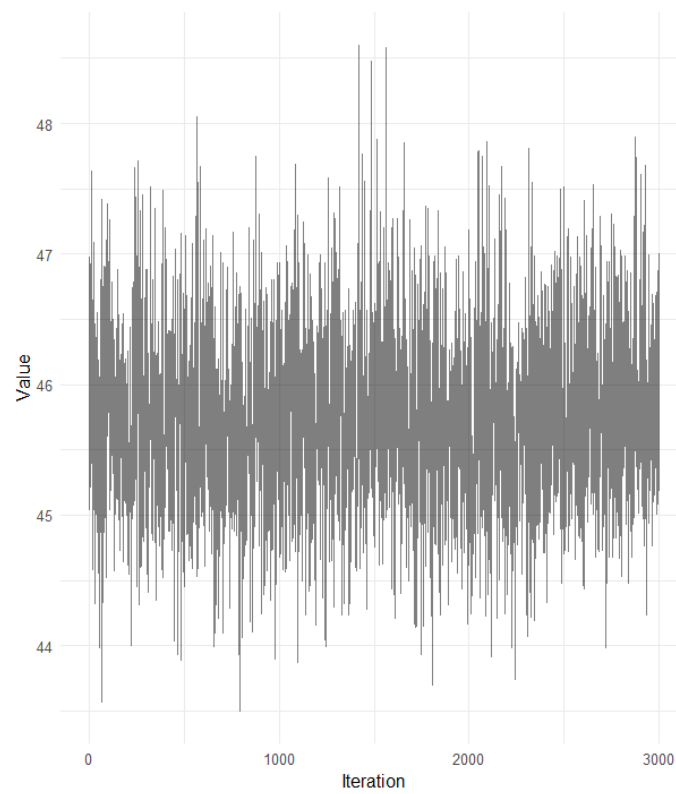


Figure B.46: Trace plot of $\text{tr}(\Lambda\Lambda^T + \Sigma)$ for the Sparse PLT model, $v = 40$.

B.2.2 UGLT Model

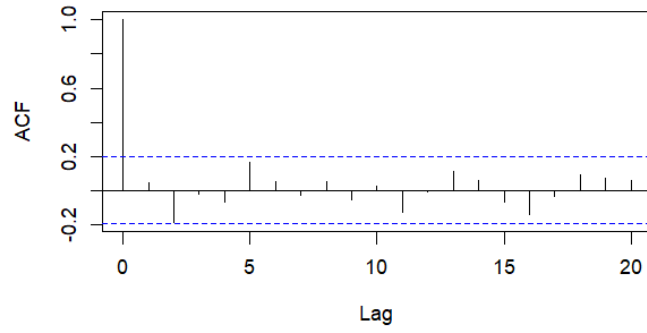


Figure B.47: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $v = 20$.

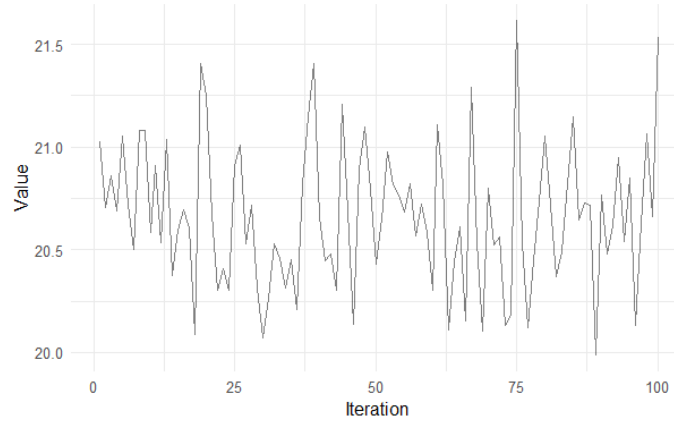


Figure B.48: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $v = 20$.

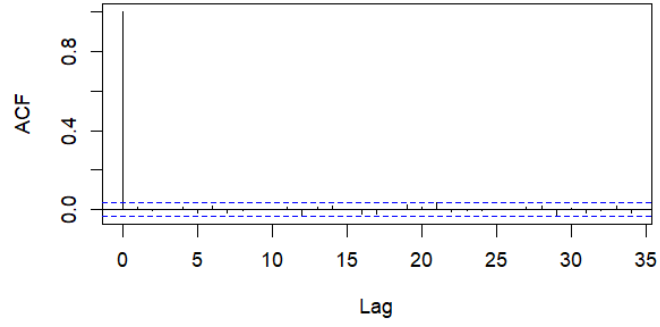


Figure B.49: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $v = 30$.

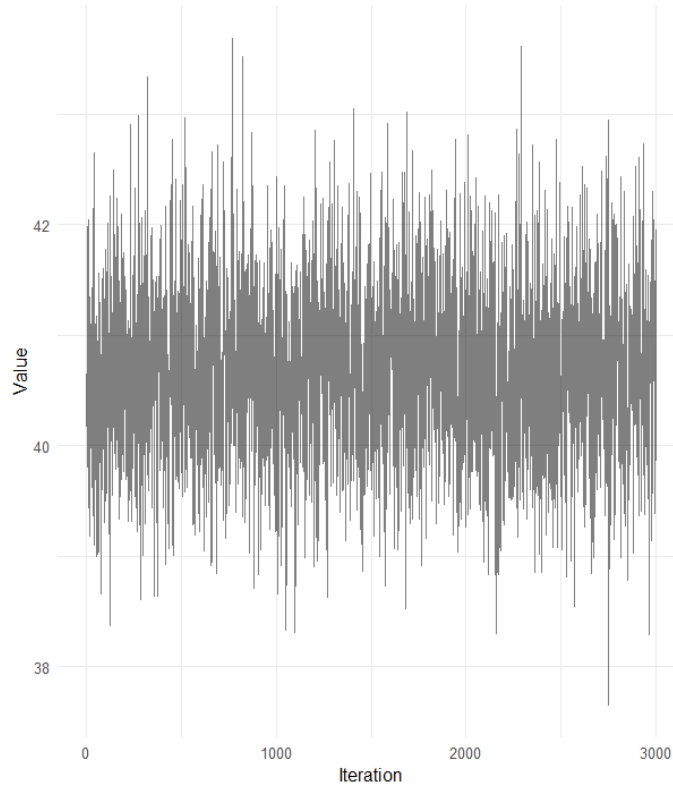


Figure B.50: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $v = 30$.

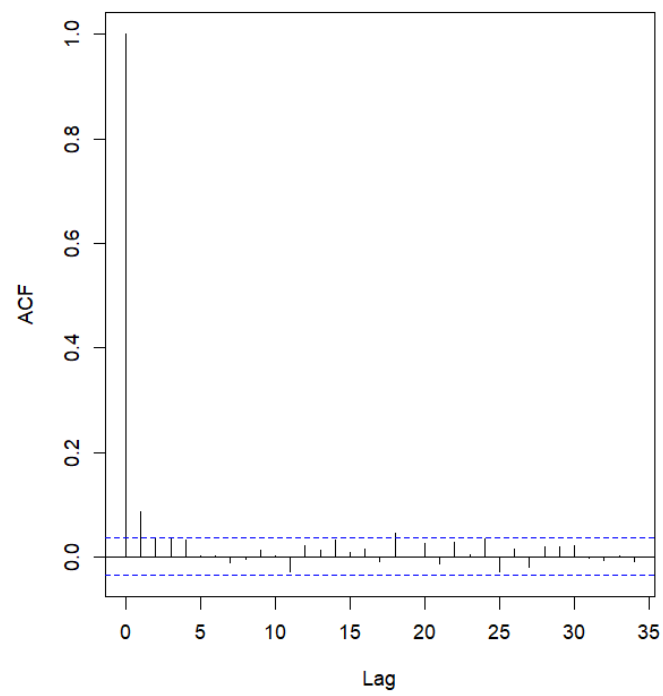


Figure B.51: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $v = 40$.

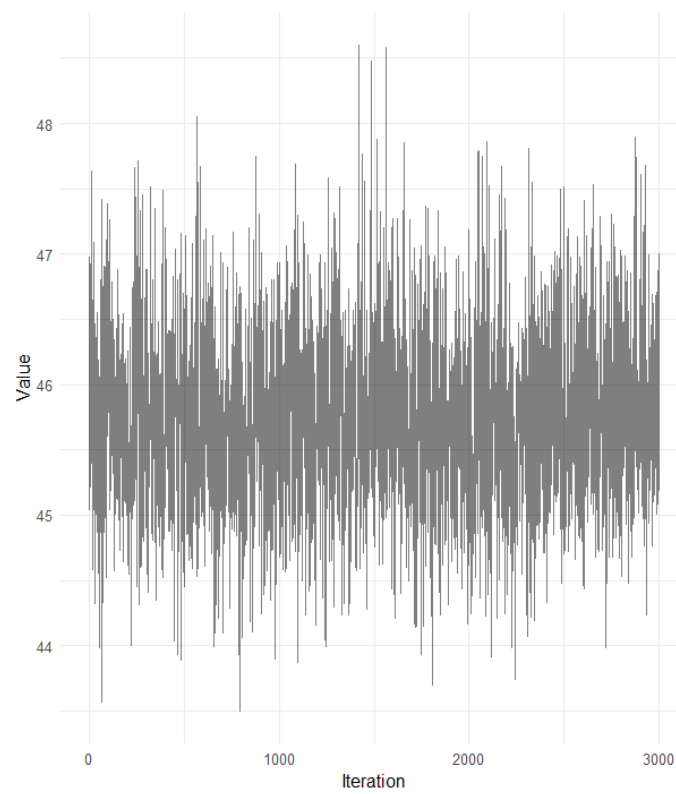


Figure B.52: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $v = 40$.

B.3 Convergence Plots for Simulation Study 3

B.3.1 Sparse PLT Model

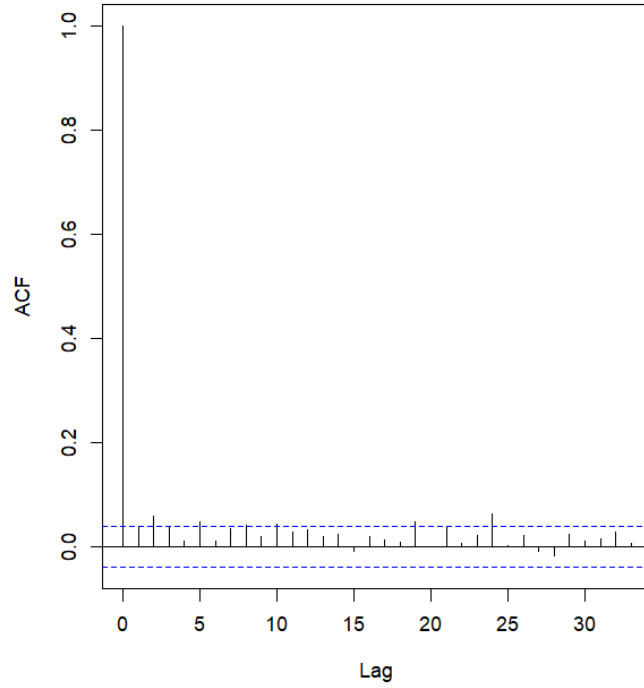


Figure B.53: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $q = 4$, UGLT structure.

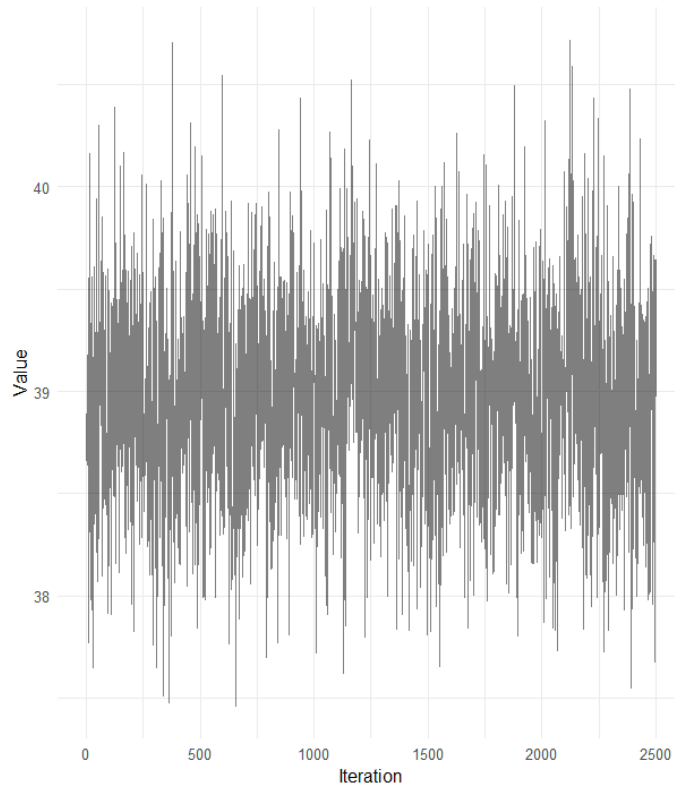


Figure B.54: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $q = 4$, UGLT structure.

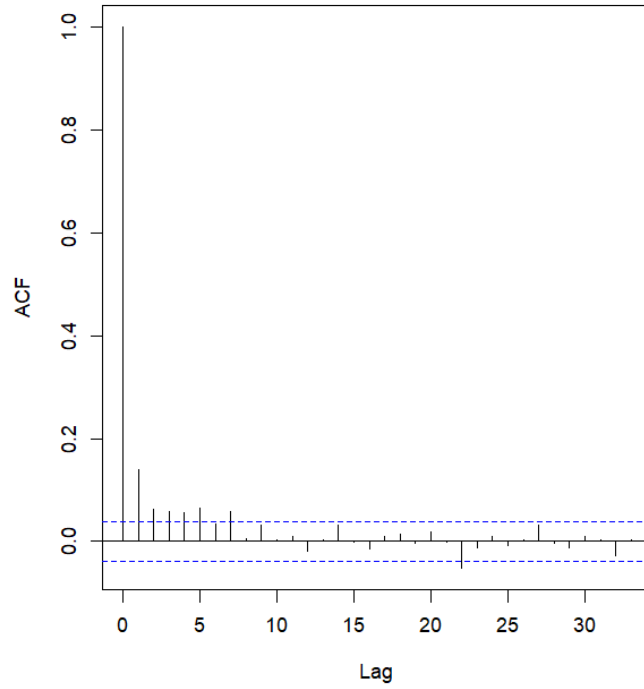


Figure B.55: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $q = 10$, UGLT structure.

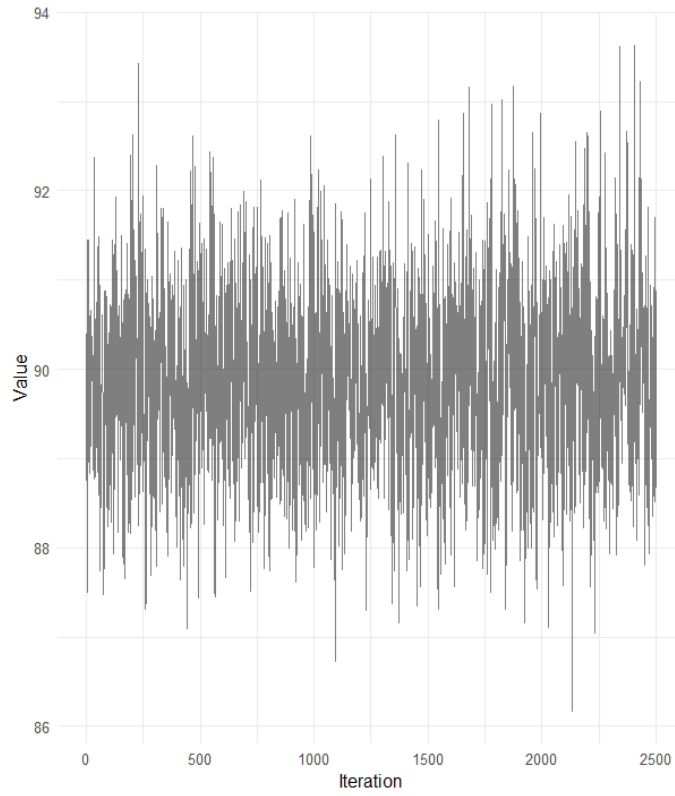


Figure B.56: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $q = 10$, UGLT structure.

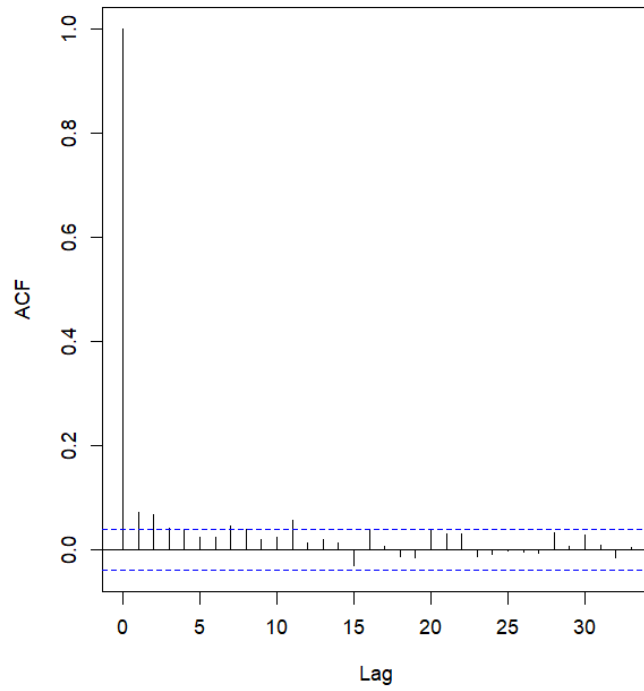


Figure B.57: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $q = 4$, PLT structure.

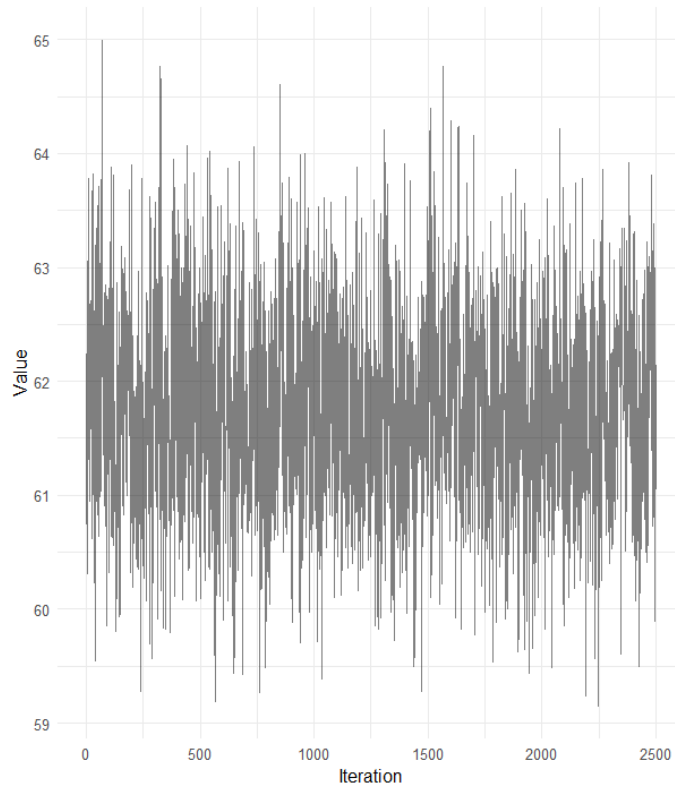


Figure B.58: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $q = 4$, PLT structure.

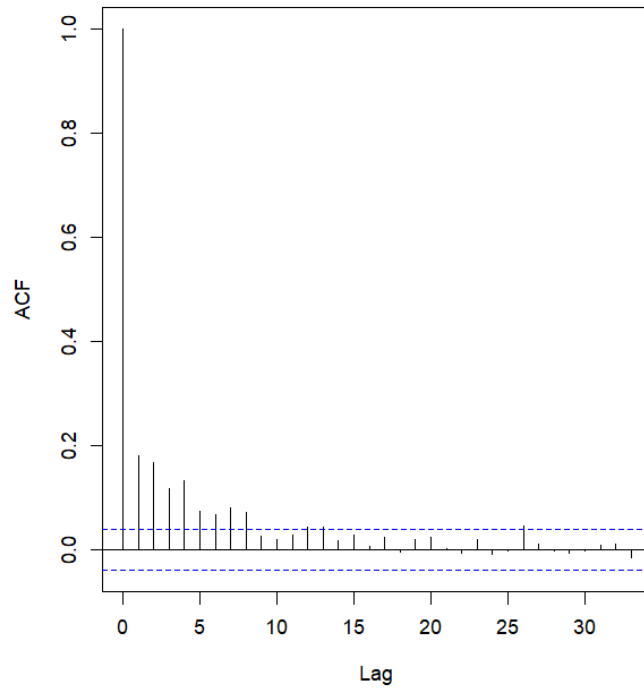


Figure B.59: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $q = 10$, PLT structure.

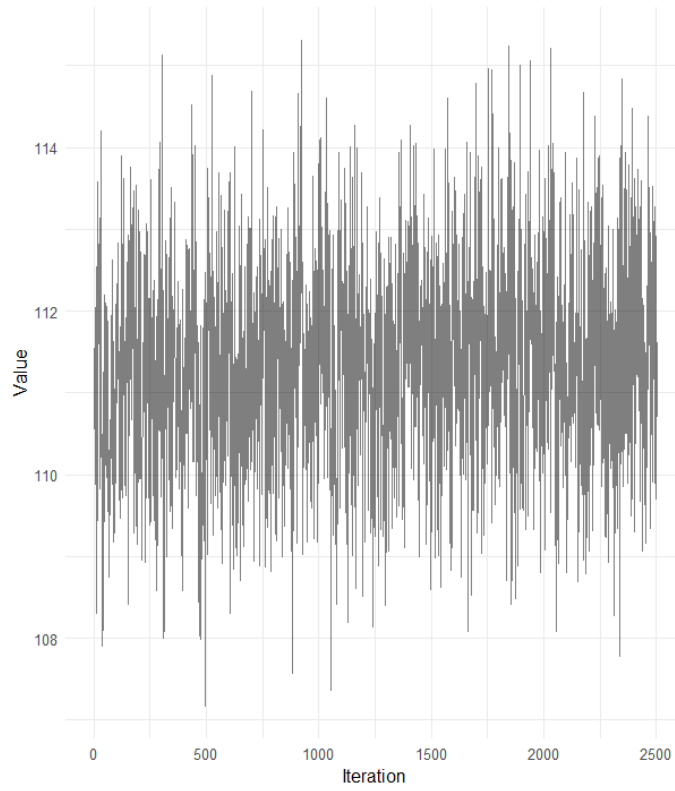


Figure B.60: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the Sparse PLT model, $q = 10$, PLT structure.

B.3.2 UGLT Model

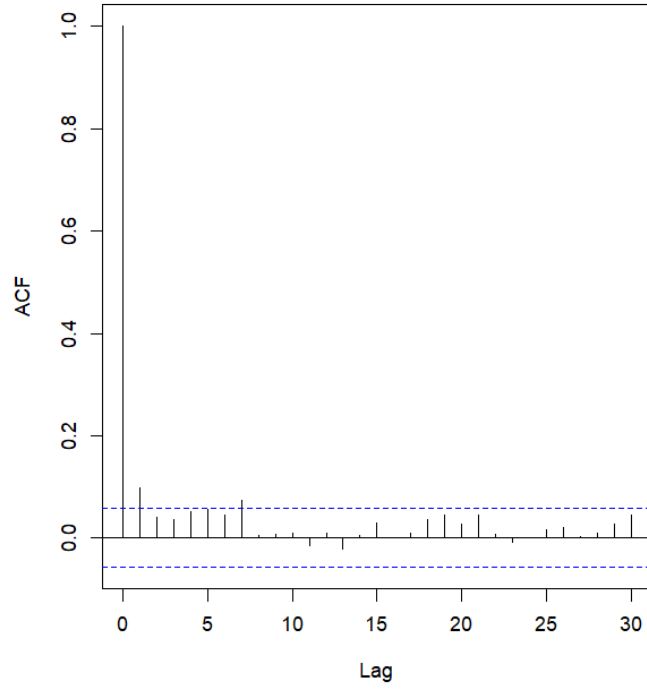


Figure B.61: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $q = 4$, UGLT structure.

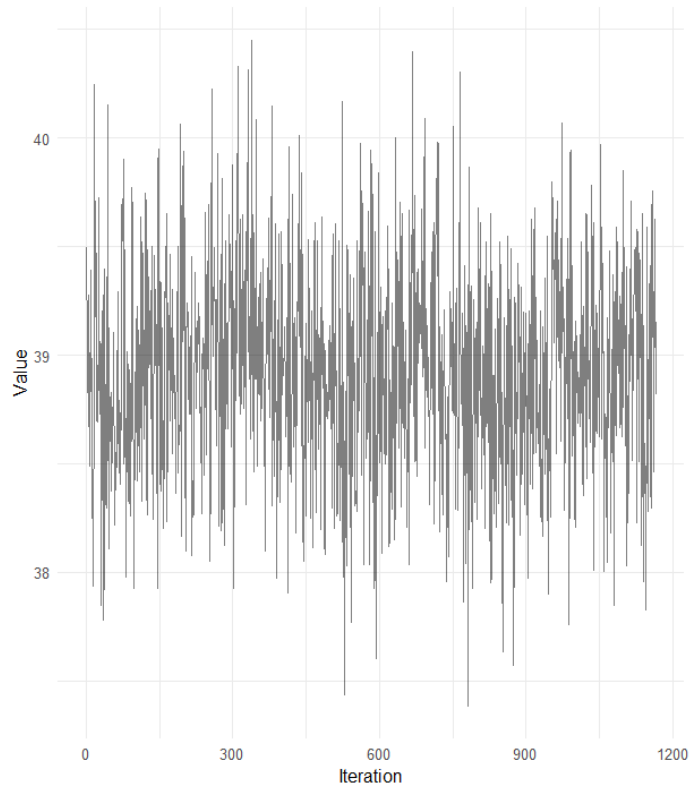


Figure B.62: Trace plot of $\text{tr}(\Lambda\Lambda^T + \Sigma)$ for the UGLT model, $q = 4$, UGLT structure.

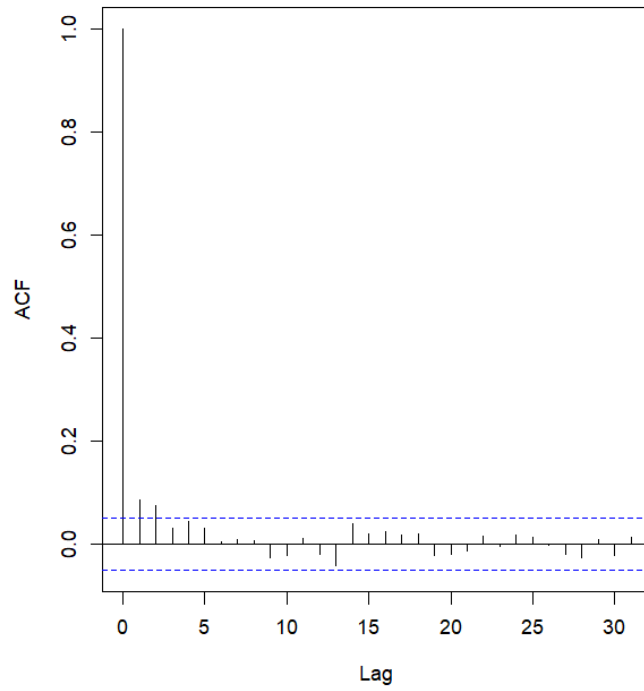


Figure B.63: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $q = 10$, UGLT structure.

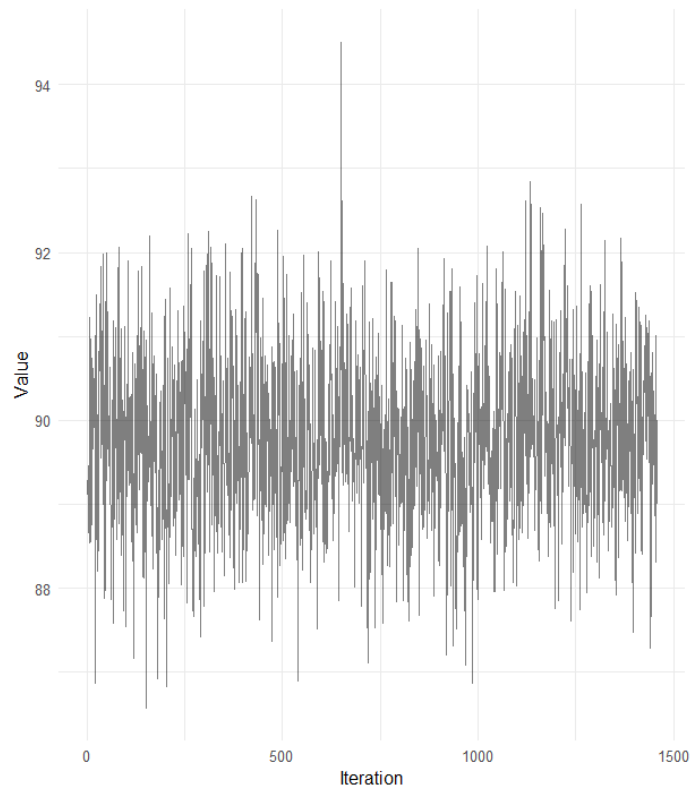


Figure B.64: Trace plot of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $q = 10$, UGLT structure.

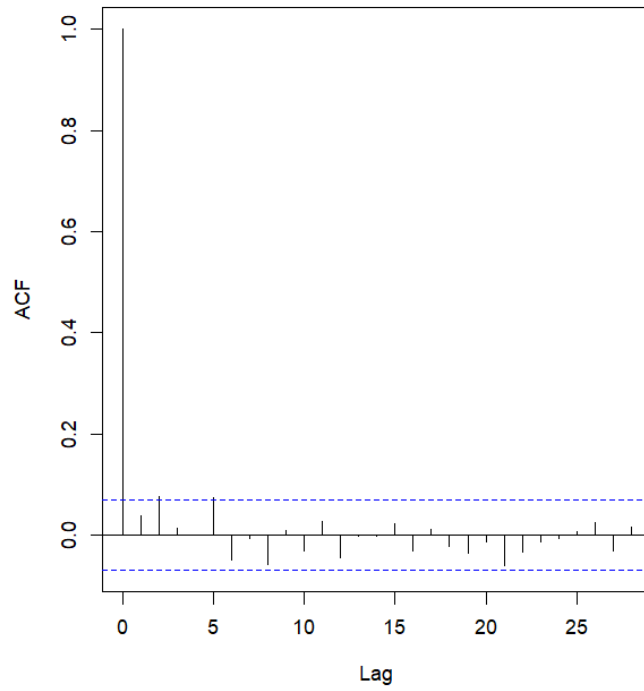


Figure B.65: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $q = 4$, PLT structure.

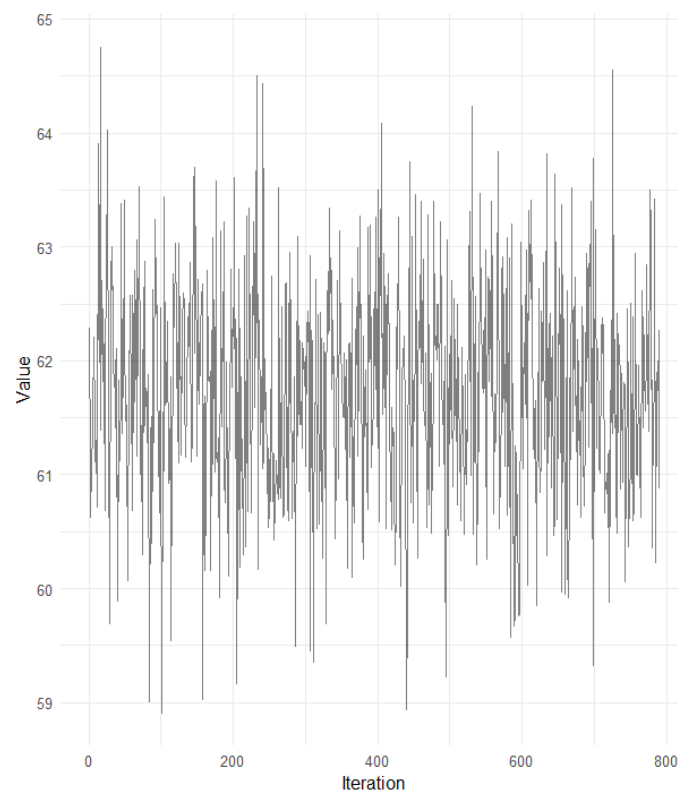


Figure B.66: Trace plot of $\text{tr}(\Lambda\Lambda^T + \Sigma)$ for the UGLT model, $q = 4$, PLT structure.

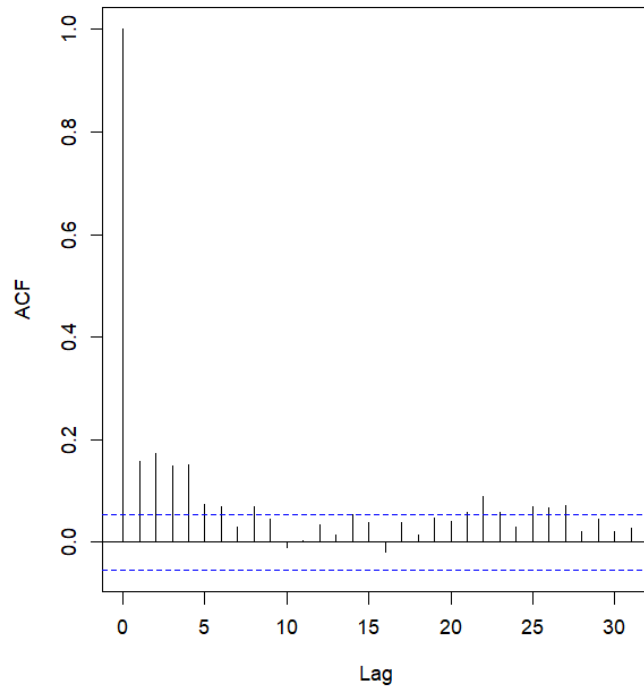


Figure B.67: ACF of $\text{tr}(\Lambda\Lambda^\top + \Sigma)$ for the UGLT model, $q = 10$, PLT structure.

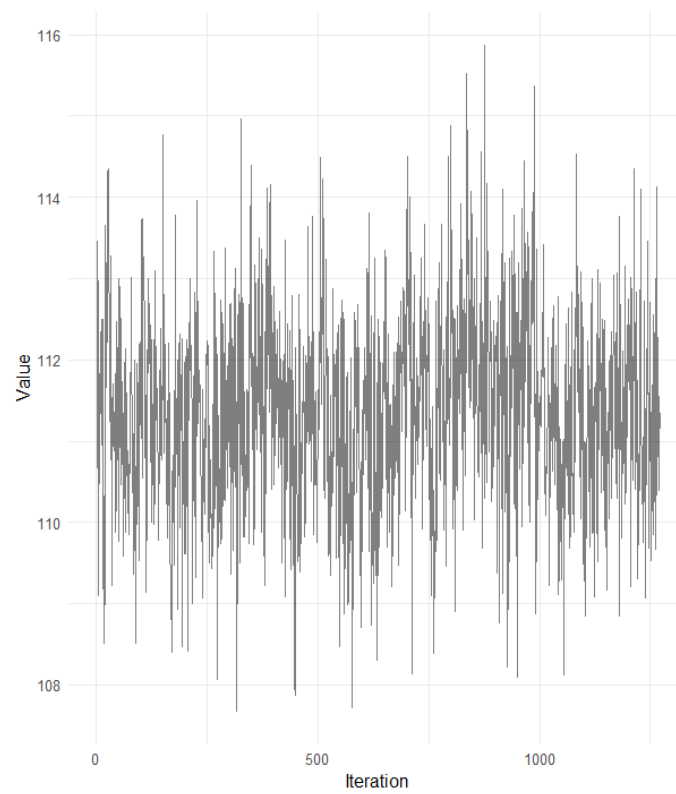


Figure B.68: Trace plot of $\text{tr}(\Lambda\Lambda^T + \Sigma)$ for the UGLT model, $q = 10$, PLT structure.

Bibliography

- T. W. Anderson, H. Rubin, et al. Statistical inference in factor analysis. In *Proceedings of the third Berkeley symposium on mathematical statistics and probability*, volume 5, pages 111–150, 1956.
- A. Bhattacharya and D. B. Dunson. Sparse bayesian infinite factor models. *Biometrika*, 98(2):291–306, 2011.
- G. Csárdi. *cli: Helpers for Developing Command Line Interfaces*, 2025. URL <https://CRAN.R-project.org/package=cli>. R package version 3.6.5.
- S. Frühwirth-Schnatter, D. Hosszejni, and H. F. Lopes. Sparse bayesian factor analysis when the number of factors is unknown (with discussion). *Bayesian Analysis*, 20(1):213–344, 2025a.
- S. Frühwirth-Schnatter, D. Hosszejni, and H. F. Lopes. Sparse bayesian factor analysis when the number of factors is unknown (with discussion). *Bayesian Analysis*, 20(1):213–344, 2025b.
- S. Frühwirth-Schnatter, D. Hosszejni, and H. F. Lopes. When it counts—econometric identification of the basic factor model based on glt structures. *Econometrics*, 11

- (4):26, Nov. 2023a. ISSN 2225-1146. doi: 10.3390/econometrics11040026. URL <http://dx.doi.org/10.3390/econometrics11040026>.
- S. Frühwirth-Schnatter, D. Hosszejni, and H. F. Lopes. When it counts—econometric identification of the basic factor model based on glt structures. *Econometrics*, 11(4):26, Nov. 2023b. ISSN 2225-1146. doi: 10.3390/econometrics11040026. URL <http://dx.doi.org/10.3390/econometrics11040026>.
- A. E. Gelfand and A. F. Smith. Sampling-based approaches to calculating marginal densities. *Journal of the American statistical association*, 85(410):398–409, 1990.
- A. Gelman, J. B. Carlin, H. S. Stern, and D. B. Rubin. *Bayesian data analysis*. Chapman and Hall/CRC, 1995.
- S. Geman and D. Geman. Stochastic relaxation, gibbs distributions, and the bayesian restoration of images. *IEEE Transactions on pattern analysis and machine intelligence*, (6):721–741, 1984.
- J. Geweke. Evaluating the accuracy of sampling-based approaches to the calculations of posterior moments. *Bayesian statistics*, 4:641–649, 1992.
- J. Geweke and G. Zhou. Measuring the pricing error of the arbitrage pricing theory. *The review of financial studies*, 9(2):557–587, 1996.
- T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American statistical Association*, 102(477):359–378, 2007.
- W. K. Hastings. Monte carlo sampling methods using markov chains and their applications. 1970.

- D. Hosszejni and S. Frühwirth-Schnatter. Cover it up! bipartite graphs uncover identifiability in sparse factor analysis. *Journal of Multivariate Analysis*, 211:105536, 2026. ISSN 0047-259X. doi: 10.1016/j.jmva.2025.105536.
- A. Jordan, F. Krüger, and S. Lerch. Evaluating probabilistic forecasts with scoringRules. *Journal of Statistical Software*, 90(12):1–37, 2019. doi: 10.18637/jss.v090.i12.
- S. Kaufmann and C. Schumacher. Identifying relevant and irrelevant variables in sparse factor models. *Journal of Applied Econometrics*, 32(6):1123–1144, 2017.
- J. Leydold and W. Hormann. *GIGrvg: Random Variate Generator for the GIG Distribution*, 2023. URL <https://CRAN.R-project.org/package=GIGrvg>. R package version 0.8.
- J. S. Liu. *Monte Carlo strategies in scientific computing*, volume 10. Springer, 2001.
- J. S. Liu and Y. N. Wu. Parameter expansion for data augmentation. *Journal of the American Statistical Association*, 94(448):1264–1274, 1999. ISSN 01621459, 1537274X. URL <http://www.jstor.org/stable/2669940>.
- H. F. Lopes and M. West. Bayesian model assessment in factor analysis. *Statistica Sinica*, pages 41–67, 2004.
- N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller. Equation of state calculations by fast computing machines. *The journal of chemical physics*, 21(6):1087–1092, 1953.
- K. Müller. *here: A Simpler Way to Find Your Files*, 2025. URL <https://CRAN.R-project.org/package=here>. R package version 1.0.2.

- T. L. Pedersen. *patchwork: The Composer of Plots*, 2025. URL <https://CRAN.R-project.org/package=patchwork>. R package version 1.3.2.
- R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2025. URL <https://www.R-project.org/>.
- V. Roy. Convergence diagnostics for markov chain monte carlo. *Annual Review of Statistics and Its Application*, 7(1):387–412, 2020.
- Statisticat and LLC. *LaplacesDemon: Complete Environment for Bayesian Inference*, 2026. URL <https://web.archive.org/web/20150206004624/http://www.bayesian-inference.com/software>. R package version 16.1.8.
- M. A. Tanner and W. H. Wong. The calculation of posterior distributions by data augmentation. *Journal of the American Statistical Association*, 82(398):528–540, 1987. ISSN 01621459, 1537274X. URL <http://www.jstor.org/stable/2289457>.
- D. A. Van Dyk and T. Park. Partially collapsed gibbs samplers: Theory and methods. *Journal of the American Statistical Association*, 103(482):790–796, 2008.
- W. N. Venables and B. D. Ripley. *Modern Applied Statistics with S*. Springer, New York, fourth edition, 2002. URL <https://www.stats.ox.ac.uk/pub/MASS4/>. ISBN 0-387-95457-0.
- H. Wickham. Reshaping data with the reshape package. *Journal of Statistical Software*, 21(12):1–20, 2007. URL <https://www.jstatsoft.org/v21/i12/>.
- H. Wickham and L. Henry. *purrr: Functional Programming Tools*, 2023. URL <https://CRAN.R-project.org/package=purrr>. R package version 1.0.2.

H. Wickham, M. Averick, J. Bryan, W. Chang, L. D. McGowan, R. François, G. Grolemund, A. Hayes, L. Henry, J. Hester, M. Kuhn, T. L. Pedersen, E. Miller, S. M. Bache, K. Müller, J. Ooms, D. Robinson, D. P. Seidel, V. Spinu, K. Takahashi, D. Vaughan, C. Wilke, K. Woo, and H. Yutani. Welcome to the tidyverse. *Journal of Open Source Software*, 4(43):1686, 2019. doi: 10.21105/joss.01686.